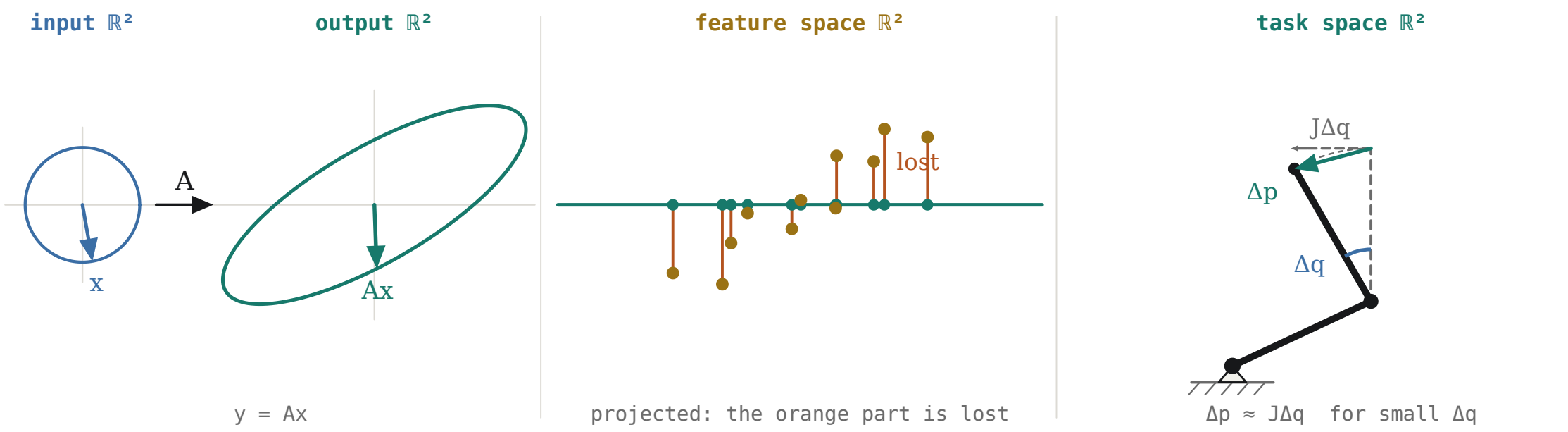
LECTURE 4 · LINEAR ALGEBRA IV · SEPTEMBER 23

# From Projections to Matrix Structure

PSD, SVD, PCA, Pseudoinverse, Jacobian and Hessian




$$y = Ax, \qquad \Delta p \approx J\,\Delta q$$

$A \in \mathbb{R}^{m\times n}$ input $x \in \mathbb{R}^n$ to output $y \in \mathbb{R}^m$

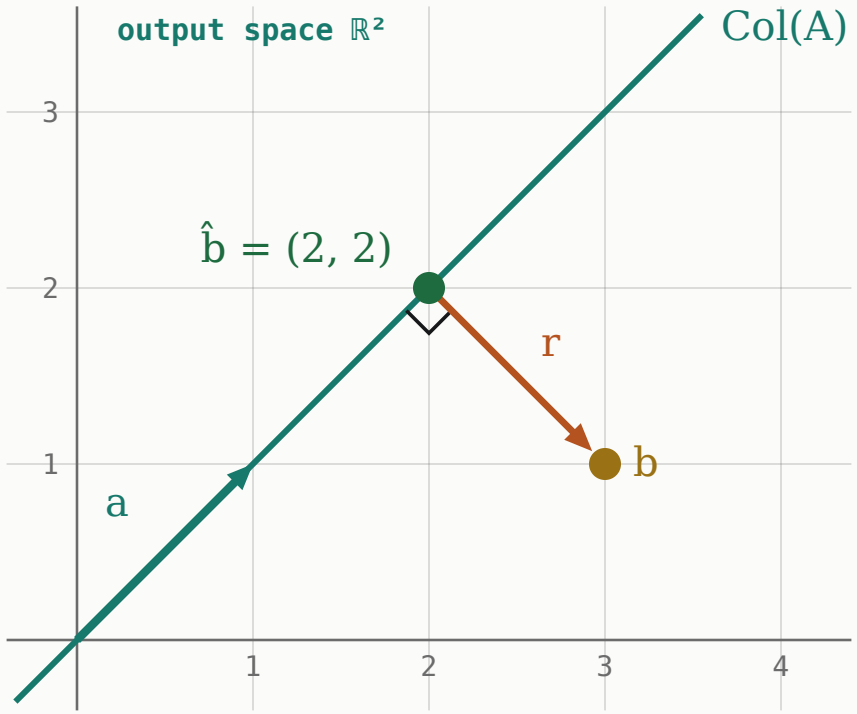
$J$ joint change to tip change; $\approx$ needs small $\Delta q$

> Which directions matter? What can we recover?

$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

# From Projections to Matrix Structure

## Recall: Least Squares Chooses the Nearest Reachable Output




$$\hat{b} = A\hat{x}, \qquad r = b - \hat{b}, \qquad A^{\mathsf{T}} r = 0$$

- $\hat{b}$ nearest reachable output
- $r$ residual, $\hat{b} \to b$
- $A^{\mathsf{T}} r = 0$ $r \perp$ every column of $A$

**WORKED EXAMPLE**

$A = [\,a\,] \in \mathbb{R}^{2\times 1}$, $\hat{x} \in \mathbb{R}$: $\hat{x} = \dfrac{a^{\mathsf{T}} b}{a^{\mathsf{T}} a} = \dfrac{4}{2} = 2$, $r = b - 2a = (1, -1)$.

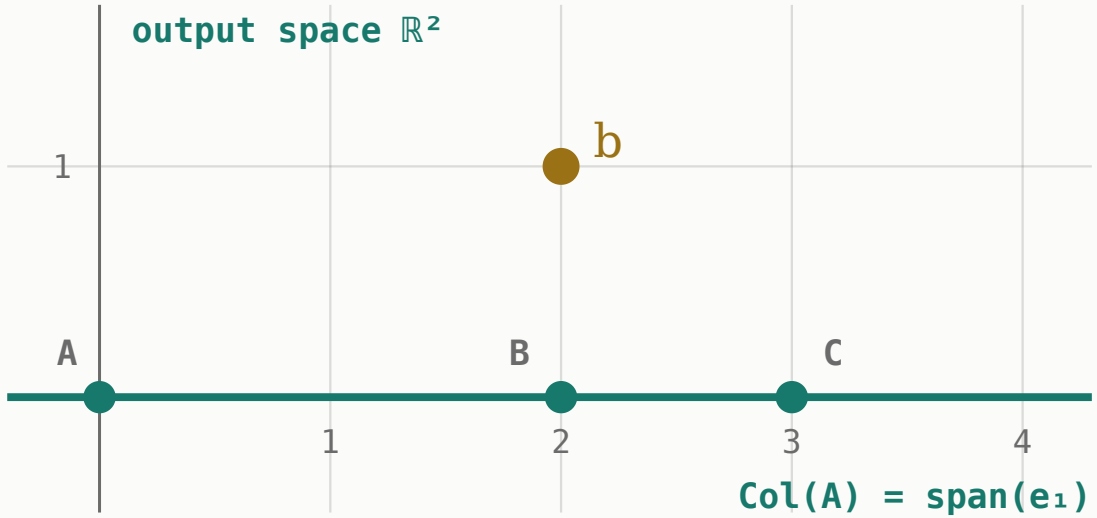
VOTE

# From Projections to Matrix Structure

## Which Candidate Output Is Closest?

$\mathrm{Col}(A) = \mathrm{span}(e_1)$ and $b = (2, 1)$. Which candidate output is closest to $b$?

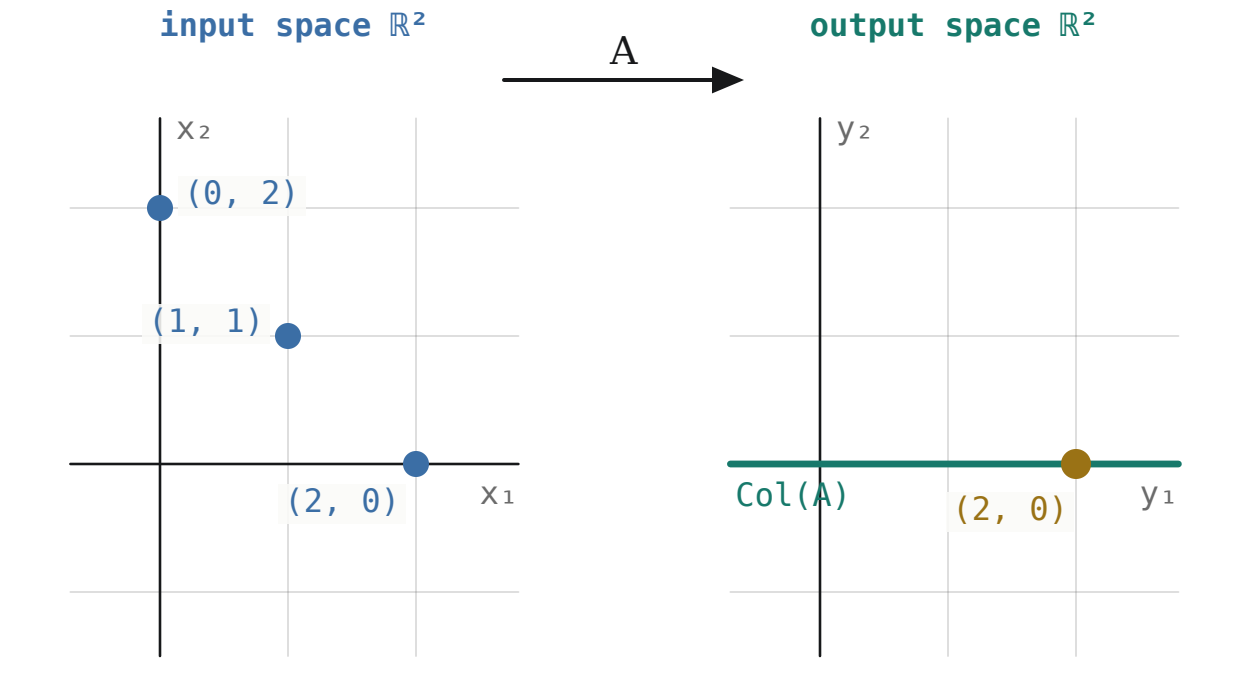
- A $(0, 0)$
- B $(2, 0)$
- C $(3, 0)$



VOTE

# From Projections to Matrix Structure

## Which Inputs Produce This Output?




$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. Which inputs give the output $(2, 0)$?

A only $(1, 1)$

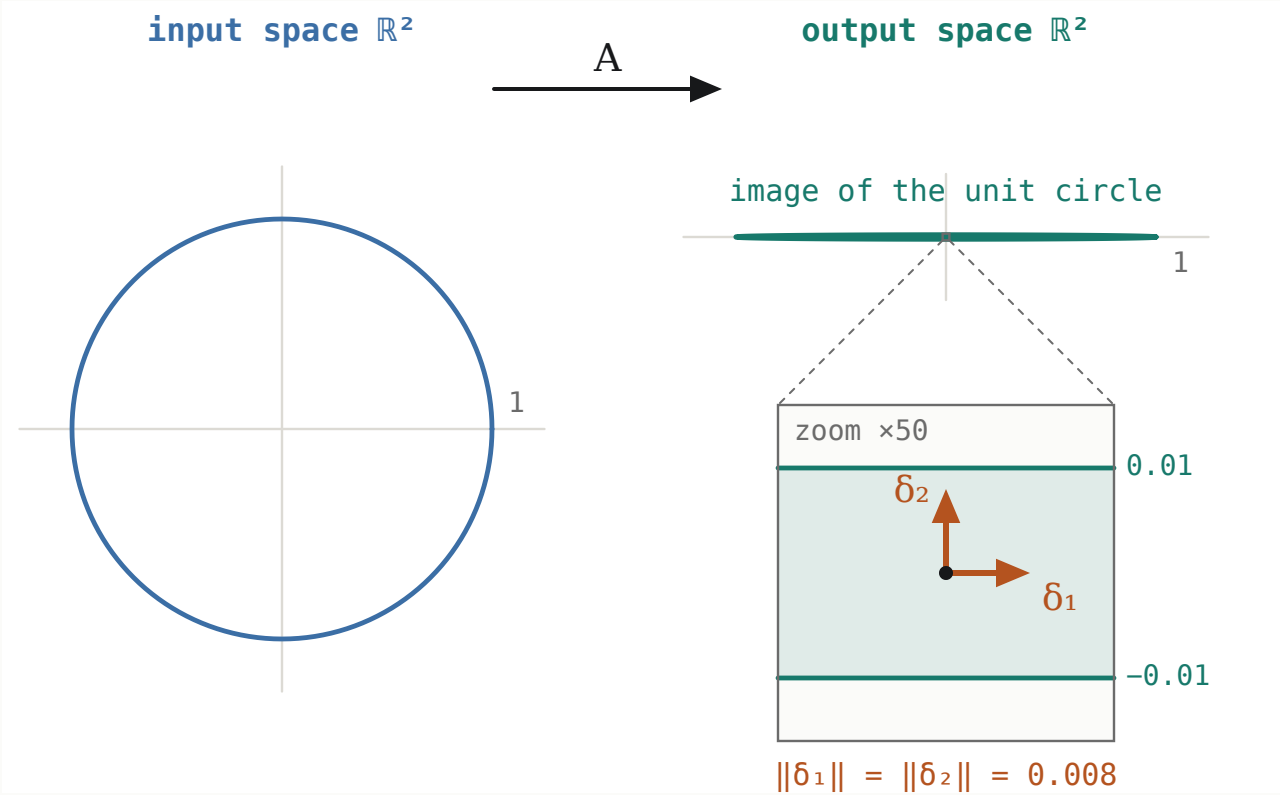
B only $(2, 0)$

C all three

VOTE

# From Projections to Matrix Structure

## Which Measurement Error Grows When We Invert?




$A = \mathrm{diag}(1, 0.01)$. Two output errors have equal length. Which grows more when we recover $x = A^{-1}y$?

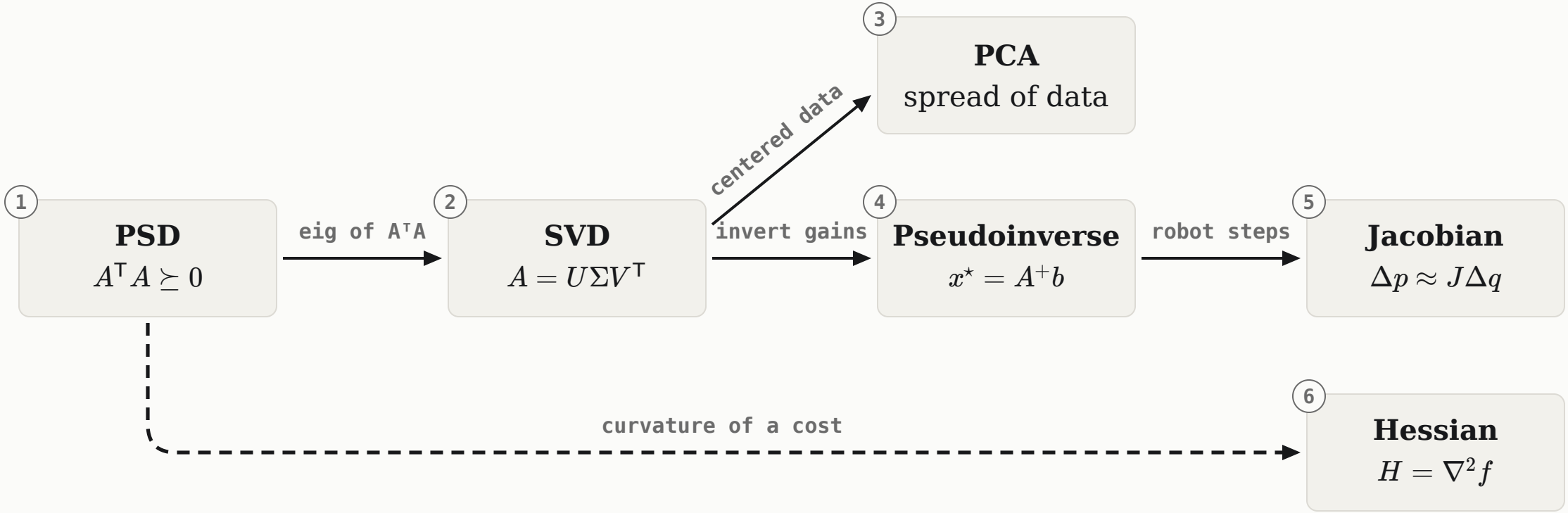
×50 zoom: read the numbers, not the lengths.

- **A** horizontal $\delta_1$
- **B** vertical $\delta_2$
- **C** equally

CONCEPT MAP

# From Projections to Matrix Structure

## One Structure, Six Tools




Why loop back: near a stationary point, a smooth cost changes like the quadratic form of its Hessian.

DEFINITION

# Positive Semidefinite Matrices

## A Quadratic Form Gives Each Displacement a Height

$$q(x) = x^\mathsf{T} H x = a x_1^2 + 2 b x_1 x_2 + c x_2^2$$

$x \in \mathbb{R}^2$ the input displacement

$H = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ real and symmetric, $H = H^\mathsf{T}$

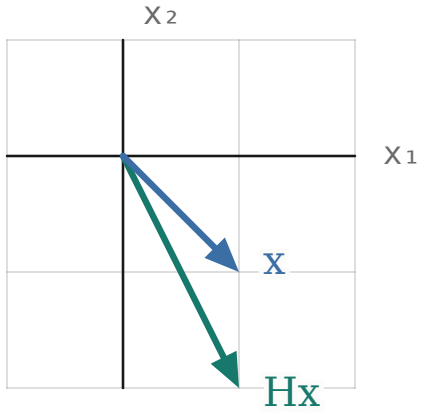
$Hx \in \mathbb{R}^2$ a vector

$q(x) \in \mathbb{R}$ one number: the height, or cost, of $x$

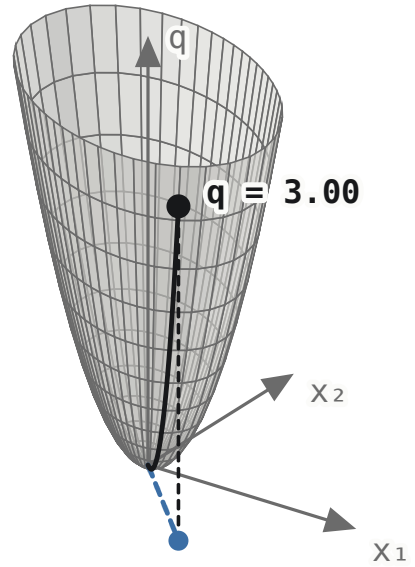
**WORKED EXAMPLE**

$H = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $x = (1, -1)$: $Hx = (1, -2)$, so $x^\mathsf{T}(Hx) = 1 + 2 = 3$. Check: $2 - 2 + 3 = 3$.

**input space $\mathbb{R}^2$**




**height q(x) over $\mathbb{R}^2$**




x = (1.00, −1.00)

Hx = (1.00, −2.00) a vector

**xᵀ(Hx) = 3.00** one number

VOTE

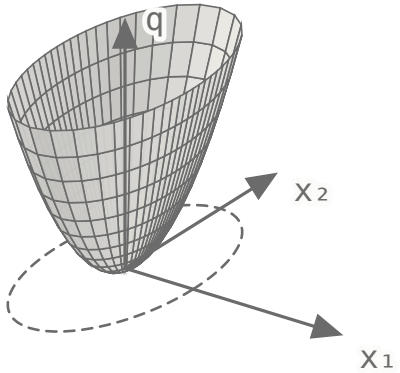
# Positive Semidefinite Matrices

## Which Surface Matches the Formula?

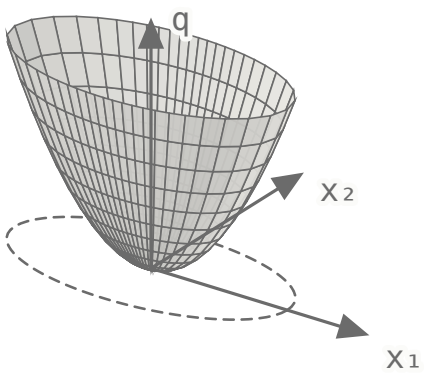
Which surface is the graph of $q(x) = 4x_1^2 + x_2^2$? Same camera, range and scales in all three.

**floor: input space $\mathbb{R}^2$, axes $x_1$, $x_2$ · height: q(x)**

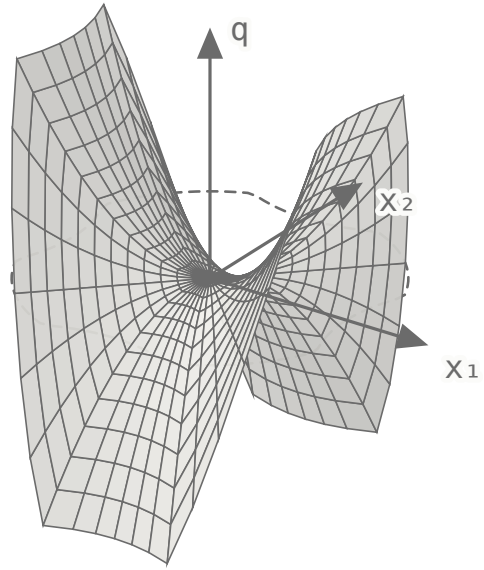
**A**




**B**




**C**




A surface A

B surface B

C surface C

DEFINITION

# Positive Semidefinite Matrices

## Positive Definite and Positive Semidefinite

$$H \succ 0 \iff x^\mathsf{T} H x > 0 \text{ for all } x \neq 0$$

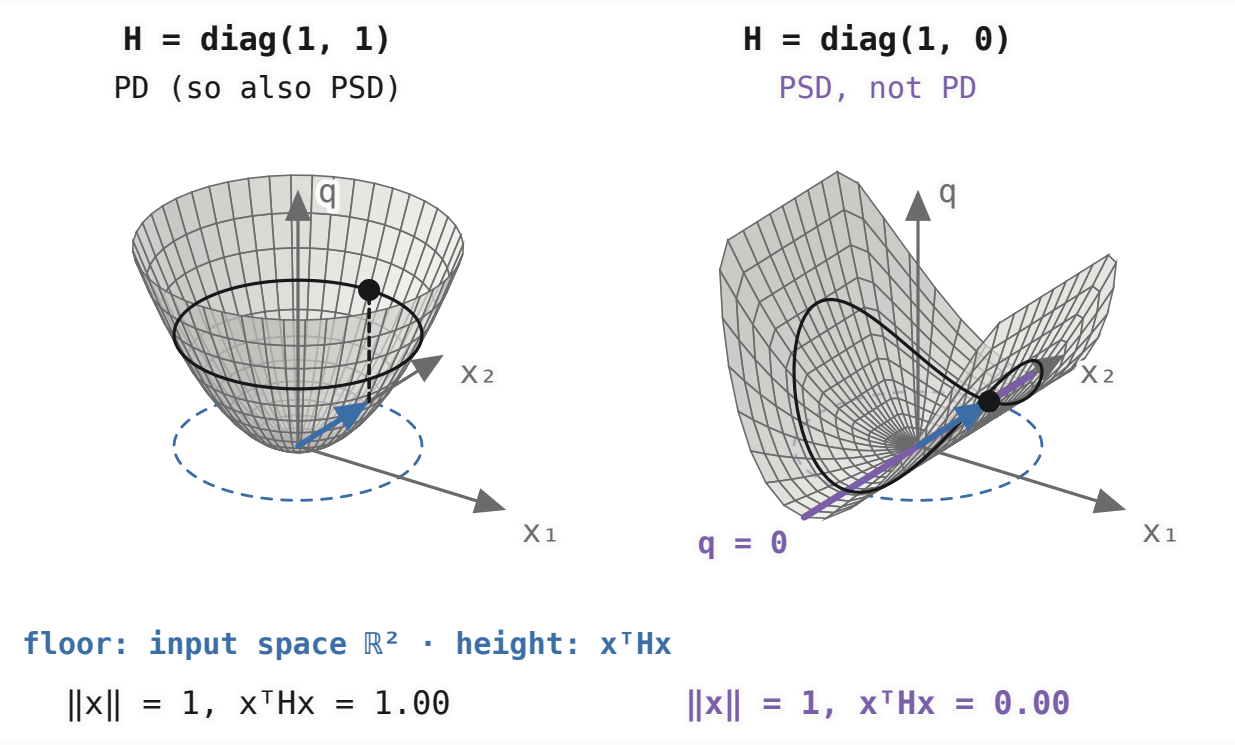
$$H \succeq 0 \iff x^\mathsf{T} H x \geq 0 \text{ for all } x$$

$H = H^\mathsf{T}$ real $n \times n$; both tests assume it

$H \succ 0$ positive definite (PD)

$H \succeq 0$ positive semidefinite (PSD)

Every PD matrix is PSD. The trough is *one* PSD matrix that is not PD; the zero matrix is another.



VOTE

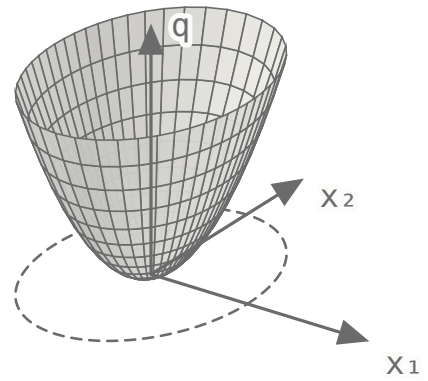
# Positive Semidefinite Matrices

## Which Matrix Is PSD but Not PD?

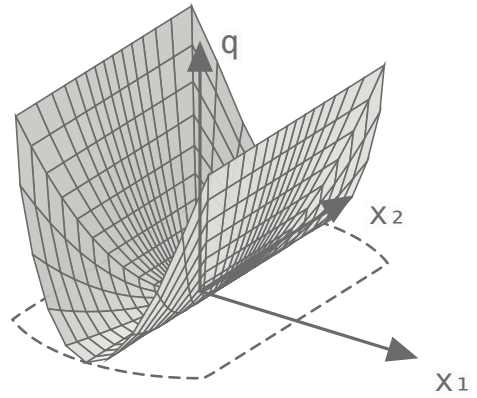
Each surface is the graph of $x^\top H x$, one camera, one scale. Which $H$ is PSD but not PD?

**floor: input space $\mathbb{R}^2$, axes $x_1$, $x_2$ · height: $q(x) = x^\top H x$**

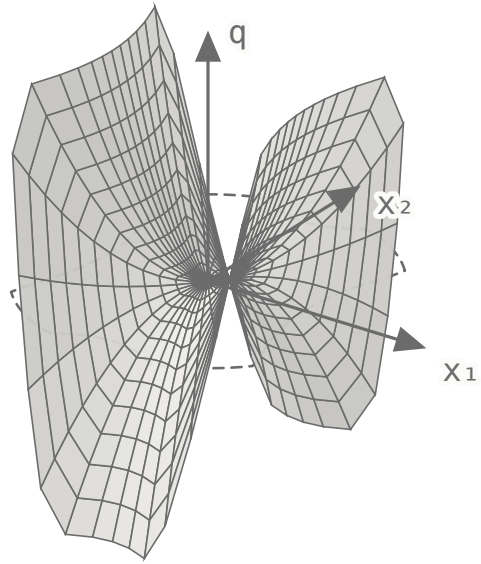
**A**




**B**




**C**




(A) $H = \mathrm{diag}(2, 1)$

(B) $H = \mathrm{diag}(2, 0)$

(C) $H = \mathrm{diag}(2, -1)$

THEOREM

# Positive Semidefinite Matrices

## Eigenvectors Rotate the Form Into Principal Axes

Spectral theorem: real symmetric $H$ has real $\lambda_i$ and orthonormal eigenvectors $v_i$.

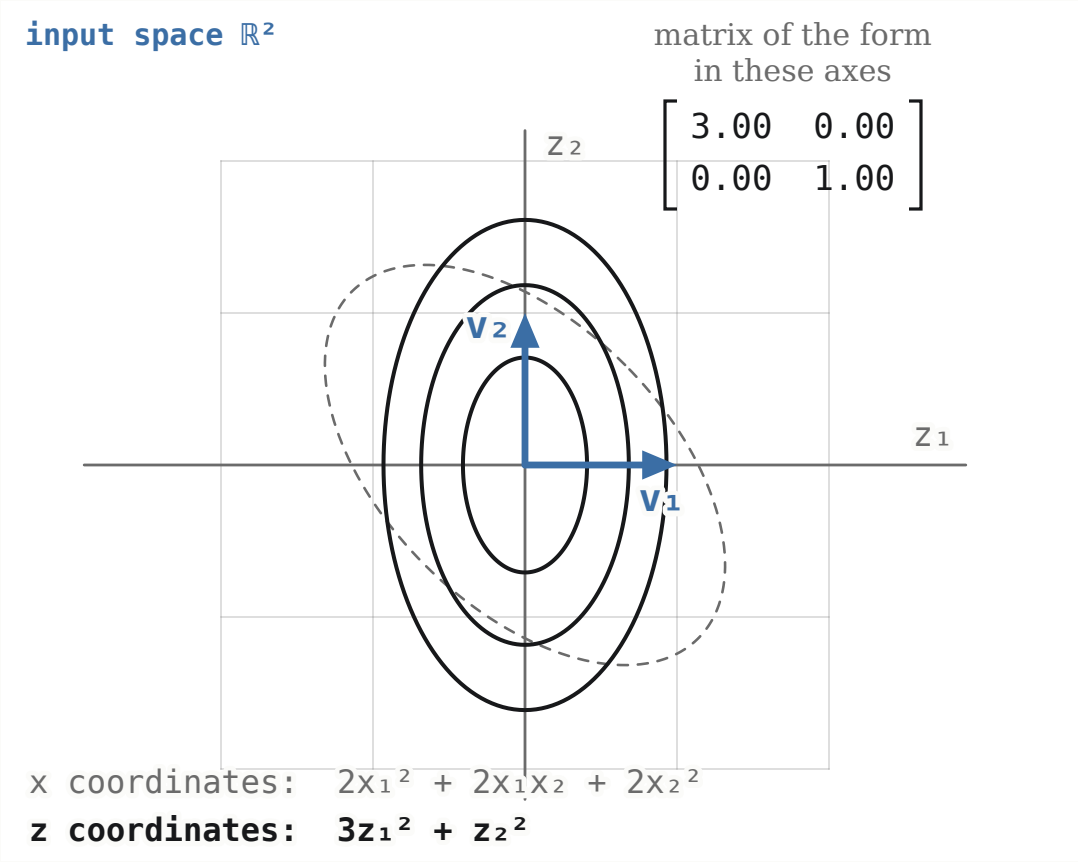
$$H = Q\Lambda Q^{\mathsf{T}}, \qquad x^{\mathsf{T}}Hx = z^{\mathsf{T}}\Lambda z = \sum_i \lambda_i z_i^2$$

$Q = [v_1 \cdots v_n]$ $Q^{\mathsf{T}}Q = I$: rotation or reflection

$\Lambda$ diagonal $\lambda_i$

$z = Q^{\mathsf{T}}x$ $z_i = v_i^{\mathsf{T}}x$

$$H \succeq 0 \iff \text{all } \lambda_i \geq 0$$
$$H \succ 0 \iff \text{all } \lambda_i > 0$$



VOTE

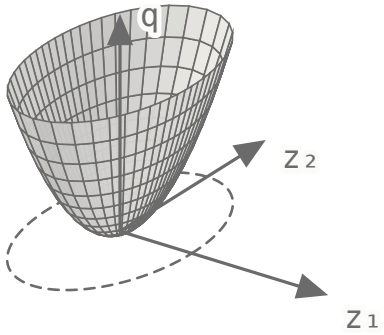
# Positive Semidefinite Matrices

## What Shape Do These Eigenvalues Make?

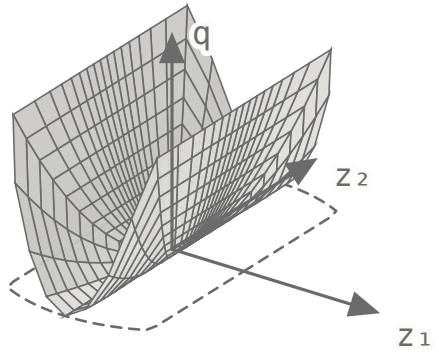
A symmetric $2 \times 2$ matrix $H$ has eigenvalues $\lambda_1 = 3$ and $\lambda_2 = -1$. Which shape is the graph of $x^\top H x$?

**floor: input space $\mathbb{R}^2$, principal coordinates $z_1$, $z_2$ · height: $x^\top Hx$**

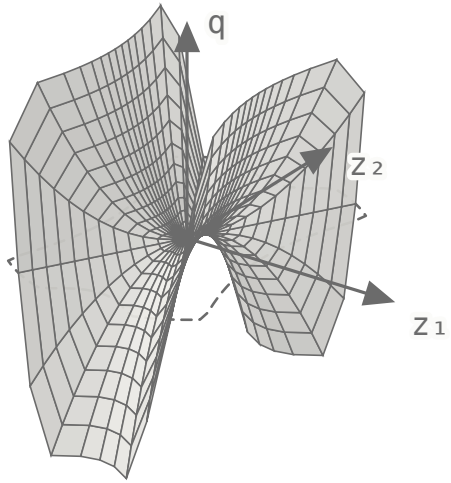
**A**




**B**




**C**




(A) bowl

(B) flat trough

(C) saddle

DERIVATION

# Positive Semidefinite Matrices

## Every Gram Matrix Is PSD

$$x^\top A^\top A x = (Ax)^\top (Ax) = \|Ax\|^2 \geq 0$$

$A \in \mathbb{R}^{m \times n}$ any shape: tall, wide or square

$A^\top A$ Gram matrix, $n \times n$: $(A^\top A)^\top = A^\top A$

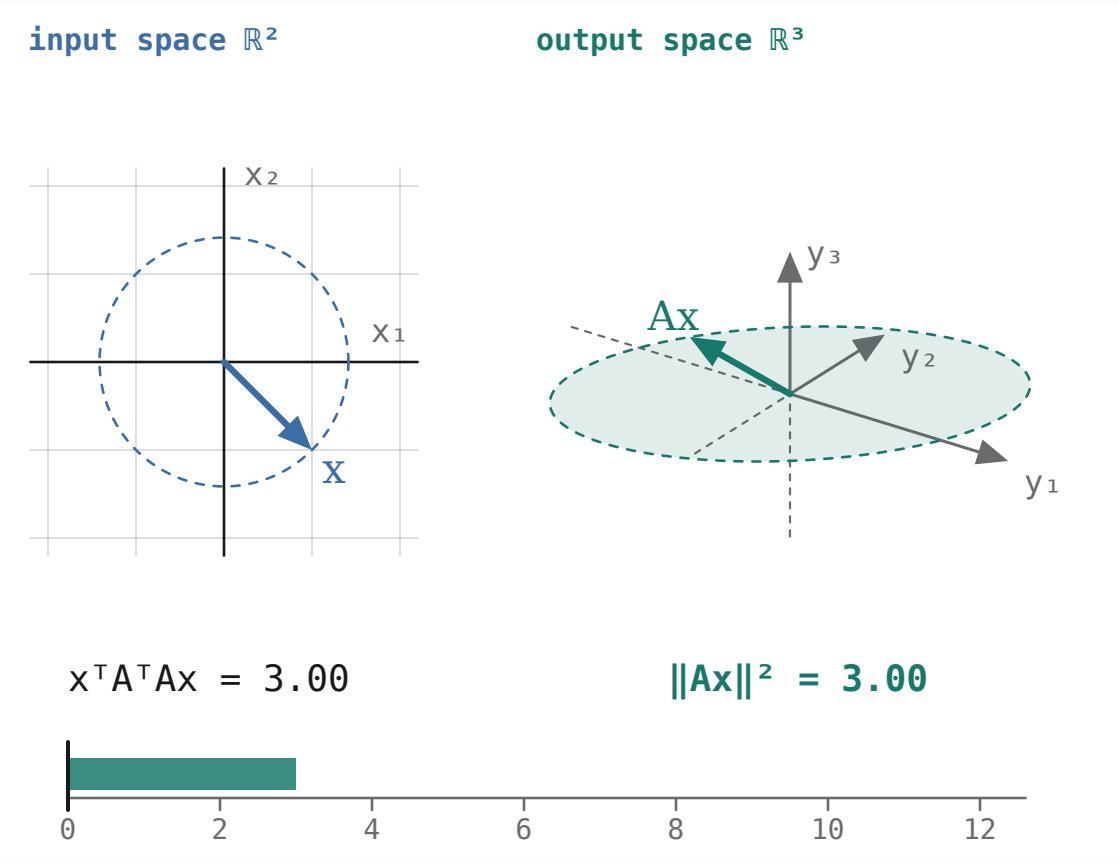
$\geq 0$ for every $x \in \mathbb{R}^n$, so $A^\top A \succeq 0$

**WORKED EXAMPLE**

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \; A^\top A = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$$

$x = (1, -1)$: $\|Ax\|^2 = \|(-1, -1, 1)\|^2 = 3$

$x^\top A^\top A x = 2 - 4 + 5 = 3$

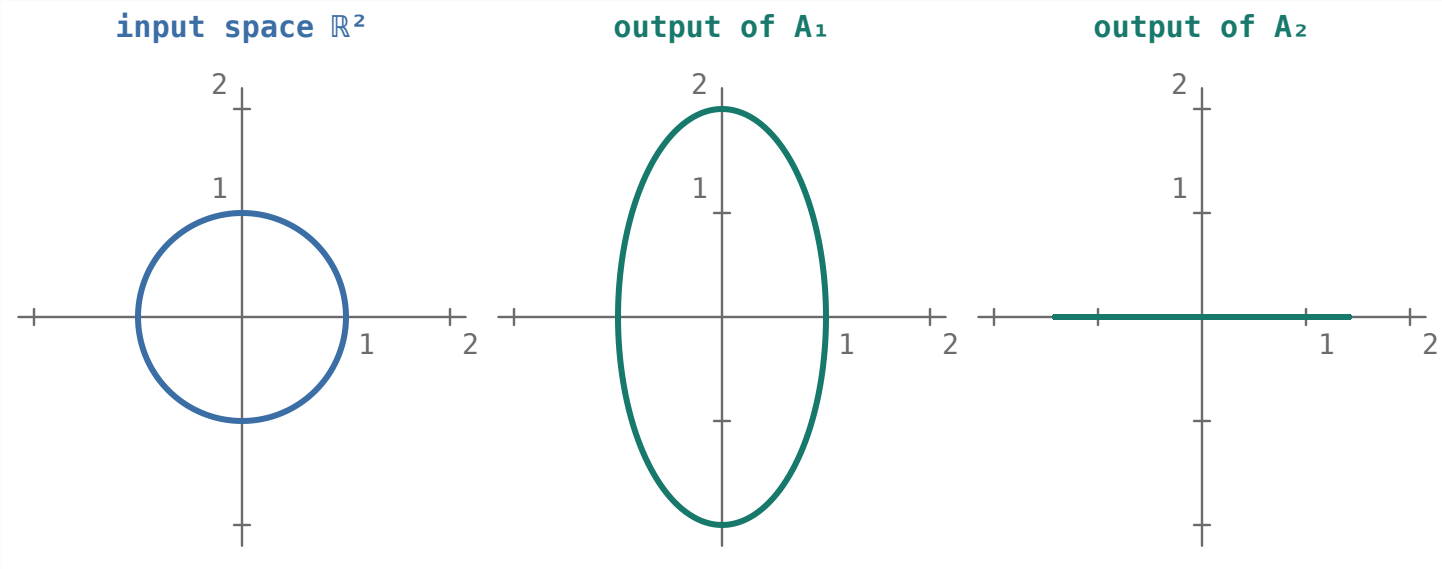


VOTE

# Positive Semidefinite Matrices

## Which Gram Matrix Is Positive Definite?

$A_1 = \mathrm{diag}(1, 2)$ and $A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ map the unit circle as shown. Which $A^\mathsf{T} A$ is PD?




- **A** only $A_1^\mathsf{T} A_1$
- **B** only $A_2^\mathsf{T} A_2$
- **C** both

PICTURE

$A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

# Positive Semidefinite Matrices

## Flat Directions Are Null Directions

$$x^\top A^\top A x = \|Ax\|^2 = 0 \iff Ax = 0$$

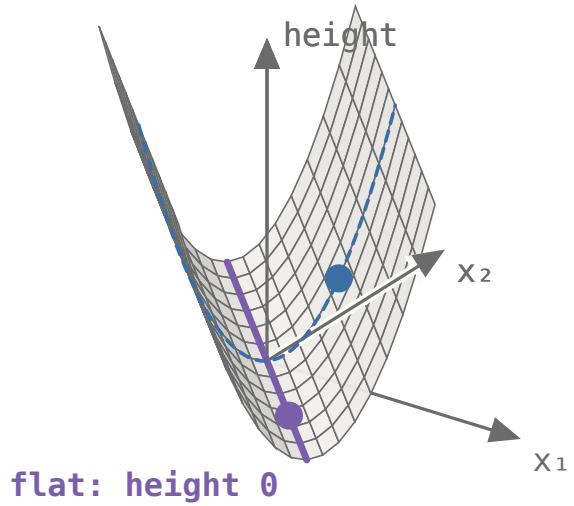
$$\mathrm{Null}(A^\top A) = \mathrm{Null}(A)$$

$\iff$ a squared length is 0 only for the zero vector

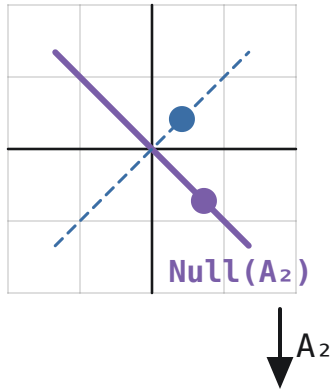
$\mathrm{Null}(A)$ all $x \in \mathbb{R}^n$ with $Ax = 0$

why $A^\top Ax = 0 \Rightarrow x^\top A^\top Ax = 0 \Rightarrow Ax = 0$; conversely $Ax = 0 \Rightarrow A^\top Ax = 0$

height $x^\top A_2^\top A_2 x = (x_1 + x_2)^2$ over the input space $\mathbb{R}^2$

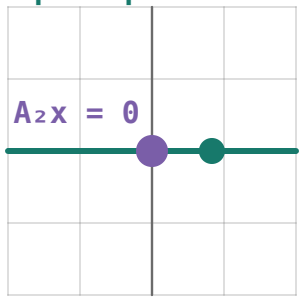



input space $\mathbb{R}^2$




$A_2$

output space $\mathbb{R}^2$



VOTE

# Positive Semidefinite Matrices

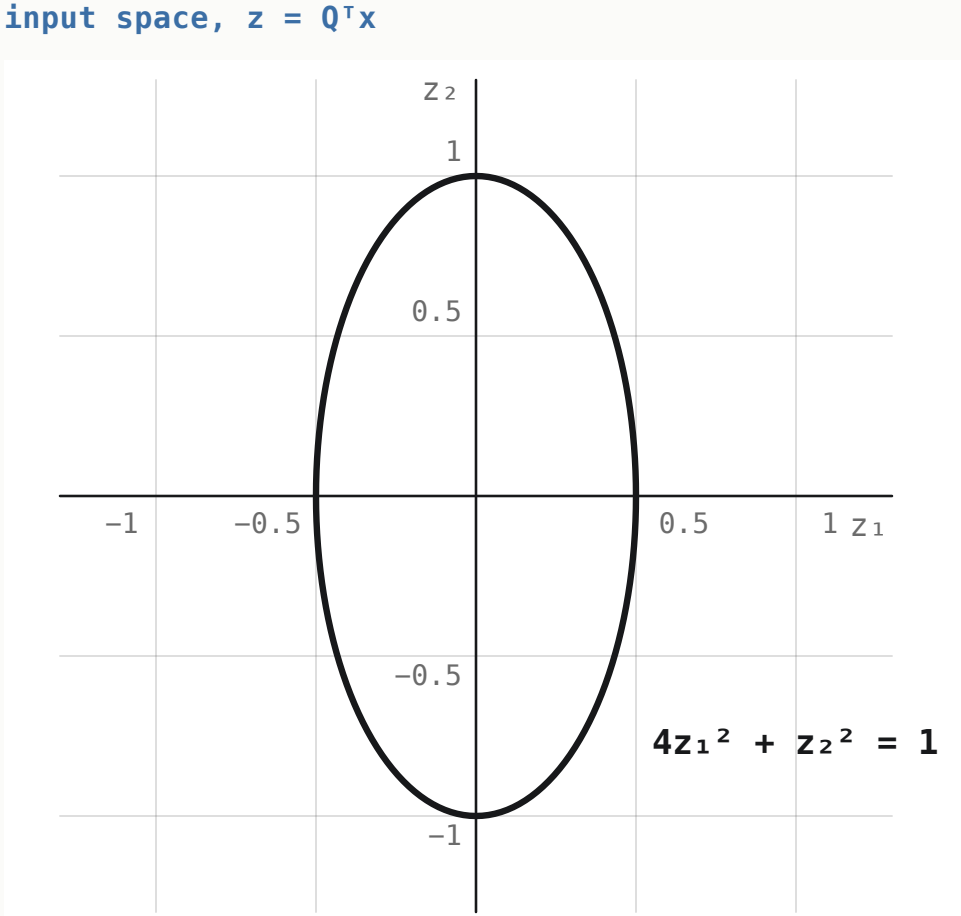
## Which Eigenvalue Sets the Long Axis?

$4z_1^2 + z_2^2 = 1$ is the level set $x^\mathsf{T} H x = 1$ in principal coordinates. Which eigenvalue sets its long axis?

A the larger

B the smaller

C neither



DEFINITION

# Positive Semidefinite Matrices

## The PSD Square Root

$$H^{1/2} = Q\Lambda^{1/2}Q^\mathsf{T}, \qquad x^\mathsf{T}Hx = \|H^{1/2}x\|^2$$

$H = Q\Lambda Q^\mathsf{T}$ PSD: $\lambda_i \geq 0$, $Q = [v_1\ v_2]$

$\Lambda^{1/2}$ $\mathrm{diag}(\sqrt{\lambda_i})$: real as $\lambda_i \geq 0$

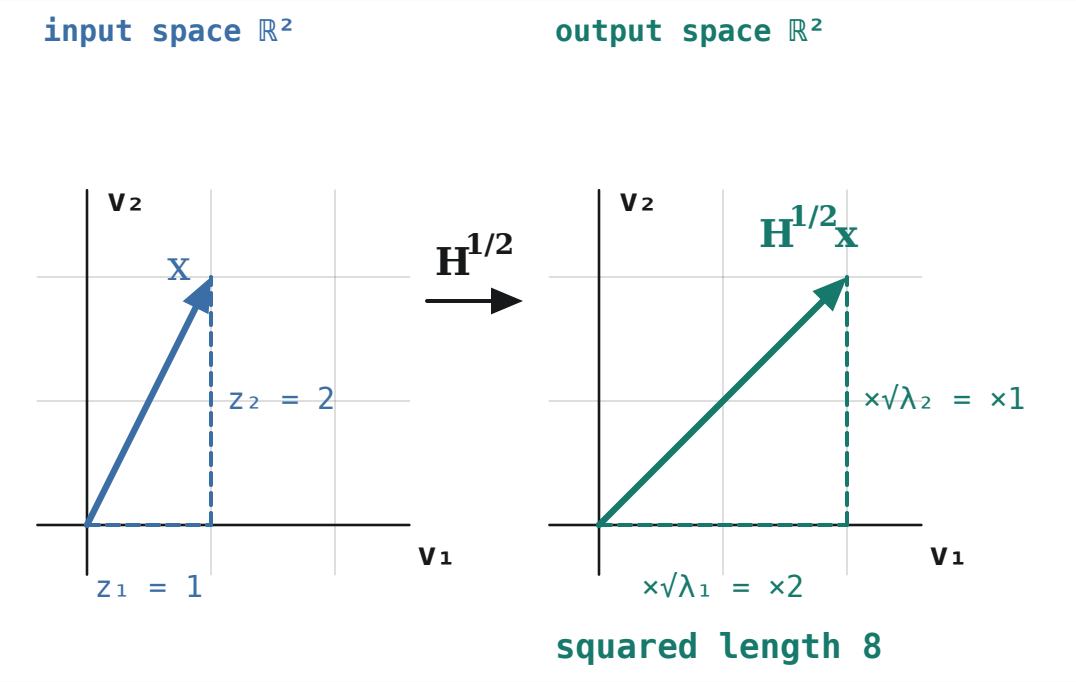
$H^{1/2}$ unique symmetric PSD: $H^{1/2}H^{1/2} = H$

**WORKED EXAMPLE**

$H = \mathrm{diag}(4, 1)$: $Q = I$, $H^{1/2} = \mathrm{diag}(2, 1)$

$x = (1, 2)$: $H^{1/2}x = (2, 2)$

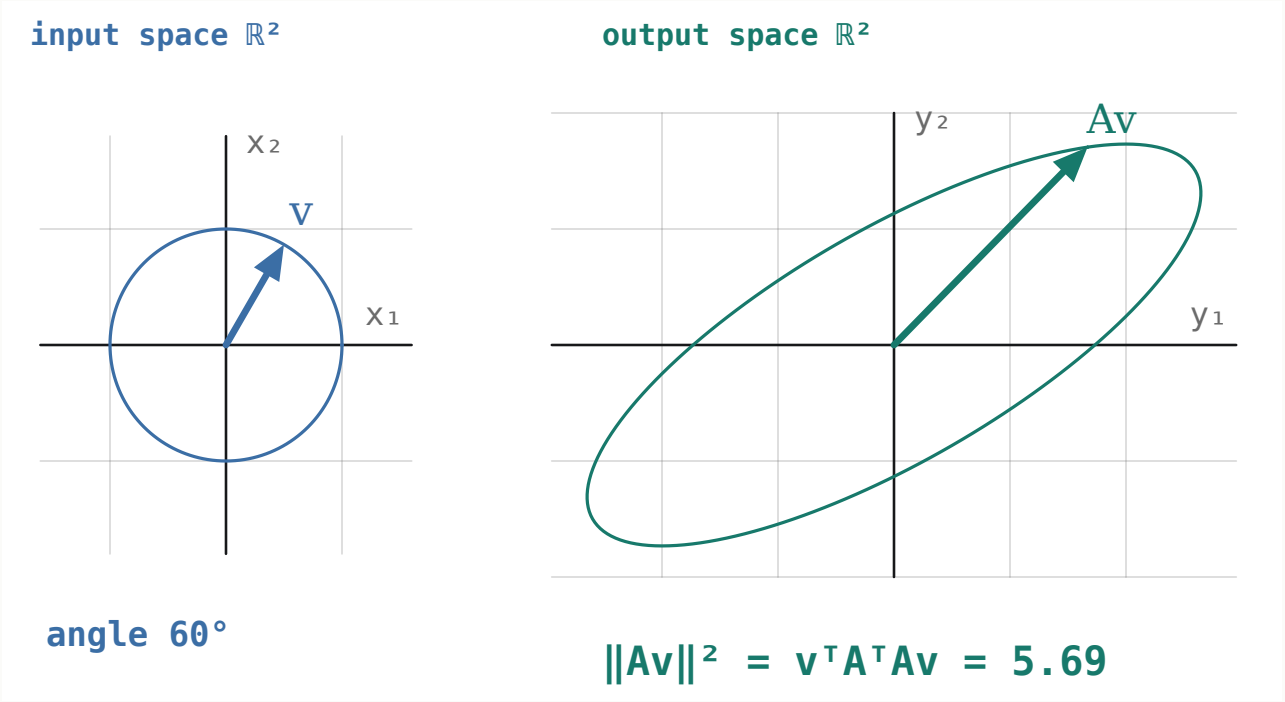
$\|H^{1/2}x\|^2 = 8 = x^\mathsf{T}Hx$



LOOKING AHEAD

# Positive Semidefinite Matrices

## Which Input Direction Stretches Most?




$$\|Av\|^2 = v^\mathsf{T} A^\mathsf{T} A v, \qquad \|v\| = 1$$

$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$, not symmetric

$v$ a unit input direction, turning

$A^\mathsf{T}A$ its Gram matrix: symmetric PSD

> Which $v$ makes $\|Av\|^2$ largest?
> Which makes it smallest?

DEFINITION

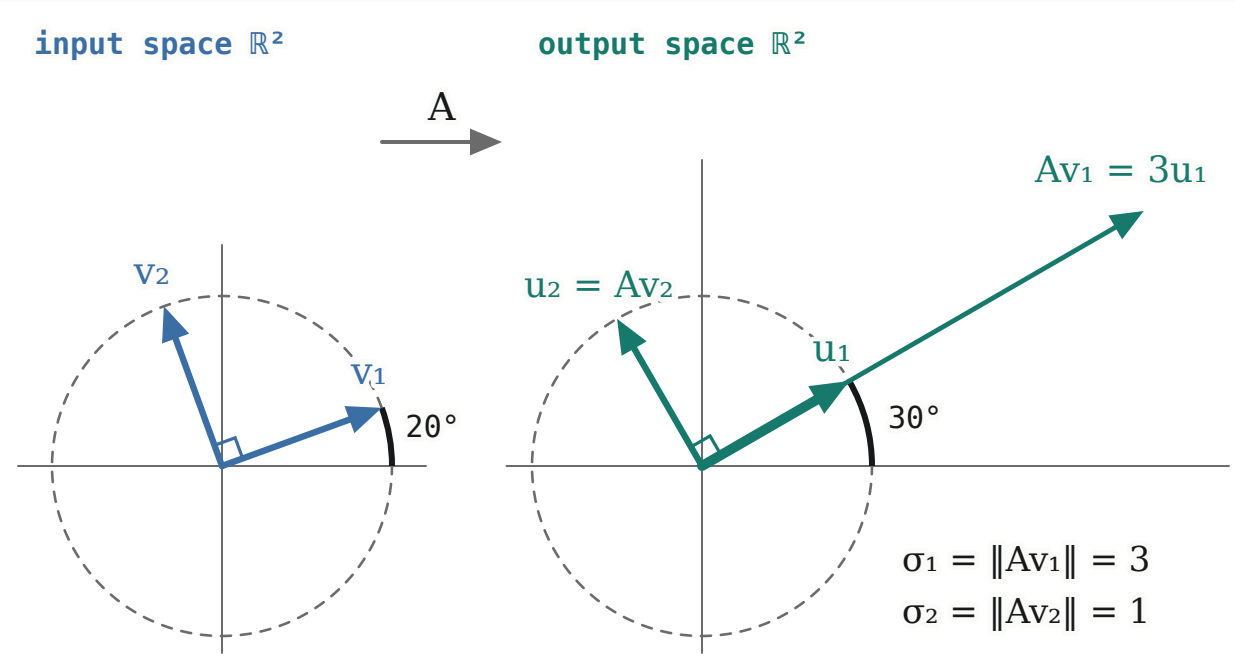
# Singular Value Decomposition

## Input Directions Paired With Output Directions

$$A = \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

$$Av_i = \sigma_i u_i, \qquad \|v_i\| = \|u_i\| = 1$$

- $A$ — $m \times n$: input $\mathbb{R}^n$ to output $\mathbb{R}^m$
- $v_i \in \mathbb{R}^n$ — input direction; the $v_i$ are orthonormal
- $u_i \in \mathbb{R}^m$ — output direction; the $u_i$ are orthonormal
- $\sigma_i \geq 0$ — length gain: $\sigma_i = \|Av_i\|$, $i \leq \min(m, n)$



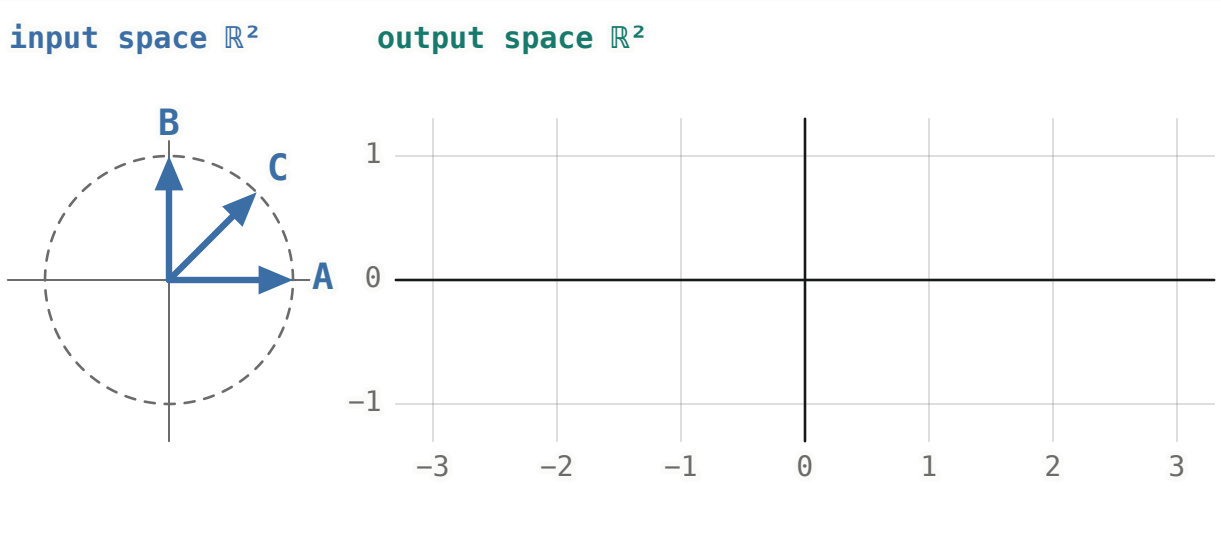
VOTE

# Singular Value Decomposition

## Which Unit Input Comes Out Longest?

$A = \mathrm{diag}(3, 1)$. Which unit input has the longest output $Ax$?

- **A** horizontal, $e_1$
- **B** vertical, $e_2$
- **C** diagonal, $(e_1 + e_2)/\sqrt{2}$



DEFINITION

# Singular Value Decomposition

## The Full SVD and Its Shapes

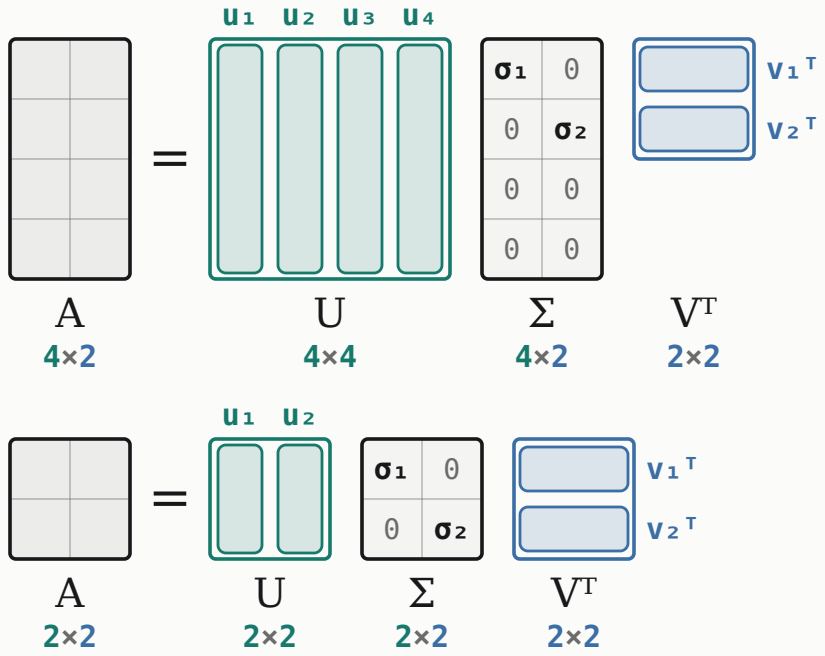
$$A_{m\times n} = U_{m\times m}\,\Sigma_{m\times n}\,V^{\mathsf{T}}_{n\times n}$$

$U^{\mathsf{T}}U = I_m$ orthonormal columns $u_i$: output $\mathbb{R}^m$

$V^{\mathsf{T}}V = I_n$ orthonormal columns $v_i$: input $\mathbb{R}^n$

$\Sigma$ $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ on the diagonal, else $0$

> Every real matrix has a full SVD.

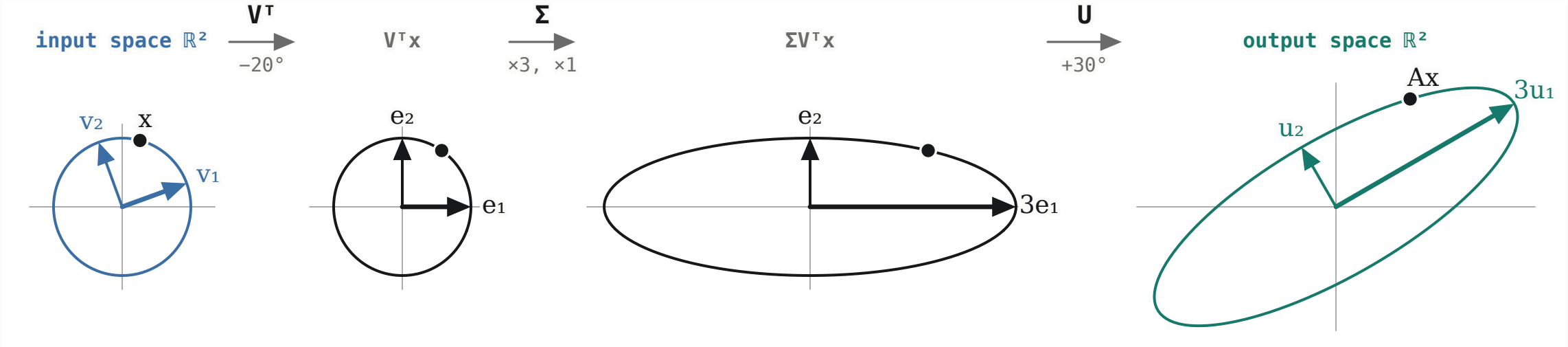


PICTURE

$$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

# Singular Value Decomposition

## Orthogonal, Diagonal, Orthogonal




$$Ax = U\left(\Sigma\left(V^{\mathsf{T}}x\right)\right)$$

- $V^{\mathsf{T}}$ orthogonal: coordinates of $x$ along $v_1, v_2$; here $R(-20^\circ)$
- $\Sigma$ stretch coordinate $i$ by $\sigma_i$; for $m \neq n$ it also drops or adds zeros
- $U$ orthogonal: coordinate $i$ goes along $u_i$; here $R(30^\circ)$
- $\det = \pm 1$ an orthogonal factor may reflect; here both rotate

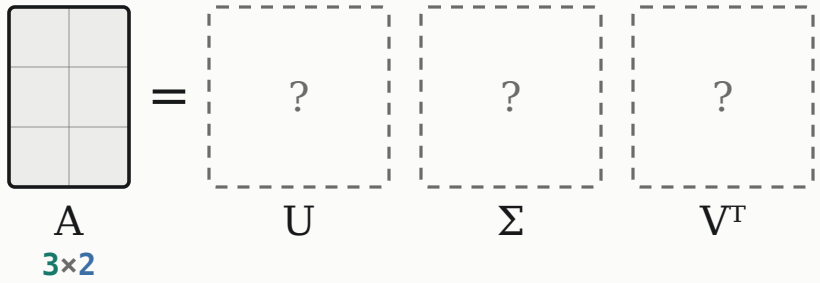
VOTE

# Singular Value Decomposition

## What Are the Shapes for a 3×2 Matrix?

$A$ is $3 \times 2$, with full SVD $A = U\Sigma V^{\mathsf{T}}$. What are the shapes of $(U, \Sigma, V^{\mathsf{T}})$?

- A (3×3, 3×2, 2×2)
- B (3×2, 2×2, 2×3)
- C (2×2, 2×3, 3×3)



PICTURE

$$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

# Singular Value Decomposition

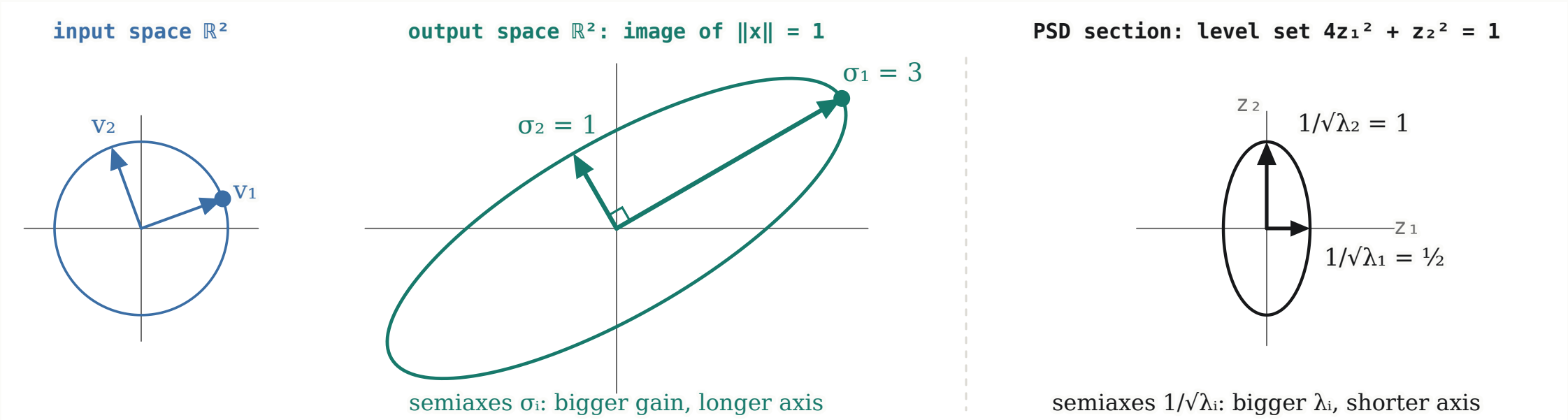
## Singular Values Are the Ellipse Semiaxes

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0, \qquad p = \min(m, n)$$

$\sigma_i$ singular values: sorted, never negative

$p$ how many there are

$\{Ax : \|x\| = 1\}$ full-rank $2 \times 2$ $A$: an ellipse with semiaxes $\sigma_i u_i$



VOTE

# Singular Value Decomposition

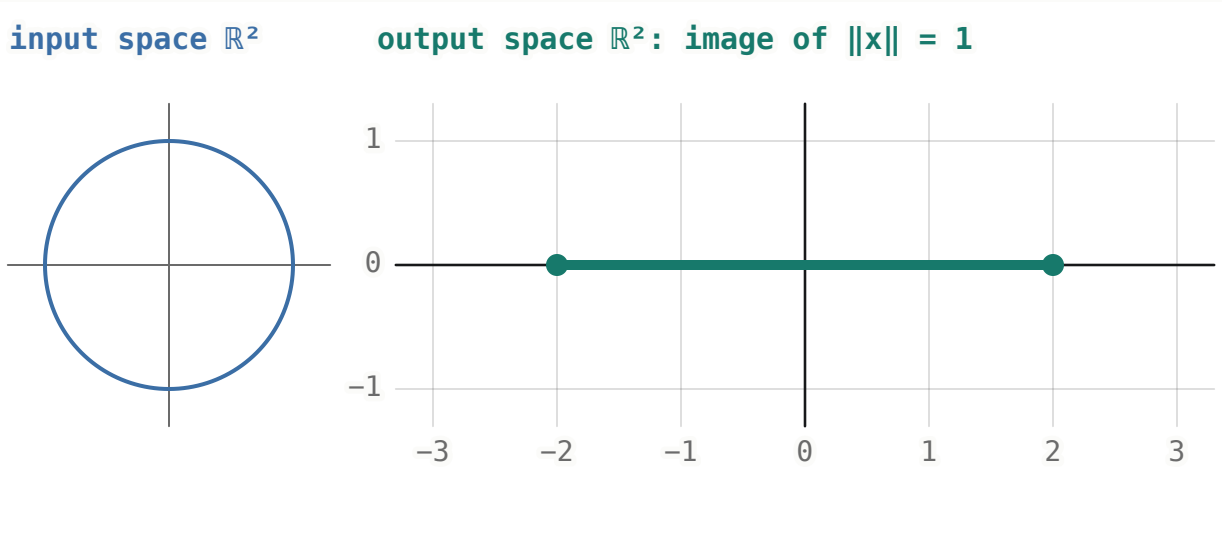
## What Are the Singular Values of This Collapse?

A $2 \times 2$ matrix maps the unit circle onto the segment from $-2e_1$ to $2e_1$. What are $(\sigma_1, \sigma_2)$?

A $(2, 1)$

B $(2, 0)$

C $(0, 0)$



DERIVATION

$$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

# Singular Value Decomposition

## Right Singular Vectors From the Gram Matrix

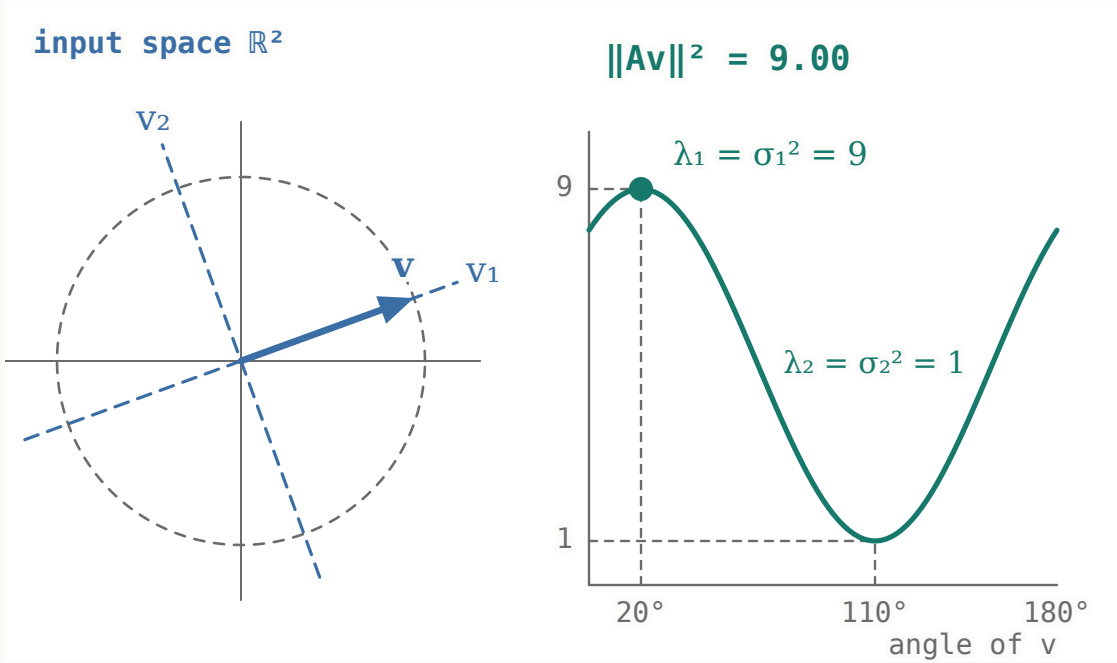
$$A^\mathsf{T} A v_i = \lambda_i v_i, \qquad \sigma_i = \sqrt{\lambda_i}$$

$\lambda_i$ $= \|Av_i\|^2 \geq 0$: eigenvalues of PSD $A^\mathsf{T} A$

$v_i$ orthonormal eigenvectors: input axes

**WORKED EXAMPLE**

$A^\mathsf{T} A \approx \begin{bmatrix} 8.06 & 2.57 \\ 2.57 & 1.94 \end{bmatrix}$ (rounded). Exact $A = R(30^\circ)\mathrm{diag}(3,1)R(-20^\circ)$: trace 10, det 9, $\lambda_i = 9, 1$, $\sigma_i = 3, 1$.



$$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

# Singular Value Decomposition

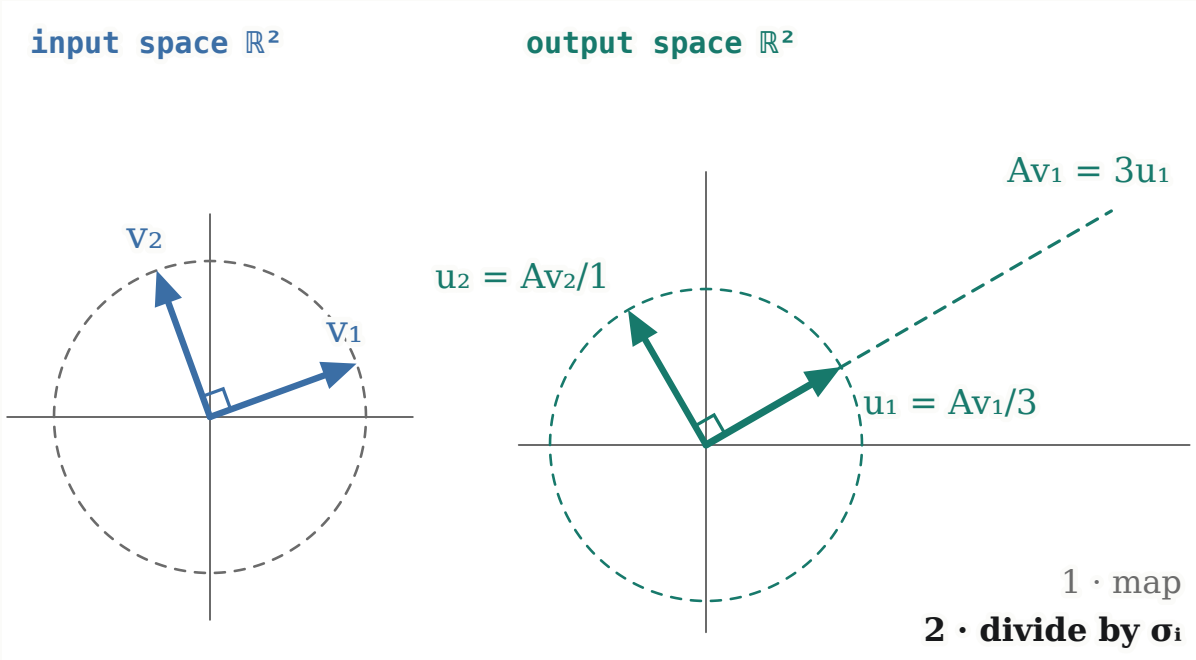
## Left Singular Vectors: Map, Then Normalize

$$u_i = \frac{Av_i}{\sigma_i} \quad (\sigma_i > 0)$$

$\sigma_i = 0$ $Av_i = 0$: pick $u_i$ to finish the basis

$$u_i^\top u_j = \frac{v_i^\top A^\top A v_j}{\sigma_i \sigma_j} = \frac{\sigma_j^2 v_i^\top v_j}{\sigma_i \sigma_j} = \delta_{ij}$$

$\delta_{ij}$ 1 if $i = j$, else 0: orthonormal $u_i$ ($\sigma_i \sigma_j > 0$)



$$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

VOTE

# Singular Value Decomposition

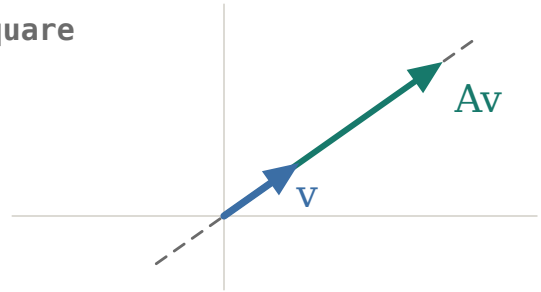
## Eigenvectors or Singular Vectors?

Which statement about the SVD is true?

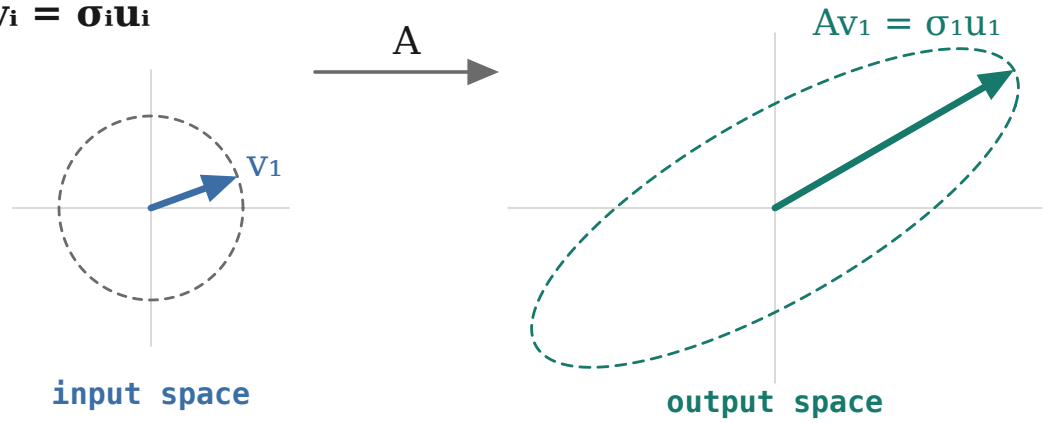
- (A) It exists only for square $A$
- (B) Input and output dimensions may differ
- (C) Singular values can be negative

**$Av = \lambda v$**

$\mathbb{R}^2 \to \mathbb{R}^2$, A square




**$Av_i = \sigma_i u_i$**



$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

# Singular Value Decomposition

## Worked Example: A Symmetric Positive Definite Matrix

**WORKED EXAMPLE**

$v_{1,2} = (1, \pm 1)/\sqrt{2}$

$A(1,1) = (3,3)$, so $Av_1 = 3v_1$

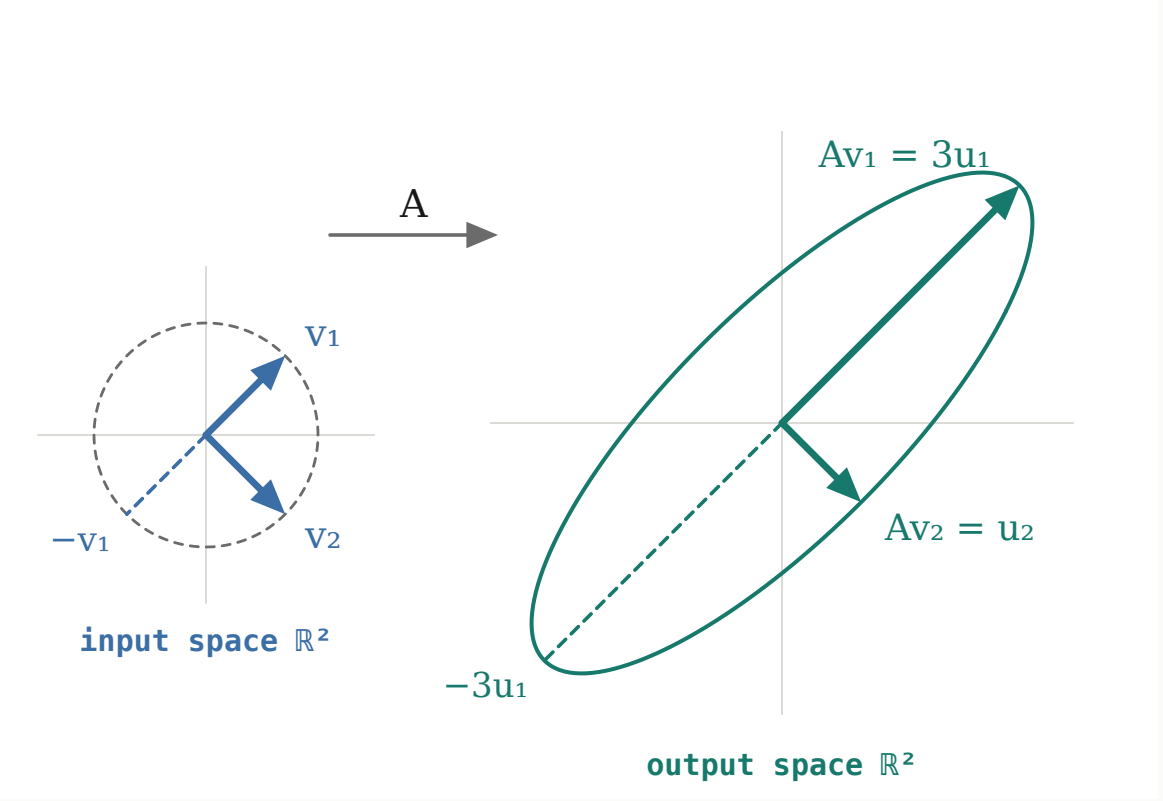
$A(1,-1) = (1,-1)$, so $Av_2 = 1\,v_2$

$\sigma = (3,1)$, $u_i = Av_i/\sigma_i = v_i$

$$A = U\Sigma V^{\mathsf{T}} = (UD)\,\Sigma\,(VD)^{\mathsf{T}}$$

$U = V$ can be chosen exactly when $A$ is symmetric PSD

$D$ $\mathrm{diag}(\pm 1, \pm 1)$: flip $u_i$ and $v_i$ together



$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

VOTE

# Singular Value Decomposition

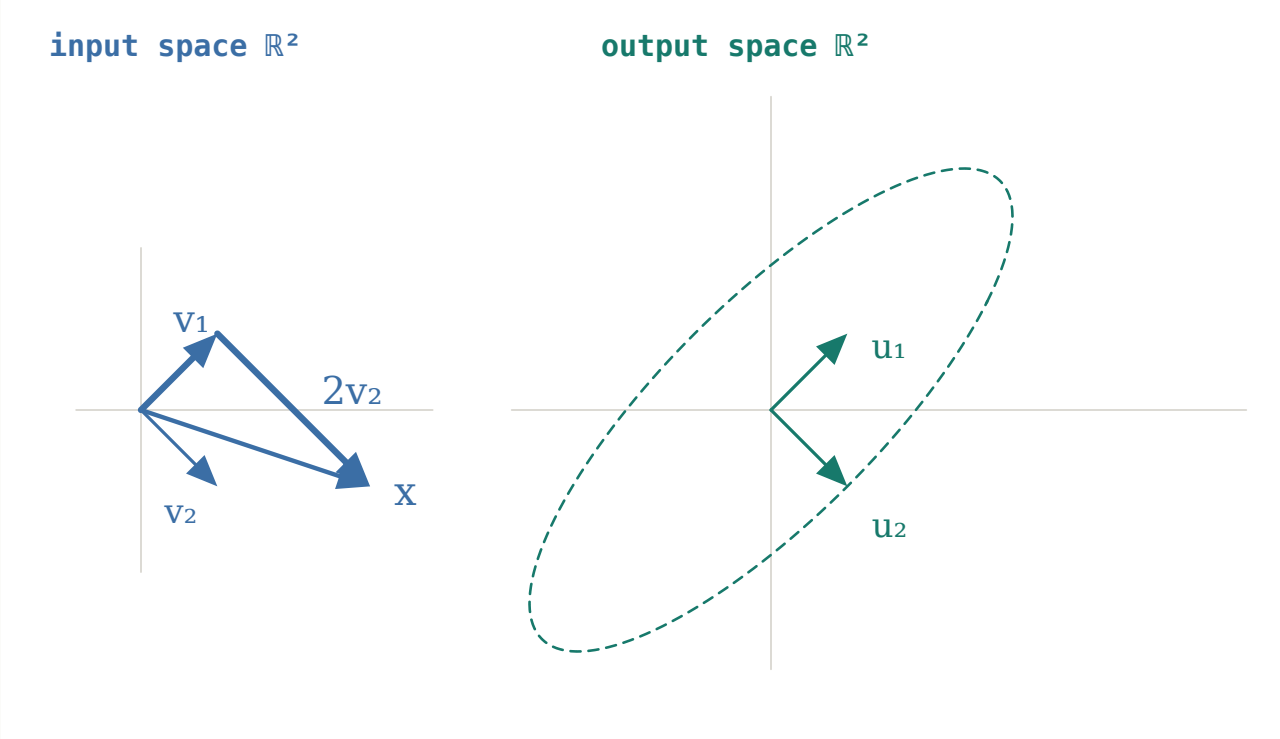
## Where Does This Combination of Inputs Go?

With $\sigma = (3, 1)$ and $u_i = v_i$ as before, let $x = v_1 + 2v_2$. What is $Ax$?

A $3u_1 + 2u_2$

B $u_1 + 6u_2$

C $3u_1 + 6u_2$



DEFINITION

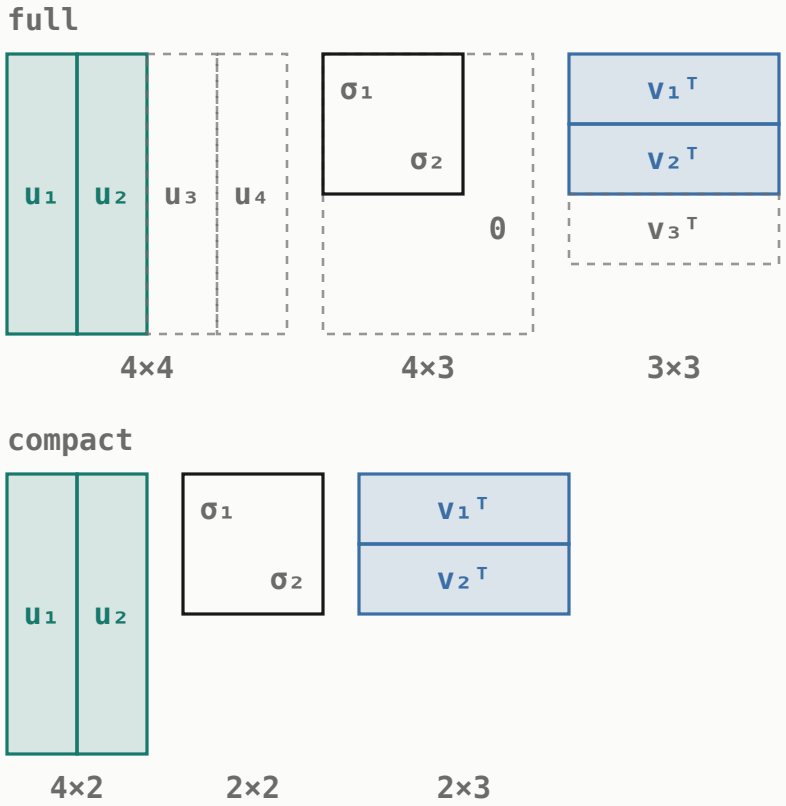
# Singular Value Decomposition

## The Compact SVD Keeps Only Rank r

$$A = U_r \Sigma_r V_r^{\mathsf{T}}, \quad U_r^{\mathsf{T}} U_r = V_r^{\mathsf{T}} V_r = I_r$$

| | |
|---|---|
| $r$ | $\operatorname{rank}(A)$: how many $\sigma_i > 0$ |
| $U_r, V_r$ | first $r$ columns: $m \times r$, $n \times r$ |
| $\Sigma_r$ | $r \times r$ diagonal, all $> 0$ |
| $U_r U_r^{\mathsf{T}}$ | $m \times m$, $\neq I_m$ if $r < m$ |

Thin (economy): $\min(m, n)$ columns, zeros kept.



PICTURE

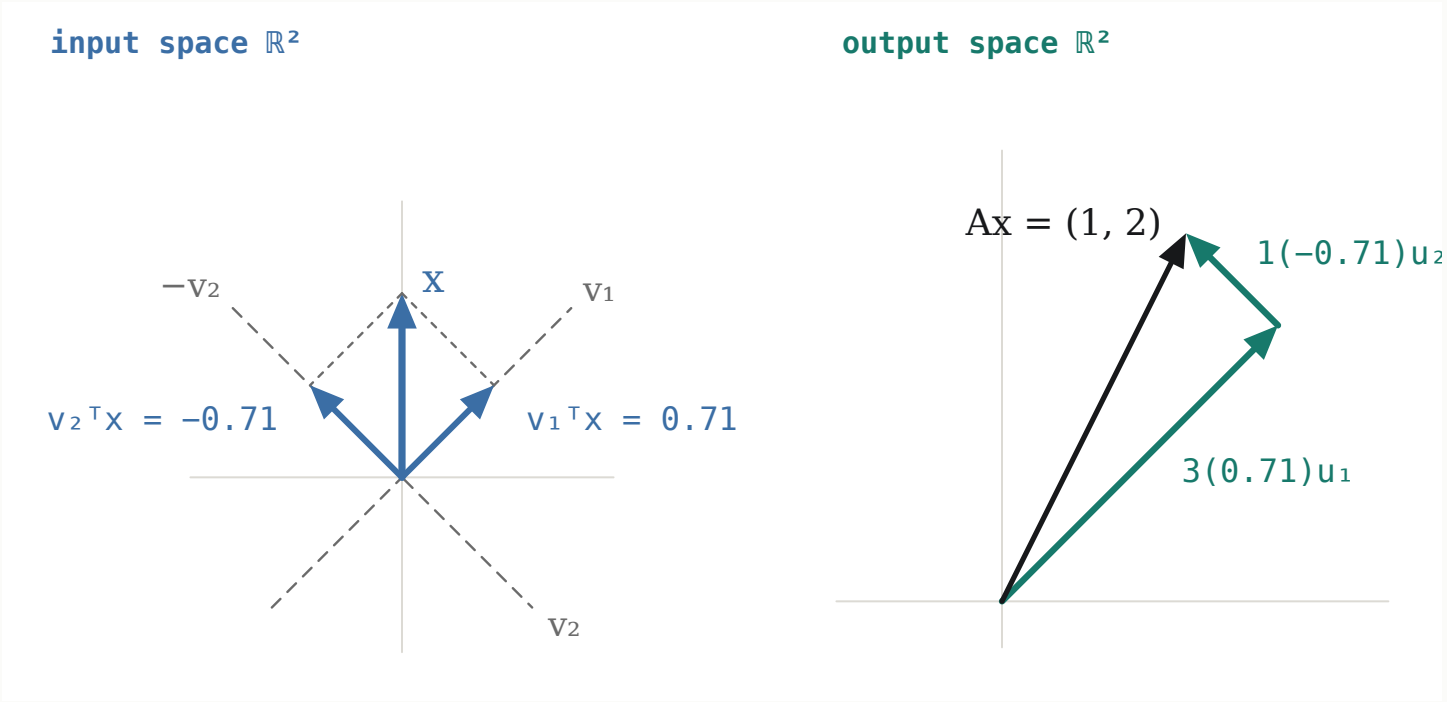
$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

# Singular Value Decomposition

## A Sum of Rank-One Maps

$$A = \sum_{i=1}^{r} \sigma_i \, u_i v_i^{\mathsf{T}} \quad \Rightarrow \quad Ax = \sum_{i=1}^{r} \sigma_i \, (v_i^{\mathsf{T}} x) \, u_i$$

- $u_i v_i^{\mathsf{T}}$ — $m \times n$, rank one: reads along $v_i$, writes along $u_i$
- $v_i^{\mathsf{T}} x$ — a signed number, the coordinate of $x$ along $v_i$
- $r$ — the number of $\sigma_i > 0$



VOTE

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

# Singular Value Decomposition

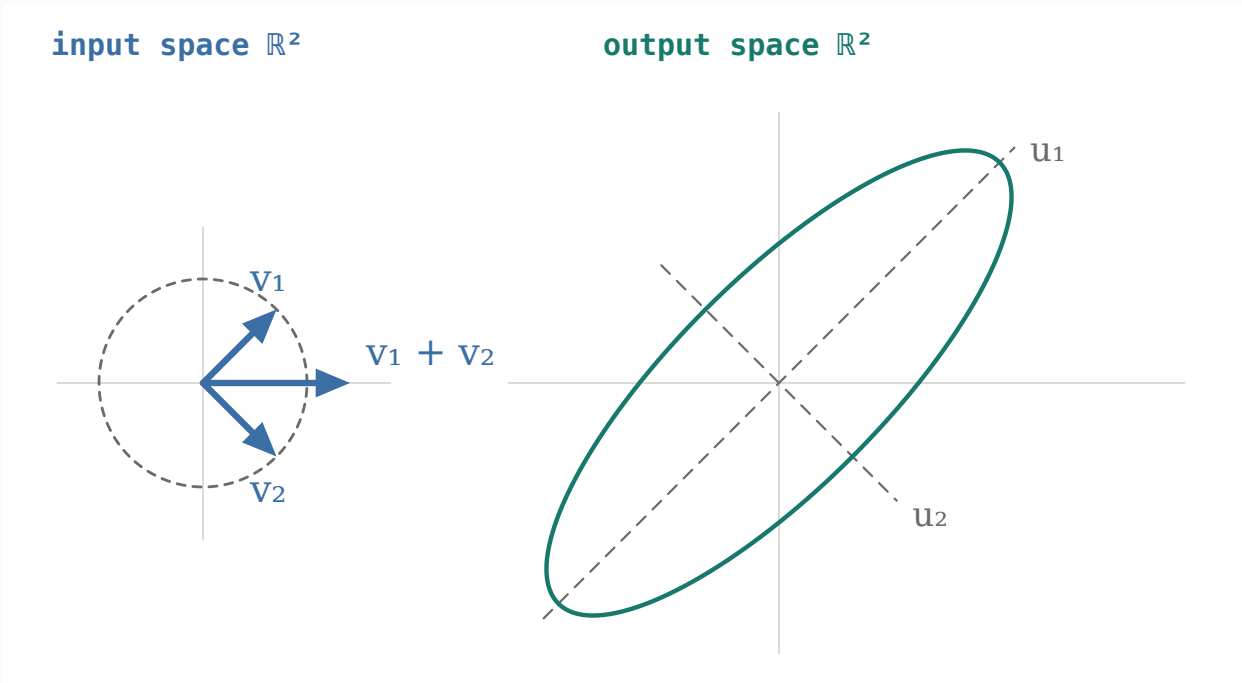
## Which Input Does Truncation Erase?

Keep one term of $A = 3u_1v_1^\mathsf{T} + u_2v_2^\mathsf{T}$:
$A_1 = 3u_1v_1^\mathsf{T}$. Which input does $A_1$ send to $0$?

A $v_1$

B $v_2$

C $v_1 + v_2$

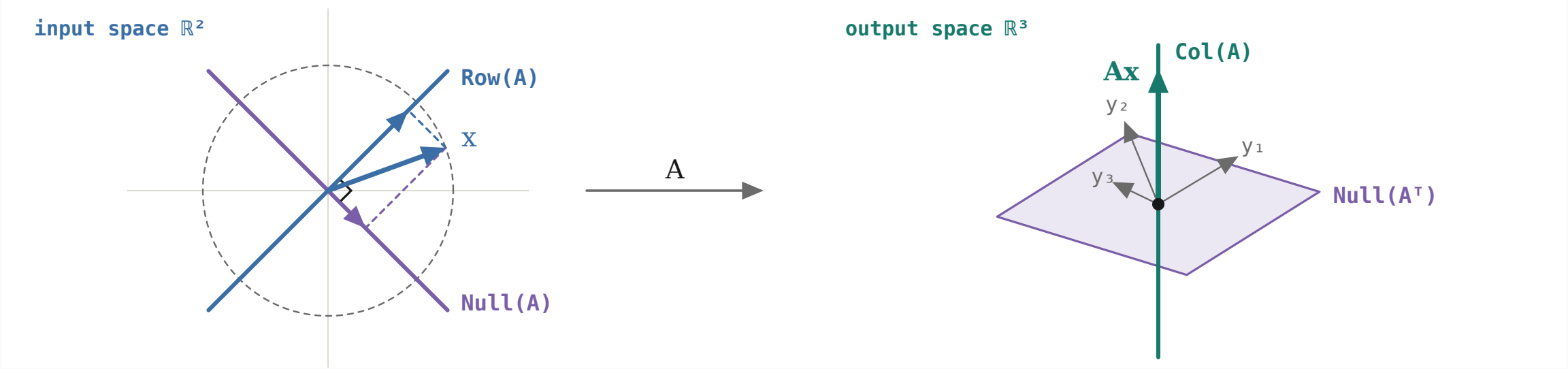


PICTURE

# Singular Value Decomposition

## The Four Subspaces From U and V

Full SVD: $U$ is $m \times m$, $V$ is $n \times n$. Example: the all-ones $3 \times 2$ matrix, rank $r = 1$.




$\mathrm{Row}(A) = \mathrm{span}(v_1, \ldots, v_r)$, dim $r$

$\mathrm{Null}(A) = \mathrm{span}(v_{r+1}, \ldots, v_n)$, dim $n - r$

$\mathrm{Col}(A) = \mathrm{span}(u_1, \ldots, u_r)$, dim $r$

$\mathrm{Null}(A^\mathsf{T}) = \mathrm{span}(u_{r+1}, \ldots, u_m)$, dim $m - r$

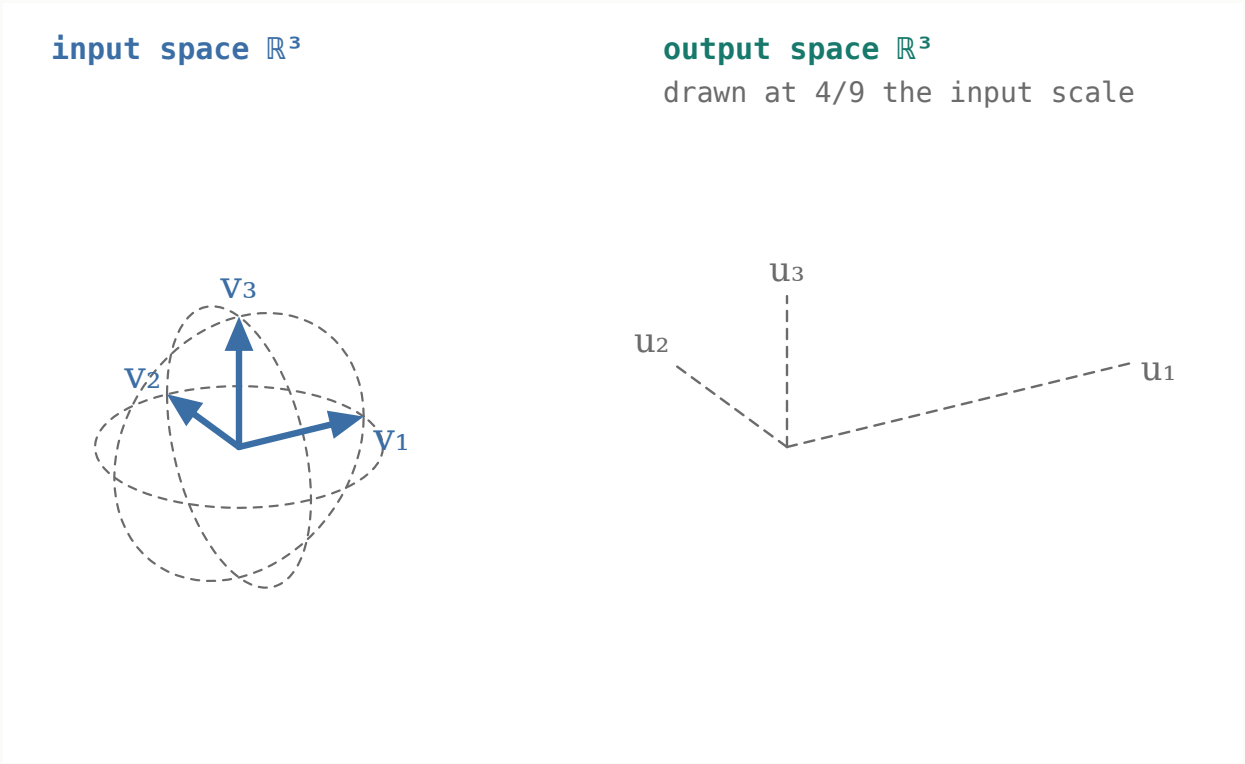
VOTE

# Singular Value Decomposition

## Rank and Nullity From Singular Values

A $3 \times 3$ matrix has singular values $5, 2, 0$. Which pair is (rank, nullity)?

- **A** $(3, 0)$
- **B** $(2, 1)$
- **C** $(1, 2)$



THEOREM

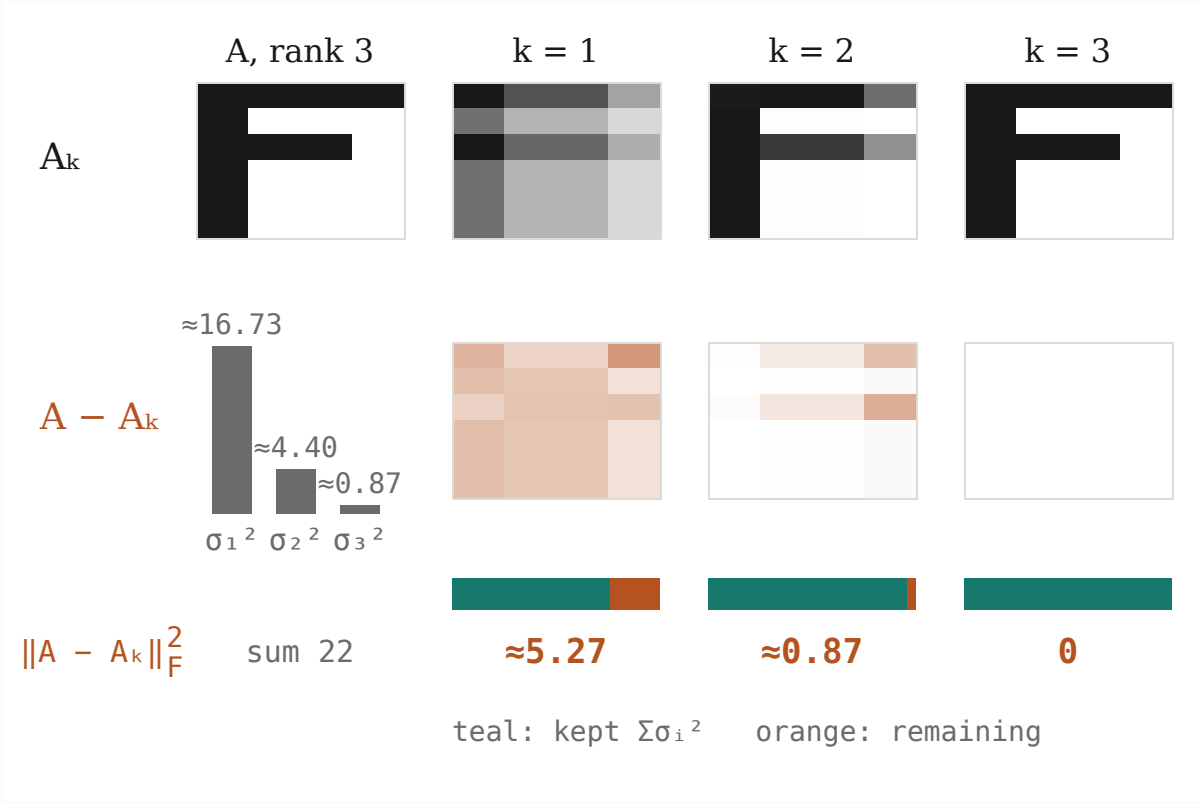
# Singular Value Decomposition

## The Best Rank-k Approximation

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^{\mathsf{T}} \in \arg\min_{\operatorname{rank} B \le k} \|A - B\|_F$$

| | |
|---|---|
| $\|M\|_F^2$ | $\sum_{i,j} M_{ij}^2$ |
| $\|A - A_k\|_F^2$ | $\sum_{i>k} \sigma_i^2$, the error |
| $k$ | $0 \le k \le r$; $A_r = A$ |
| $B$ | any matrix with rank $\le k$ |
| Eckart–Young | no proof; may not be unique if $\sigma_k = \sigma_{k+1} > 0$ |

Next: each row of $A$ is one data sample.



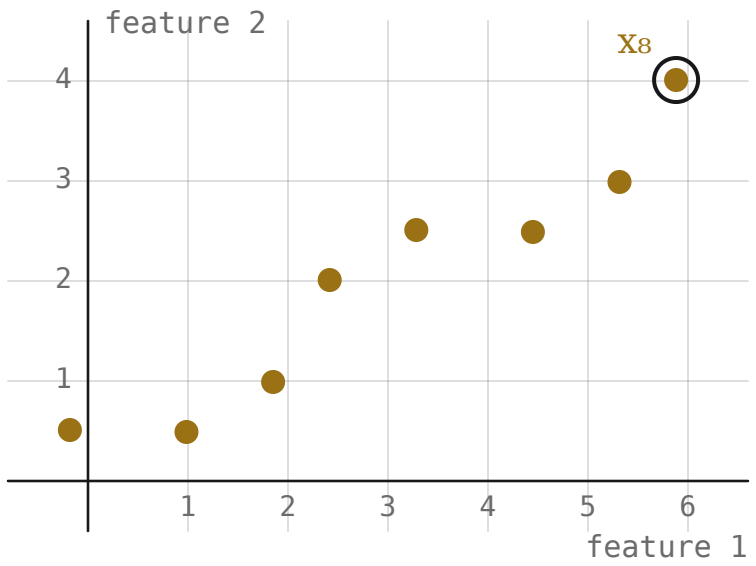
DEFINITION

# Principal Component Analysis

## Data as a Matrix: One Row per Sample

data matrix X, 8 × 2

| | feature 1 | feature 2 |
|---|---|---|
| $x_1^\top$ | −0.18 | 0.51 |
| $x_2^\top$ | 0.98 | 0.49 |
| $x_3^\top$ | 1.85 | 0.99 |
| $x_4^\top$ | 2.42 | 2.01 |
| $x_5^\top$ | 3.28 | 2.51 |
| $x_6^\top$ | 4.45 | 2.49 |
| $x_7^\top$ | 5.32 | 2.99 |
| $x_8^\top$ | 5.88 | 4.01 |




$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_N^\top \end{bmatrix} \in \mathbb{R}^{N \times d}$$

$x_j \in \mathbb{R}^d$ sample $j$: one point in feature space

$N$ rows = samples; here $N = 8$

$d$ columns = features; here $d = 2$

> Sixteen numbers, but a thin cloud. Could one number per sample nearly do?

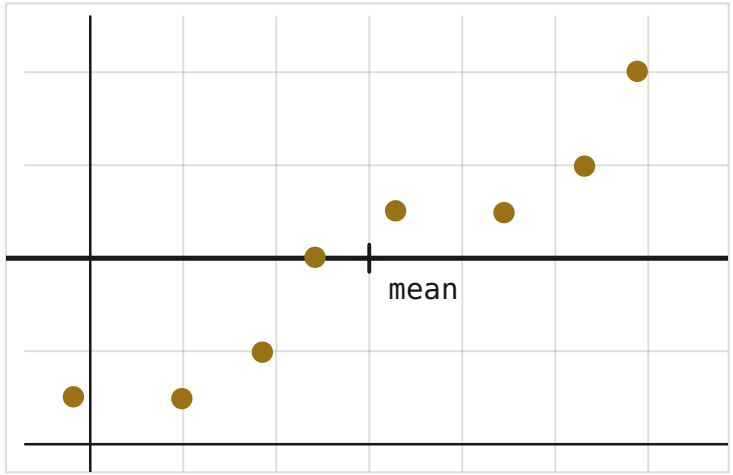
VOTE

# Principal Component Analysis

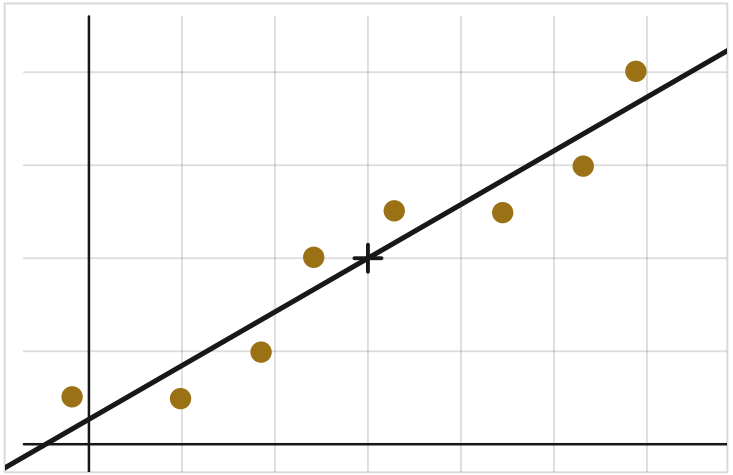
## Which Line Best Summarizes the Cloud?

Which line through the mean gives the smallest sum of squared perpendicular distances?

**A · horizontal**

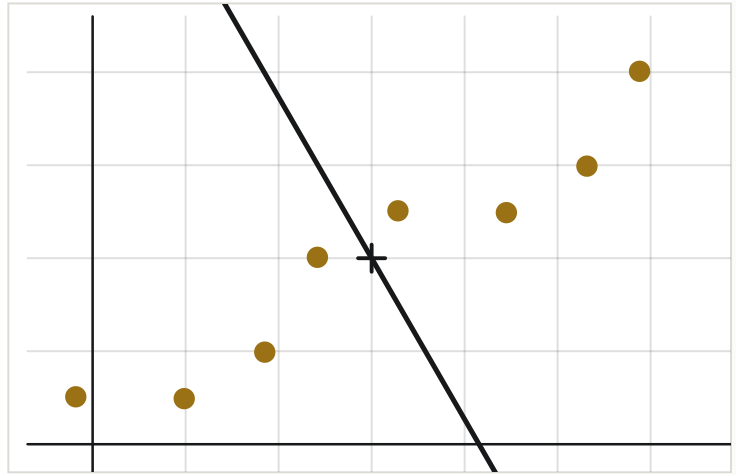



**B · long axis**



**C · short axis**

feature space $\mathbb{R}^2$



A the horizontal line

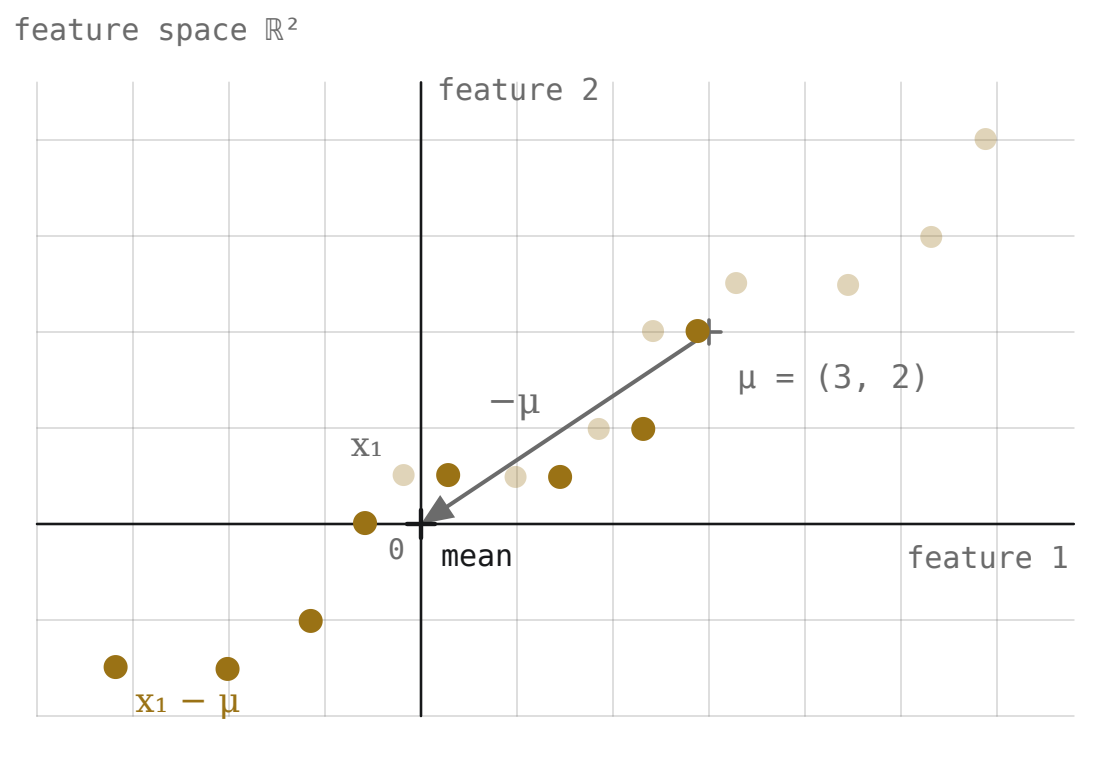
B the long-axis line

C the short-axis line

DEFINITION

# Principal Component Analysis

## Centering Moves the Mean to the Origin




$$\mu = \frac{1}{N}\sum_{j=1}^{N} x_j, \qquad X_c = X - \mathbf{1}\mu^{\mathsf{T}}$$

$\mu \in \mathbb{R}^d$ mean sample, in feature space

$\mathbf{1} \in \mathbb{R}^N$ all ones; $\mathbf{1}\mu^{\mathsf{T}}$ is $N \times d$, each row $\mu^{\mathsf{T}}$

$X_c$ $N \times d$, row $j$ is $(x_j - \mu)^{\mathsf{T}}$; columns sum to $0$

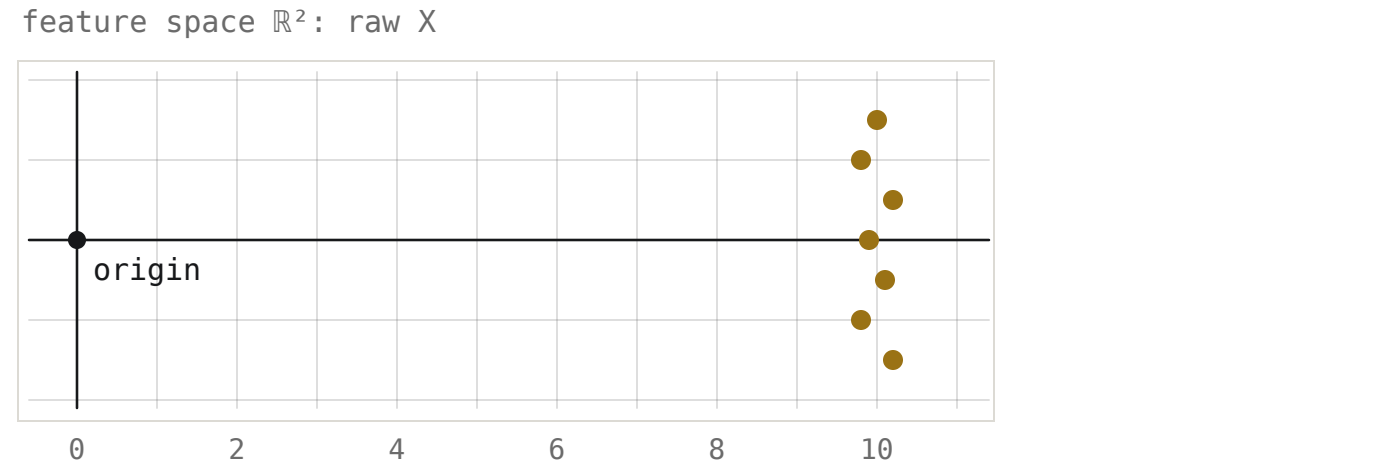
**WORKED EXAMPLE**

Row sums $(24, 16)$: $\mu = (3, 2)$; $x_1 \approx (-0.18, 0.51)$, $x_1 - \mu \approx (-3.18, -1.49)$

VOTE

# Principal Component Analysis

## What Does an Uncentered SVD Find?




Uncentered $X$: what does its first right singular vector $v_1$ mostly reflect?

A. the cloud’s offset from the origin

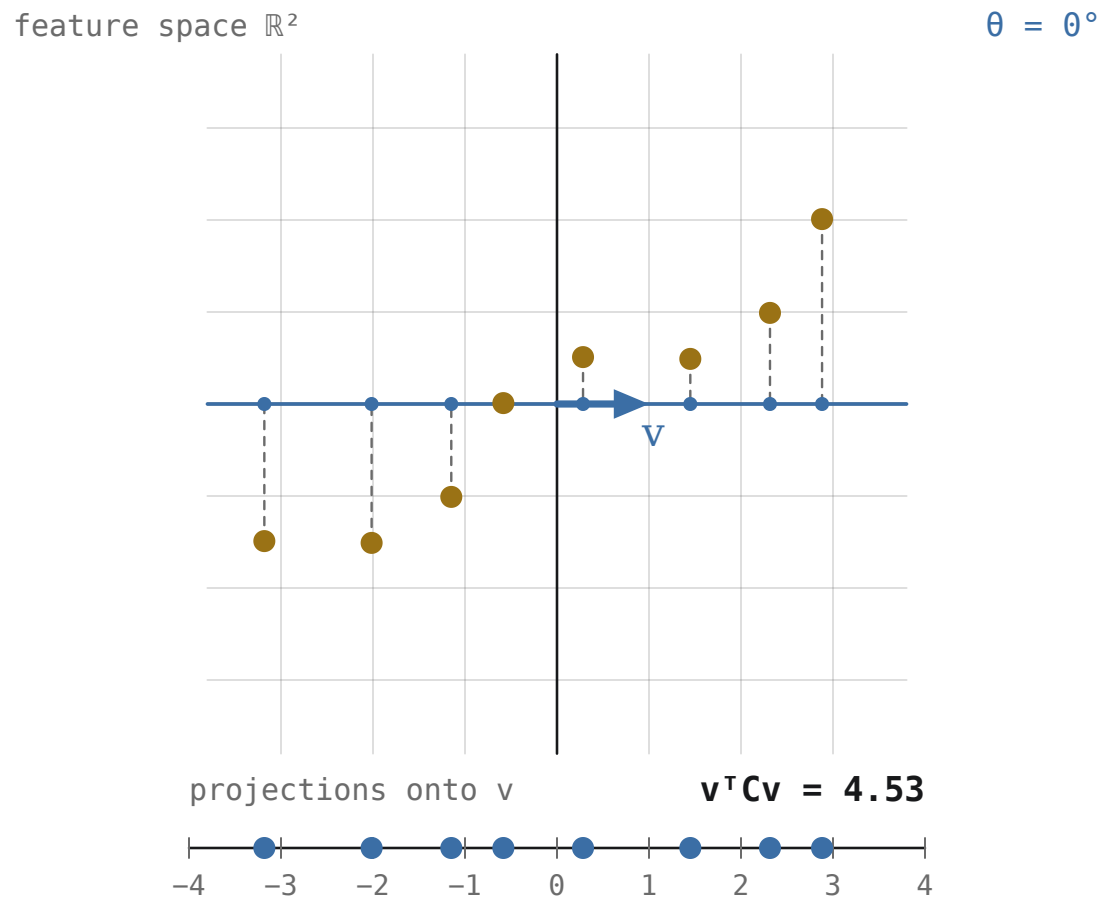
B. the largest variation around the mean, guaranteed

C. the number of samples $N$

DEFINITION

# Principal Component Analysis

## Sample Covariance and Directional Variance




$$C = \frac{X_c^\top X_c}{N-1}, \qquad \frac{1}{N-1}\sum_j (x_{c,j}^\top v)^2 = v^\top C v$$

| | |
|---|---|
| $x_{c,j}^\top$ | row $j$ of $X_c$ |
| $v \in \mathbb{R}^d$ | unit direction |
| left side | variance of the $N$ projections (mean $0$): sum of squares over $N-1$ |
| $C$ | $d \times d$, symmetric, PSD; $N > 1$ |

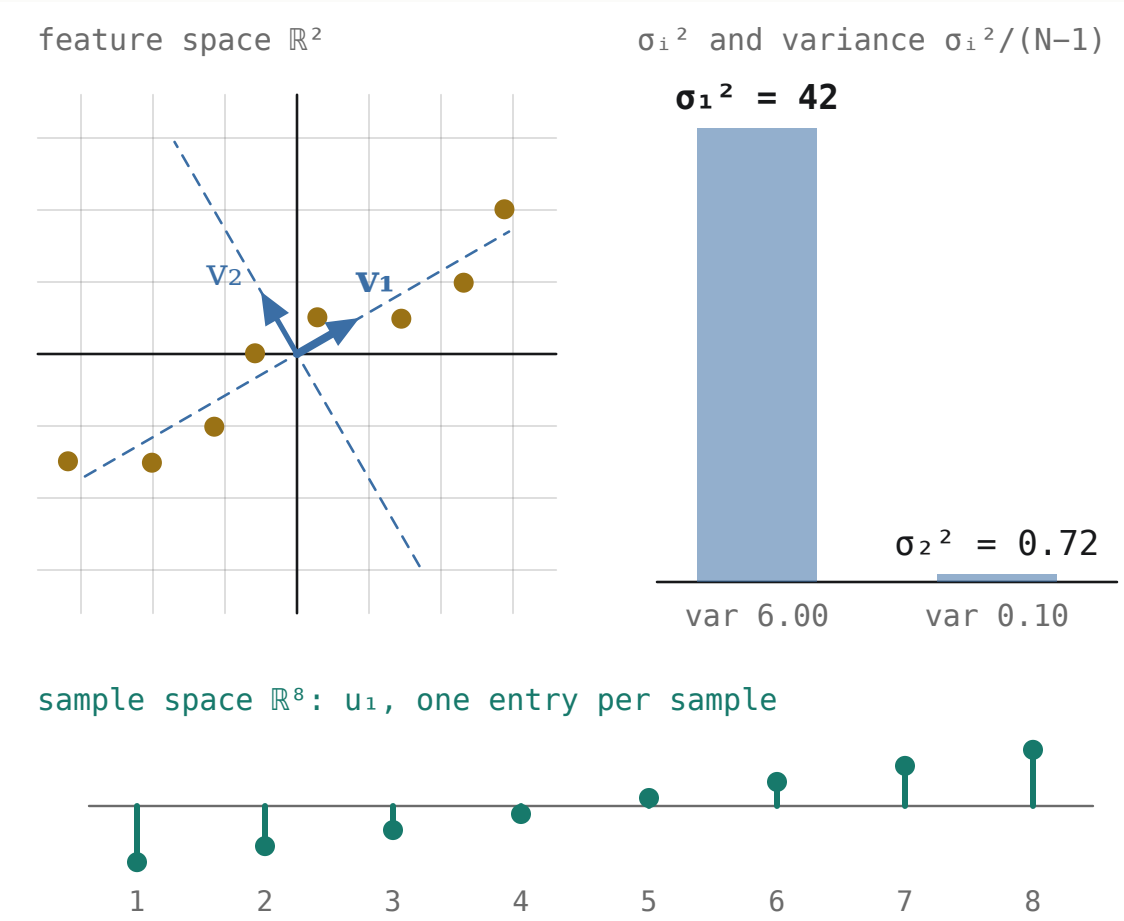
**WORKED EXAMPLE**

$v = e_1$: $v^\top C v = C_{11} = 31.68/7 \approx 4.53$, the variance of feature 1.

$\mu = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

# Principal Component Analysis

## The SVD of Centered Data Gives PCA




$$X_c = U\Sigma V^\top, \qquad Cv_i = \frac{\sigma_i^2}{N-1} v_i$$

why $X_c^\top X_c = V\Sigma^\top \Sigma V^\top$

$v_i \in \mathbb{R}^d$ feature space: principal directions; $v_1$ maximizes $v^\top C v$ over unit $v$

$u_i \in \mathbb{R}^N$ sample space: $u_i = X_c v_i / \sigma_i$ if $\sigma_i > 0$; one entry per sample

$\sigma_i^2$ $= \|X_c v_i\|^2$; variance along $v_i$ is $\sigma_i^2/(N-1)$

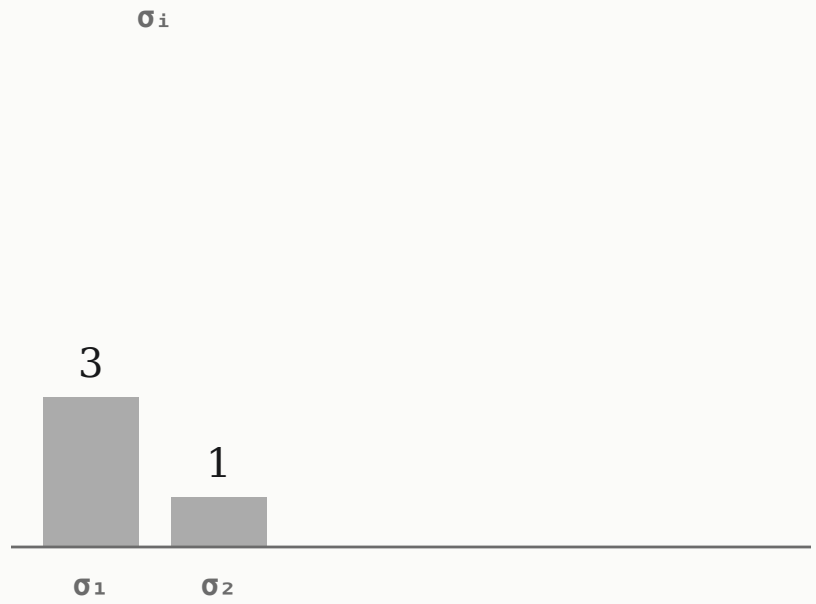
VOTE

# Principal Component Analysis

## How Much Variance Does One Component Keep?

A new small example: centered data have singular values $\sigma_1 = 3$, $\sigma_2 = 1$. What share of the total variance does the first component keep?

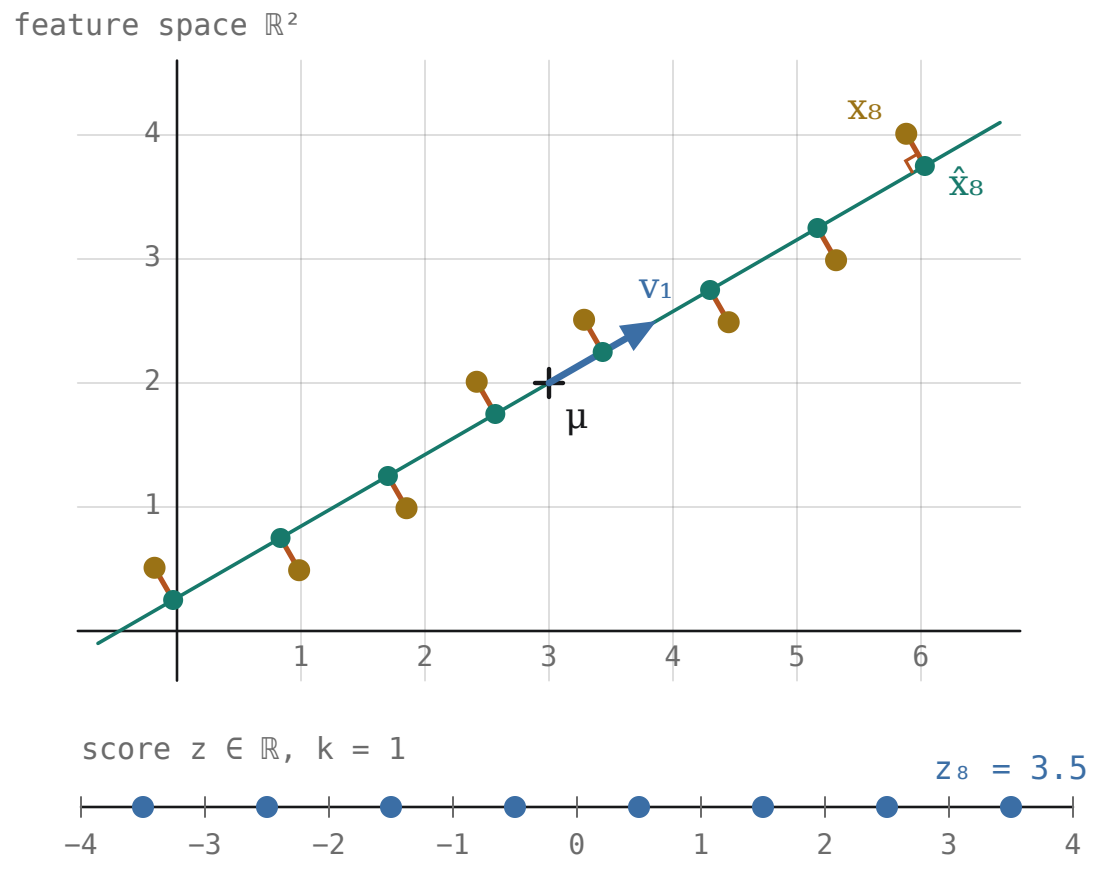
- A 75%
- B 90%
- C 50%



DEFINITION

# Principal Component Analysis

## Coordinates, Then Reconstruction




$$Z = X_c V_k, \quad \hat{X} = Z V_k^{\mathsf{T}} + \mathbf{1}\mu^{\mathsf{T}}$$

- $V_k$ first $k$ columns of $V$, $d \times k$, orthonormal
- $Z$ $N \times k$ scores, one row per sample
- $\hat{X}$ $N \times d$; $Z V_k^{\mathsf{T}} = X_c V_k V_k^{\mathsf{T}}$: best rank-$k$ approximation of $X_c$

**WORKED EXAMPLE**

$x_8 \approx (5.88, 4.01)$: $z_8 = (x_8 - \mu)^{\mathsf{T}} v_1 \approx 3.5$, $\hat{x}_8 = \mu + 3.5 v_1 \approx (6.03, 3.75)$, error $0.3$

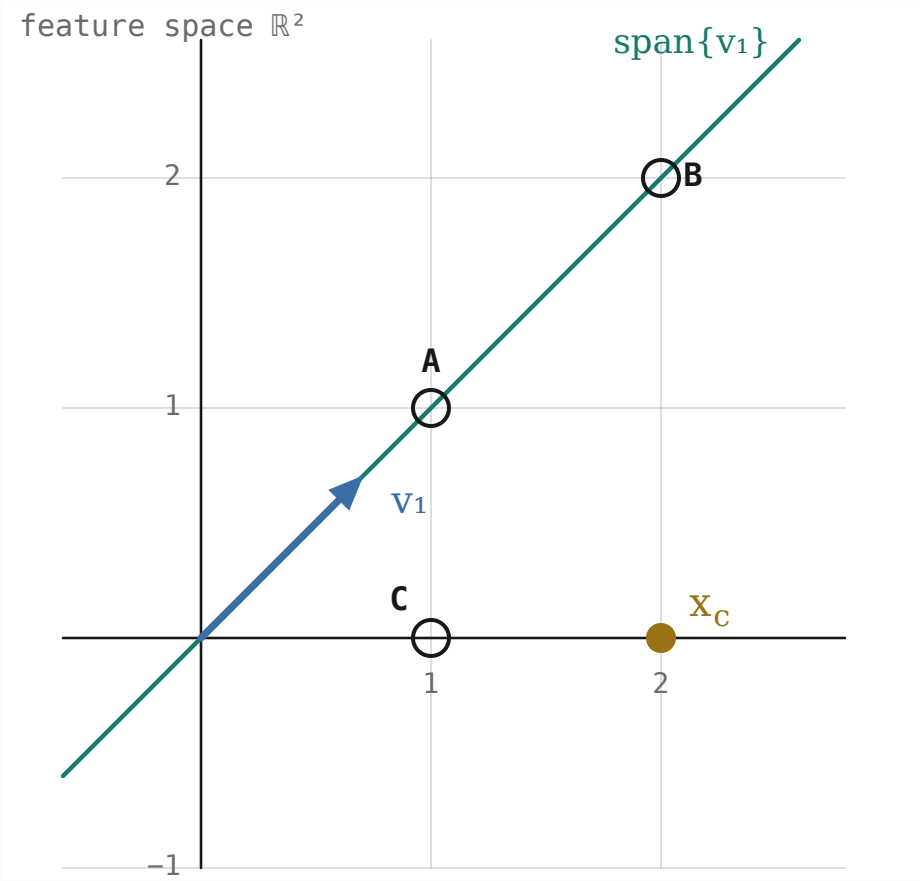
VOTE

# Principal Component Analysis

## Score and Reconstruction of One Point

$\mu = 0$, centered point $x_c = (2, 0)$, first direction $v_1 = (1, 1)/\sqrt{2}$. Which pair is its score $z$ and reconstruction $\hat{x}$?

- **A** $z = \sqrt{2},\ \hat{x} = (1, 1)$
- **B** $z = 2,\ \hat{x} = (2, 2)$
- **C** $z = 1,\ \hat{x} = (1, 0)$



IN PRACTICE

# Principal Component Analysis

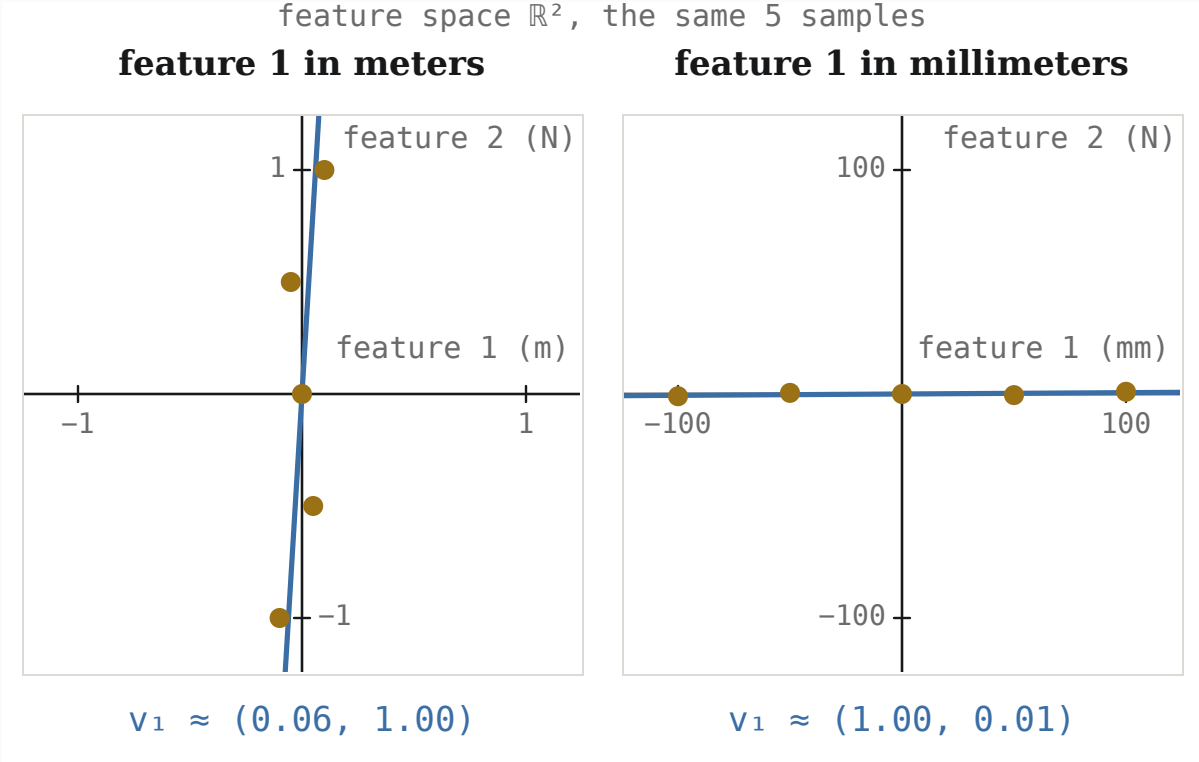
## Units, Scale and What Variance Means

$$\text{kept}_k = \frac{\sum_{i \le k} \sigma_i^2}{\sum_i \sigma_i^2}$$

$\sigma_i$ singular values of $X_c$

$\text{kept}_k$ share in $k$ components; undefined if $X_c = 0$

- Same samples, new units, new $v_1$.
- Center first. Standardizing is a modeling choice, not an automatic cure.
- Large variance need not mean task relevance.



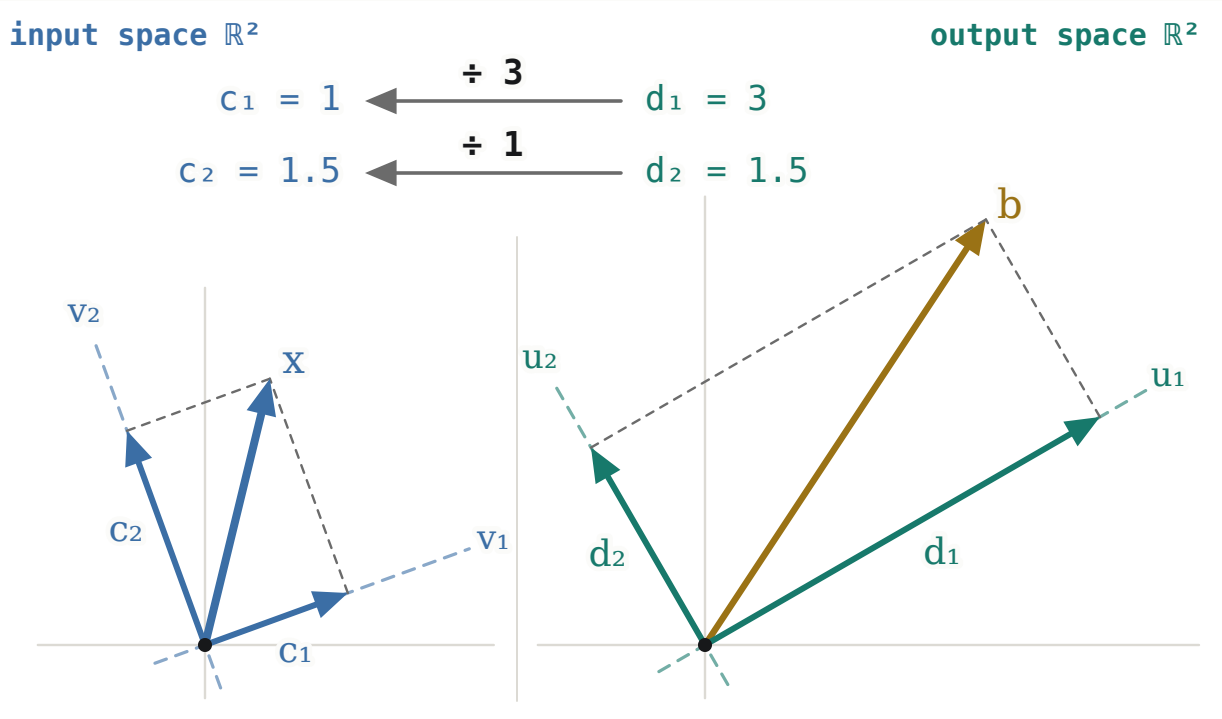
$$A \approx \begin{bmatrix} 2.61 & 0.42 \\ 1.11 & 1.33 \end{bmatrix}$$

# Pseudoinverse and Conditioning

## Reversing the SVD One Coordinate at a Time

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \end{aligned} \iff \begin{aligned} \sigma_1 c_1 &= d_1 \\ \sigma_2 c_2 &= d_2 \end{aligned}$$

- $c = V^\mathsf{T} x$ — $x$ along $v_1, v_2$
- $d = U^\mathsf{T} b$ — $b$ along $u_1, u_2$
- $\sigma_i > 0$ — one per row: $c_i = d_i / \sigma_i$
- $\sigma_i = 0 \neq d_i$ — $0 = d_i$: no solution
- $\sigma_i = 0 = d_i$ — $c_i$ free: take 0



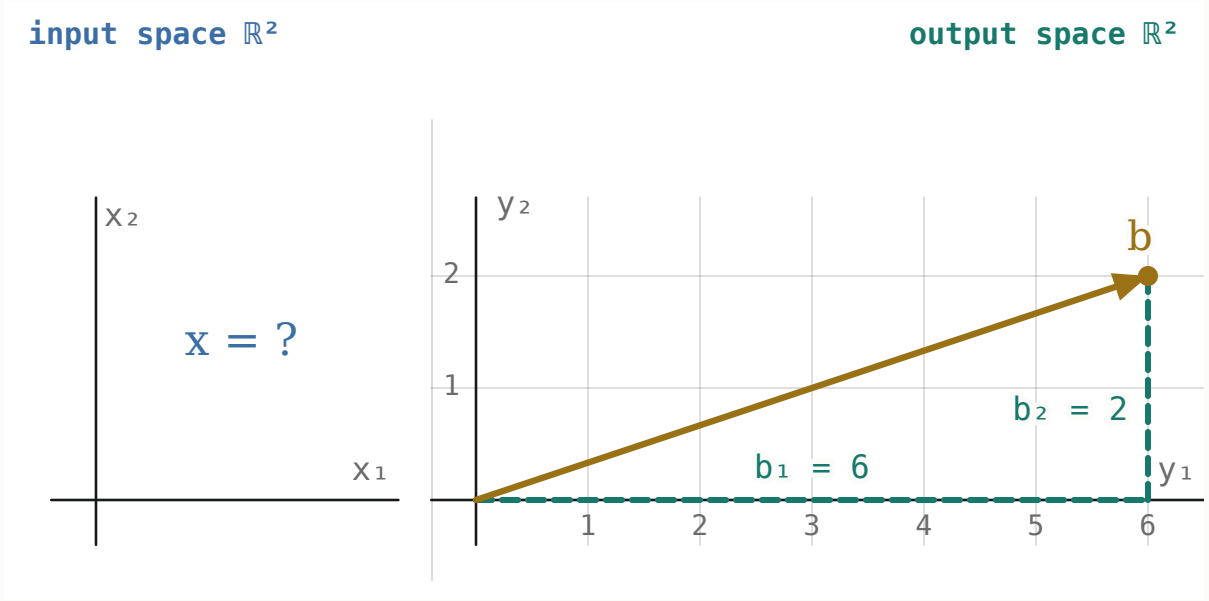
VOTE

# Pseudoinverse and Conditioning

## Which Input Produces This Output?

$A = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$, $b = (6, 2)$. Which $x$ gives $Ax = b$?

- **A** $x = (18, 2)$
- **B** $x = (2, 2)$
- **C** $x = (2, 6)$



DEFINITION

# Pseudoinverse and Conditioning

## The Pseudoinverse

$$A^+ = V_r \Sigma_r^{-1} U_r^\mathsf{T} = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^\mathsf{T}$$

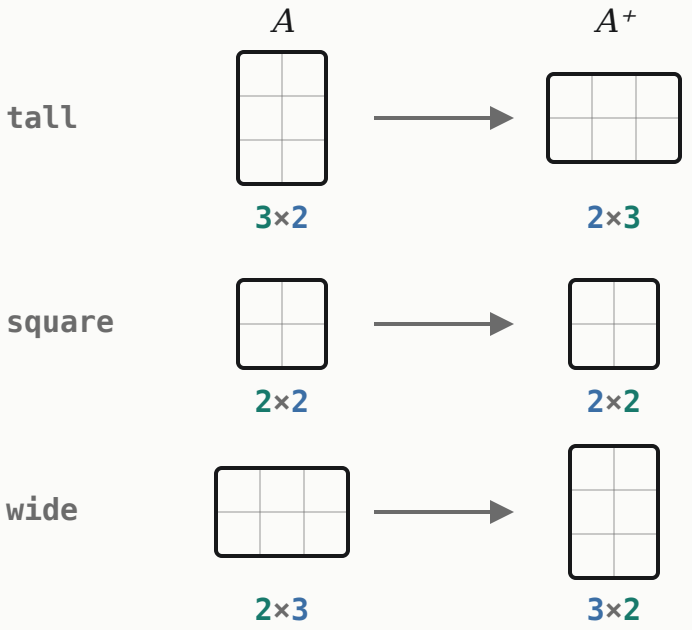
$A \in \mathbb{R}^{m \times n}$ $A^+ \in \mathbb{R}^{n \times m}$; rank $r$

$\sigma_i > 0$ inverted; a zero gain is not: what is lost stays lost

$r = 0$ $A^+ = 0$

**WORKED EXAMPLE**

$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 2 \end{bmatrix}$, $\sigma = (2, 1)$: $A^+ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$



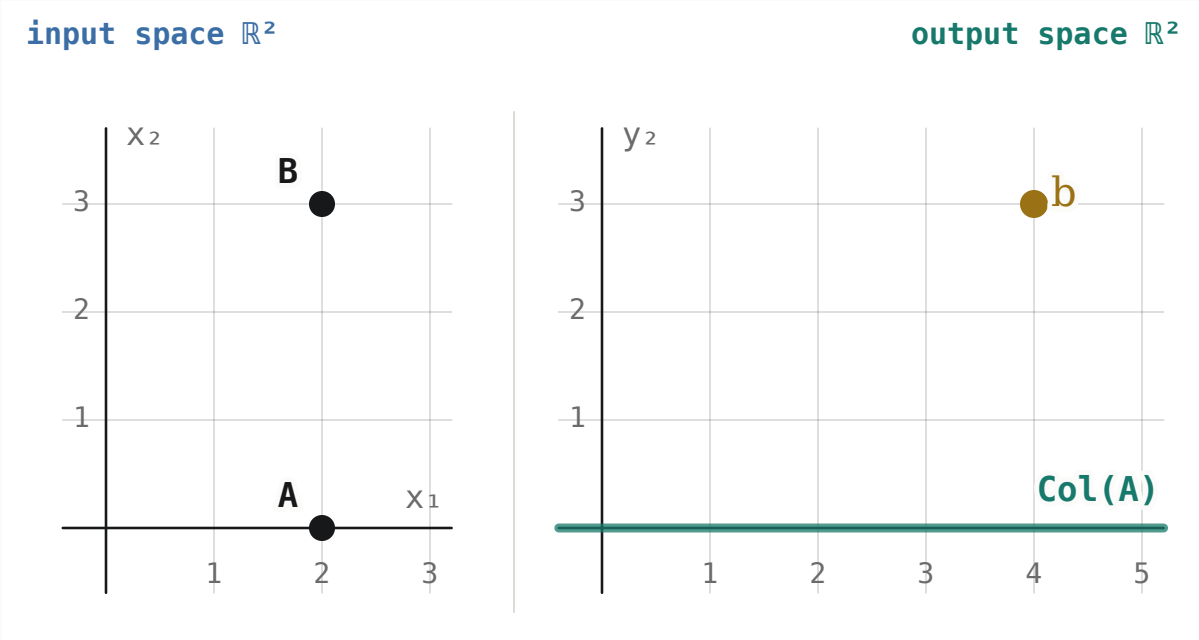
VOTE

# Pseudoinverse and Conditioning

## What Does the Pseudoinverse Return?

$A = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, $b = (4, 3)$. What is $A^{+}b$?

- A $(2, 0)$
- B $(2, 3)$
- C none: no exact solution



PICTURE

# Pseudoinverse and Conditioning

## Two Projectors: Output Side and Input Side

$$AA^{+} = U_r U_r^{\mathsf{T}}, \qquad A^{+}A = V_r V_r^{\mathsf{T}}$$

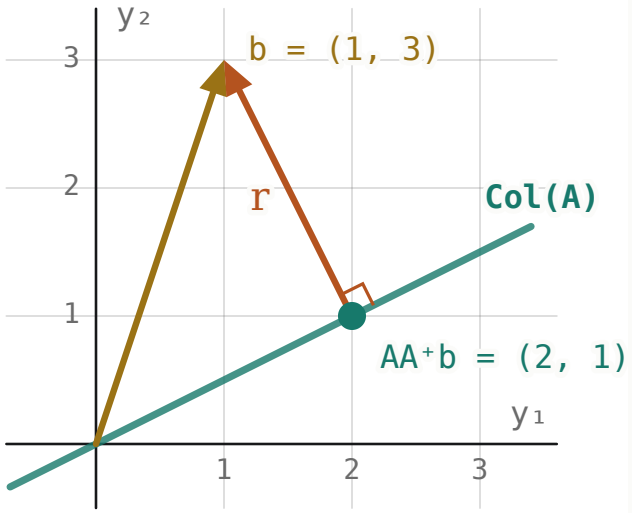
$AA^{+}$ $m \times m$, projects onto $\mathrm{Col}(A)$

$A^{+}A$ $n \times n$, projects onto $\mathrm{Row}(A)$

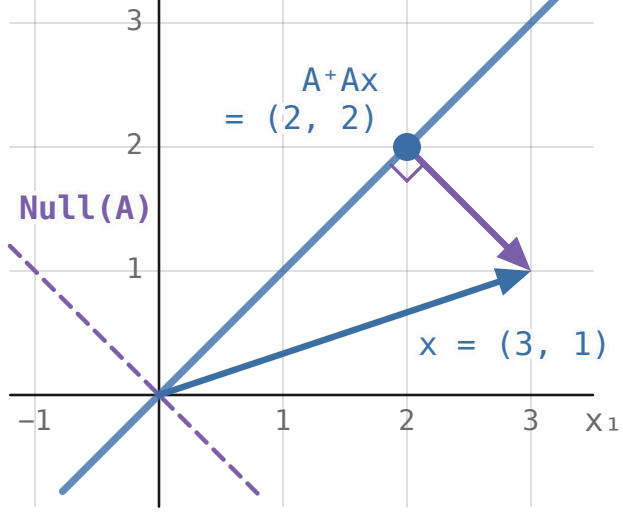
$\neq I$ unless $r = m$, resp. $r = n$

Here $A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$, rank 1.

output space $\mathbb{R}^m$, $m = 2$




input space $\mathbb{R}^n$, $n = 2$



DEFINITION

# Pseudoinverse and Conditioning

## Minimum-Norm Least Squares

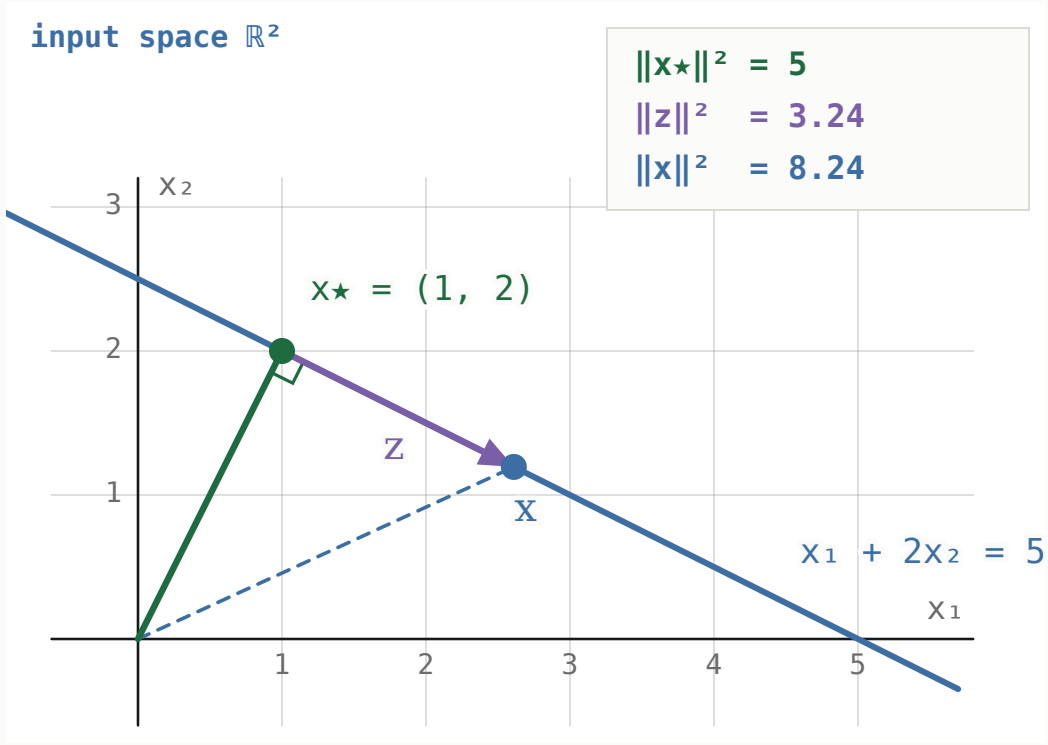
First minimize the residual $\|b - Ax\|$; then take the smallest norm among tied minimizers.

$$x = x^\star + z: \quad \|x\|^2 = \|x^\star\|^2 + \|z\|^2$$

- $x$ any minimizer; $z \in \mathrm{Null}(A)$
- $x^\star = A^+ b$ in $\mathrm{Row}(A)$, so $x^\star \perp z$
- $\|x\|^2$ squared norm: least only at $z = 0$

**WORKED EXAMPLE**

$A = [1\ \ 2]$, $b = 5$: $A^+ = (1, 2)/5$, $x^\star = (1, 2)$, $\|x^\star\|^2 = 5$



VOTE

# Pseudoinverse and Conditioning

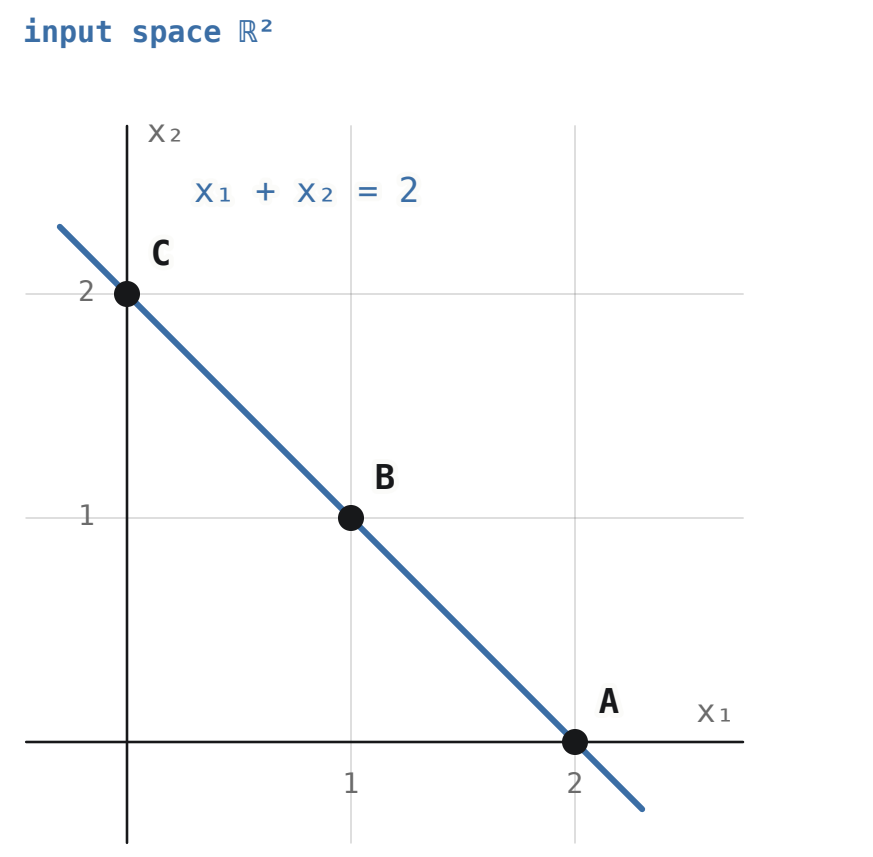
## Which Solution Has the Smallest Norm?

All three solve $x_1 + x_2 = 2$. Which has the smallest norm $\|x\|$?

A $(2, 0)$

B $(1, 1)$

C $(0, 2)$



PICTURE

# Pseudoinverse and Conditioning

## Small Gains Amplify Noise

$$\delta x = \sum_{\sigma_i > 0} \frac{u_i^\top \delta b}{\sigma_i} v_i$$

$\delta x \in \mathbb{R}^n$ input error (mm)

$\delta b \in \mathbb{R}^m$ output error (mm)

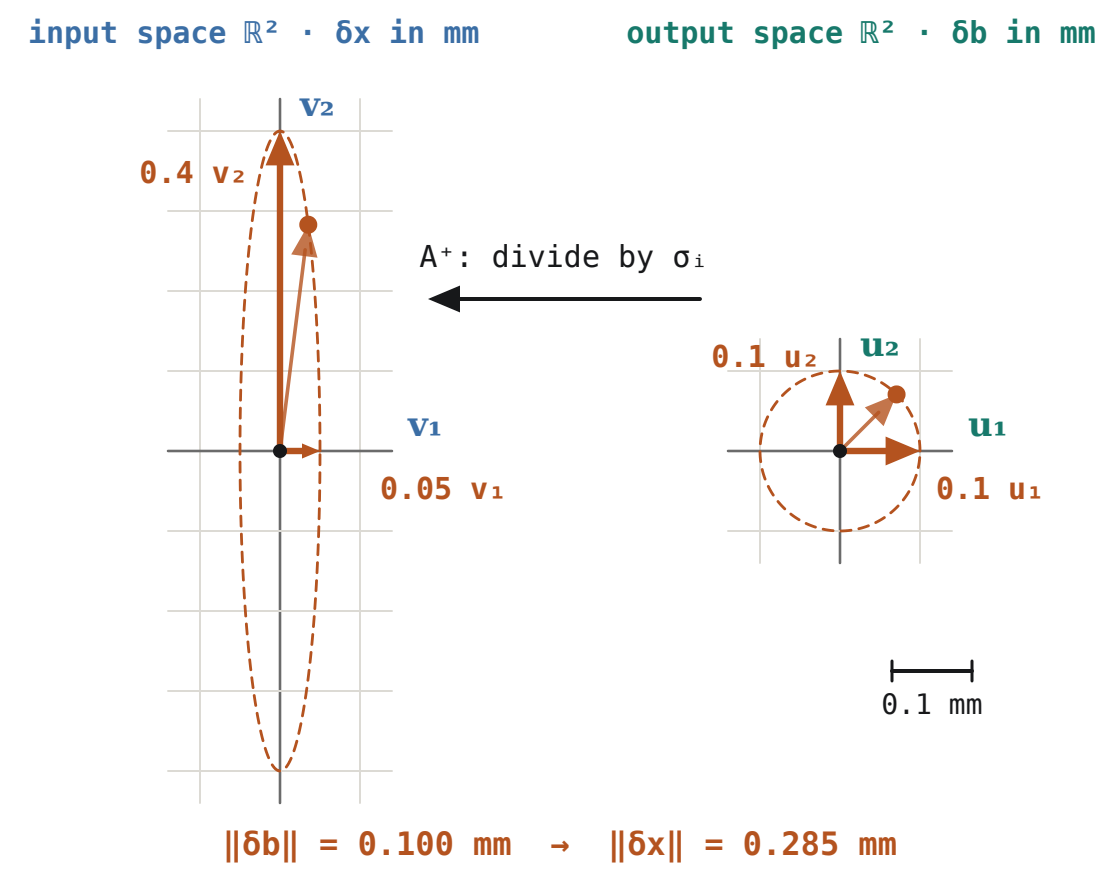
$A$ fixed $x^\star + \delta x = A^+(b + \delta b)$: only $b$ moves

**WORKED EXAMPLE**

$A = \mathrm{diag}(2, 0.25)$, $\|\delta b\| = 0.1$ mm.

Along $u_1$: $0.1/2 = 0.05$ mm along $v_1$.

Along $u_2$: $0.1/0.25 = 0.4$ mm along $v_2$: 8 times.



VOTE

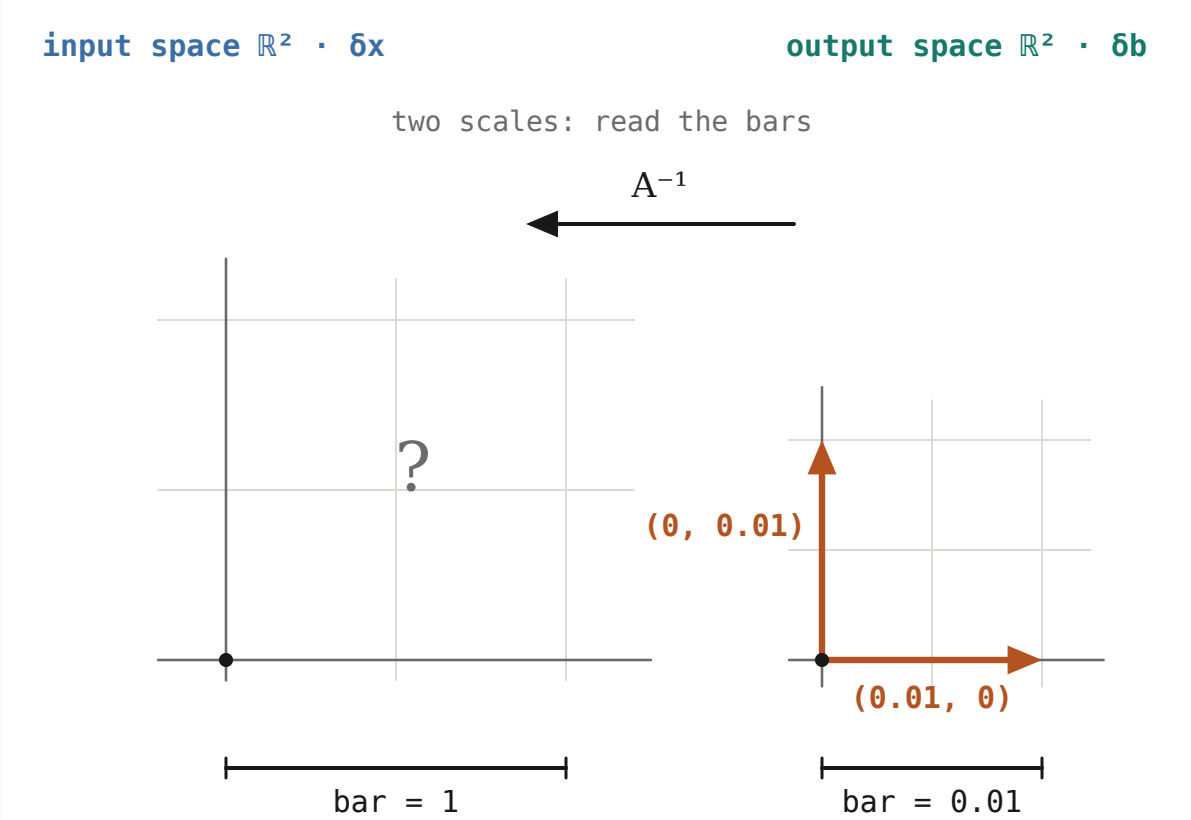
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0.01 \end{bmatrix}$$

# Pseudoinverse and Conditioning

## Which Perturbation Moves the Input More?

Both output errors have length $0.01$. Which gives the larger input error $\|\delta x\|$?

- **A** $\delta b = (0.01,\ 0)$
- **B** $\delta b = (0,\ 0.01)$
- **C** the same for both



DEFINITION

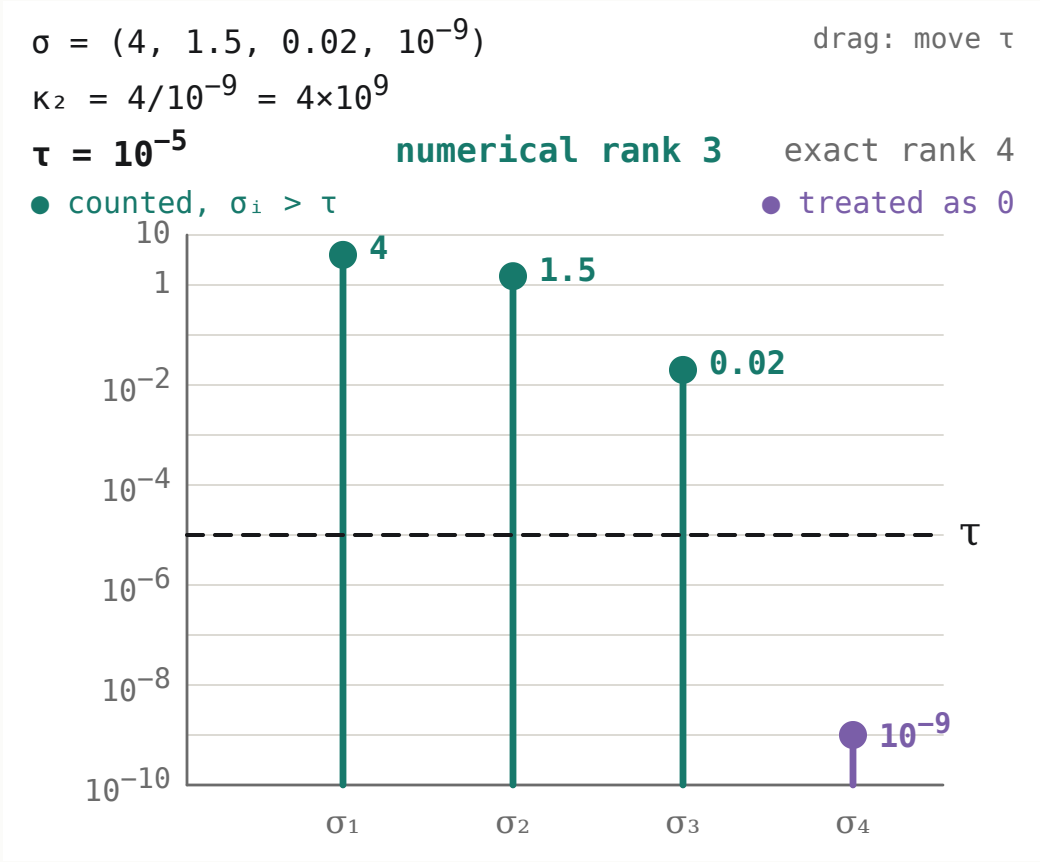
# Pseudoinverse and Conditioning

## Condition Number and Numerical Rank

$$\|A\|_2 = \sigma_{\max}, \quad \kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}, \quad r_\tau = \#\{i : \sigma_i > \tau\}$$

| | |
|---|---|
| $\|A\|_2, r_\tau$ | any $A$, $\tau \geq 0$ |
| $\kappa_2$ | full rank; else $\infty$ by convention |
| $1/\sigma_{\min}$ | absolute; $\kappa_2$ relative, worst case |
| scope | exact $Ax = b$, only $b$ moves |

Conditioning is the problem's, stability the algorithm's.

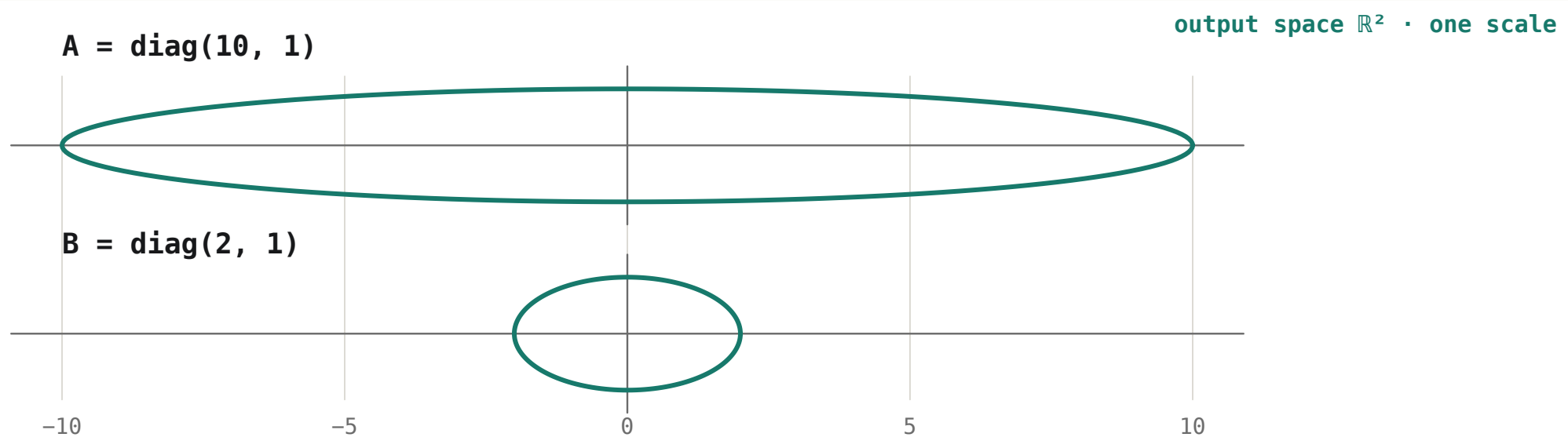


VOTE

# Pseudoinverse and Conditioning

## Which Matrix Is Worse Conditioned?

Image ellipses of $A$ and $B$ at one scale. Which is worse conditioned?




A $A = \mathrm{diag}(10, 1)$

B $B = \mathrm{diag}(2, 1)$

C equally conditioned

DEFINITION

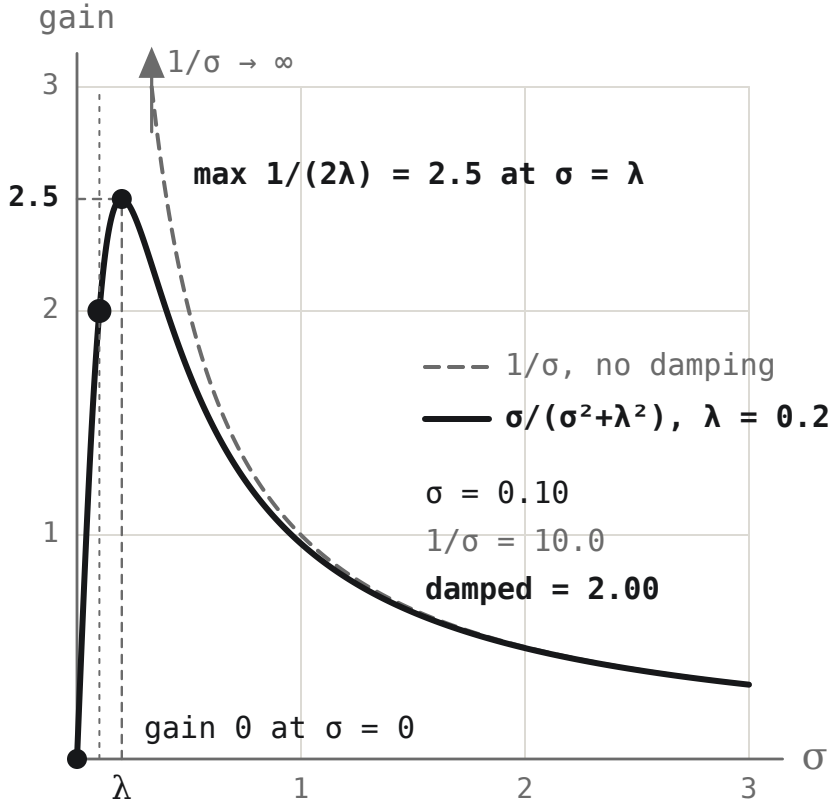
# Pseudoinverse and Conditioning

## Damped Least Squares

$$x_\lambda = \arg\min_x \left( \|Ax - b\|^2 + \lambda^2 \|x\|^2 \right) = \sum_{\sigma_i > 0} \frac{\sigma_i \, u_i^\top b}{\sigma_i^2 + \lambda^2} v_i$$

- $\lambda > 0$ damping, a number we choose
- $x_\lambda$ solves $(A^\top A + \lambda^2 I)x = A^\top b$: always unique
- $\sigma_i/(\sigma_i^2 + \lambda^2)$ damped gain; $0$ when $\sigma_i = 0$

> Bounded amplification: every damped gain is at most $1/(2\lambda)$.



VOTE

# Pseudoinverse and Conditioning

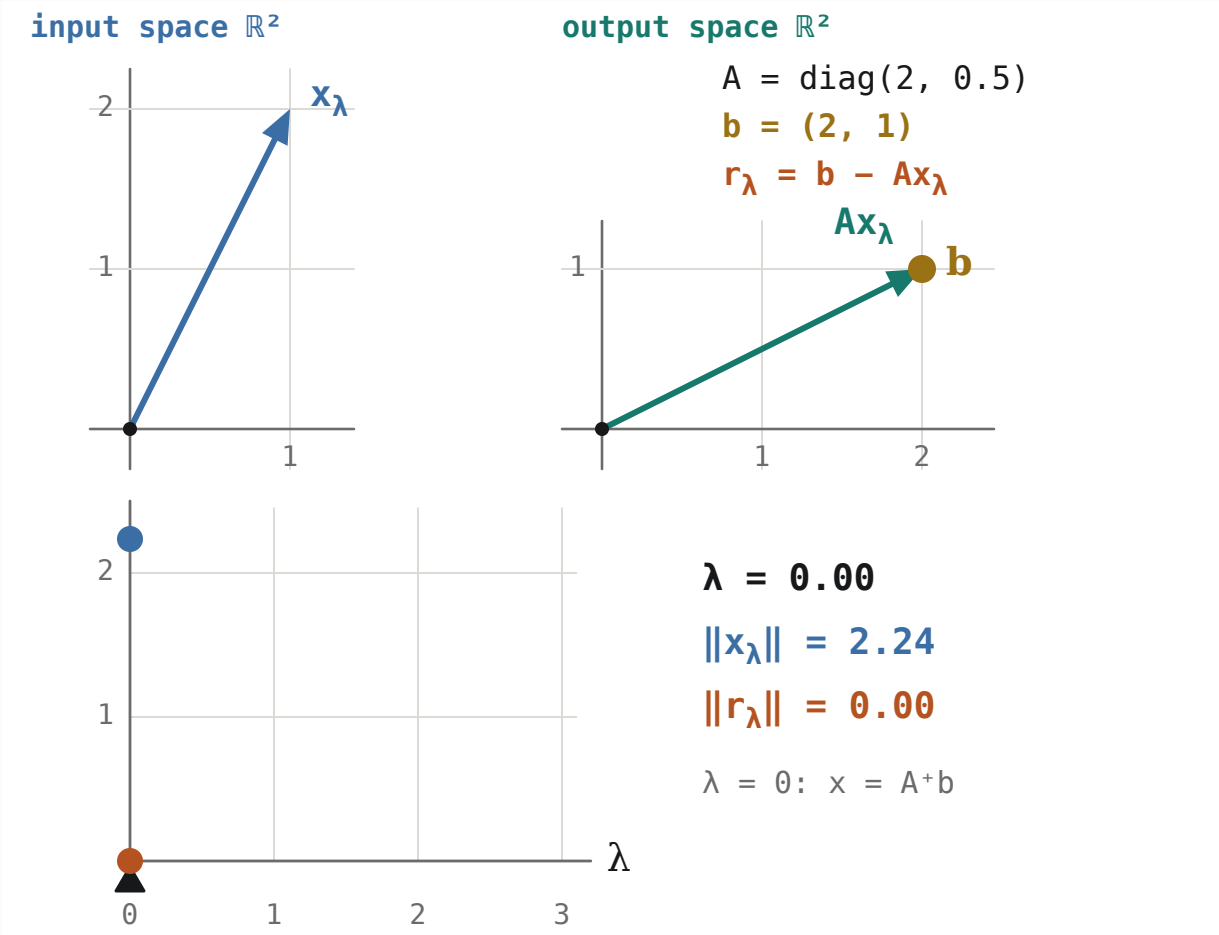
## What Happens as Damping Grows?

Fix $A, b$. As $\lambda$ grows $0 \to 3$, what do $\|x_\lambda\|$ and $\|r_\lambda\|$ do?

A $\|x_\lambda\|$ nonincreasing, $\|r_\lambda\|$ nondecreasing

B $\|x_\lambda\|$ increases, $\|r_\lambda\|$ always decreases

C $b$ stays exactly attained



IN PRACTICE

# Pseudoinverse and Conditioning

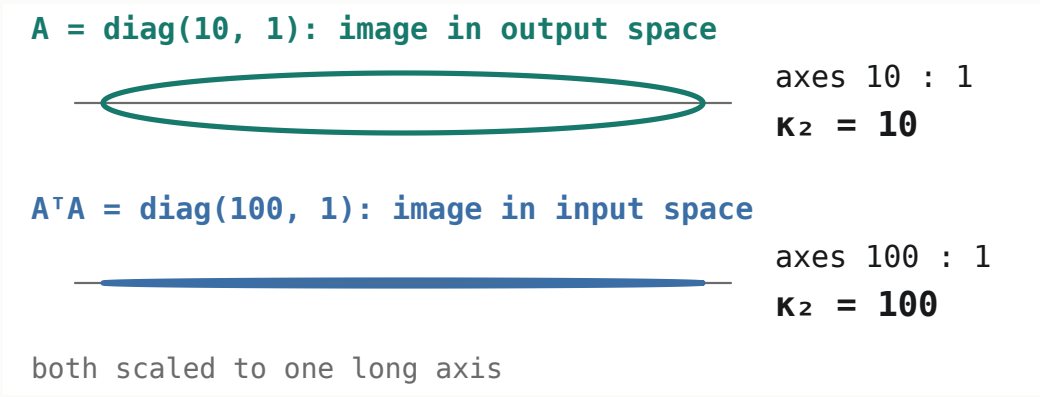
## Choosing a Solver

| SOLVER | NEEDS | GIVES |
|---|---|---|
| thin QR, Lecture 3 | full column rank | $R\hat{x} = Q^\top b$ |
| SVD | any $A$ | $A^+ b$, minimum norm |
| damped | $\lambda > 0$ | $x_\lambda$ (new objective), gains $\leq 1/(2\lambda)$ |

$$\kappa_2(A^\top A) = \kappa_2(A)^2$$

$A$, $r = n$ $\kappa_2 = \sigma_1/\sigma_n$

$A^\top A$ gains $\sigma_i^2$




In code, solve systems, never invert. QR exists for every $A$.

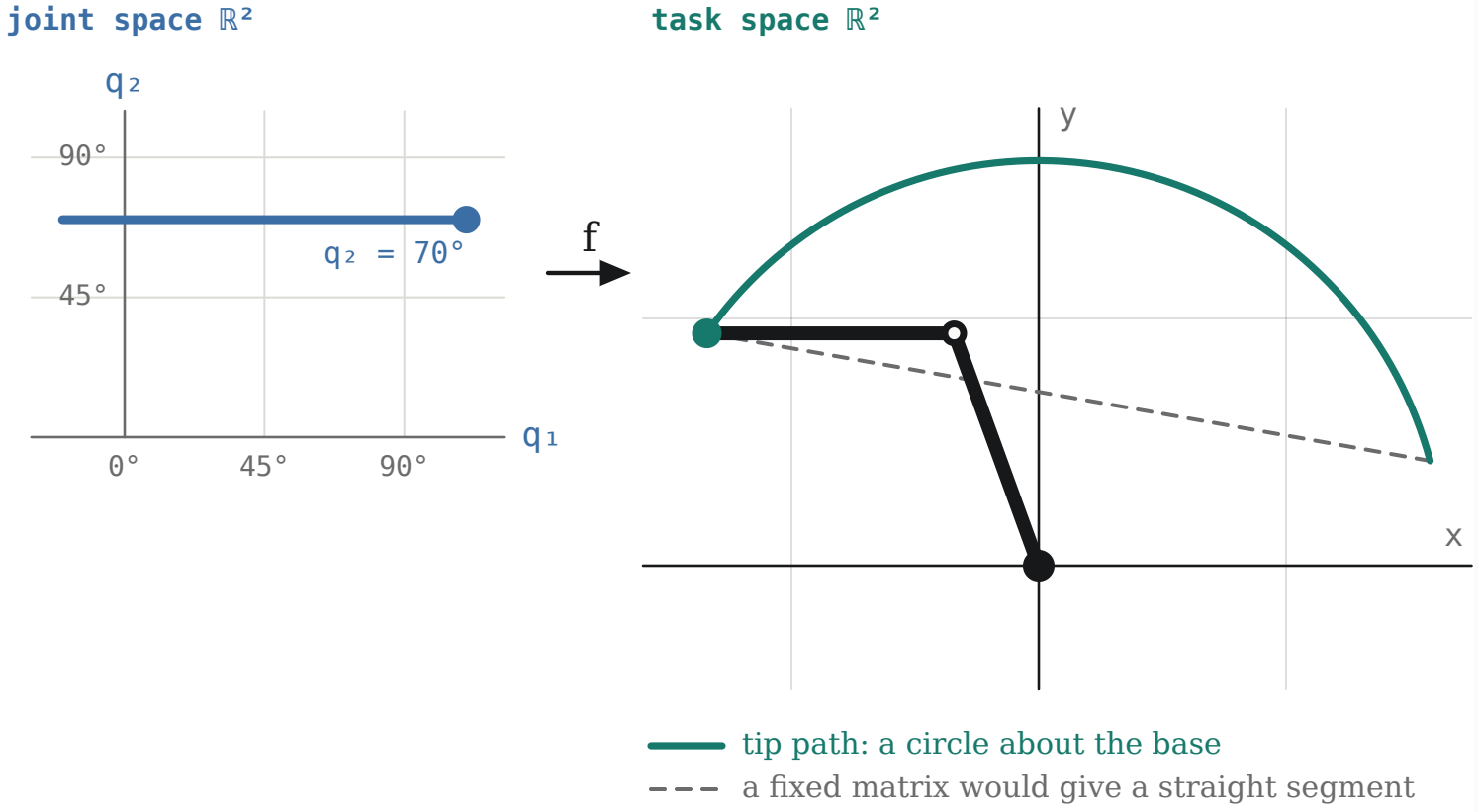
PICTURE

# Jacobians and Robot Motion

## Forward Kinematics Is Nonlinear

$$p = f(q), \qquad q \in \mathbb{R}^n, \;\; p \in \mathbb{R}^m$$

- $q$ joint vector: $n$ angles in radians here; a prismatic joint would contribute a length
- $p$ tip position only: $m$ task coordinates, no orientation
- $f$ forward kinematics: sines and cosines of $q$, not a matrix



VOTE

# Jacobians and Robot Motion

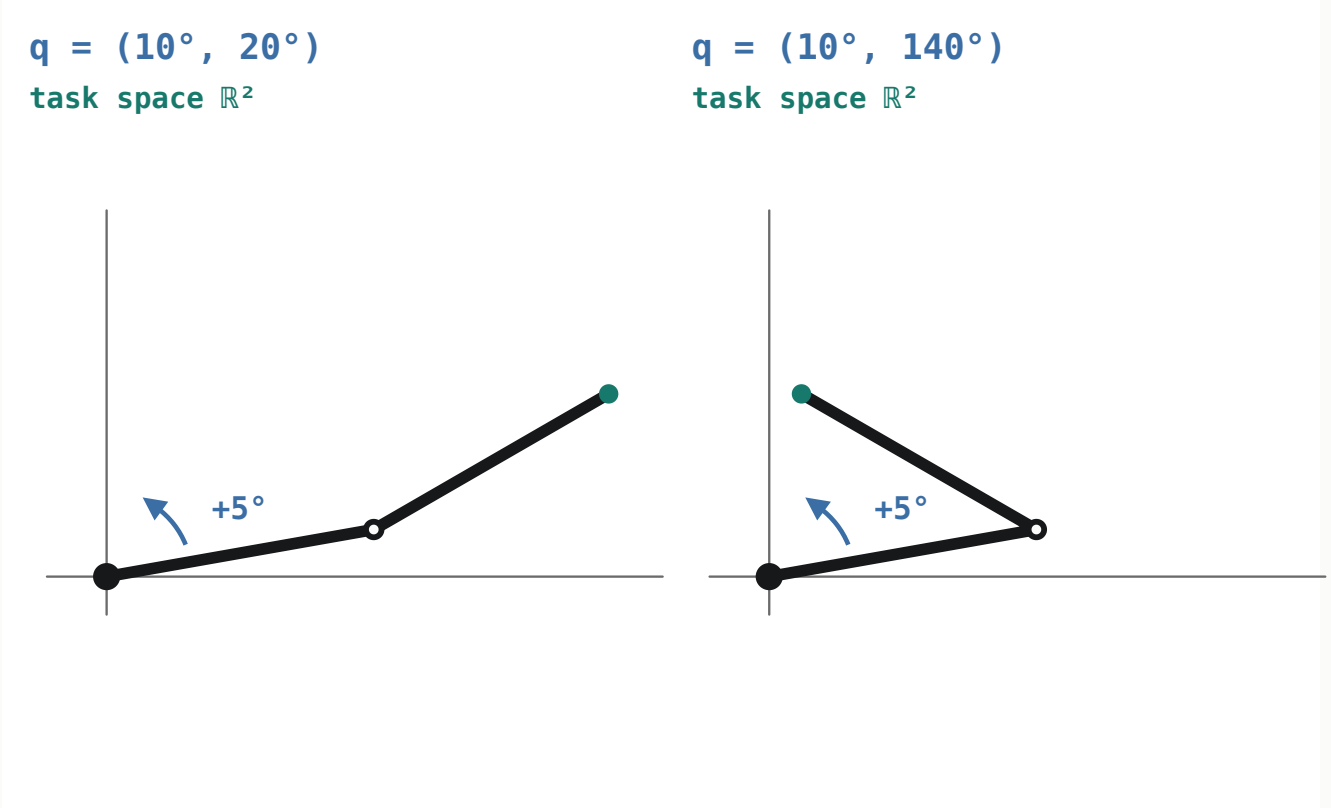
## Does the Same Joint Change Move the Tip the Same Way?

Both shoulders turn $+5°$, elbows fixed. The tip changes $\Delta p$ are…

A always identical

B generally pose-dependent

C determined only by joint count



RECALL

# Jacobians and Robot Motion

## The Derivative as a Local Gain

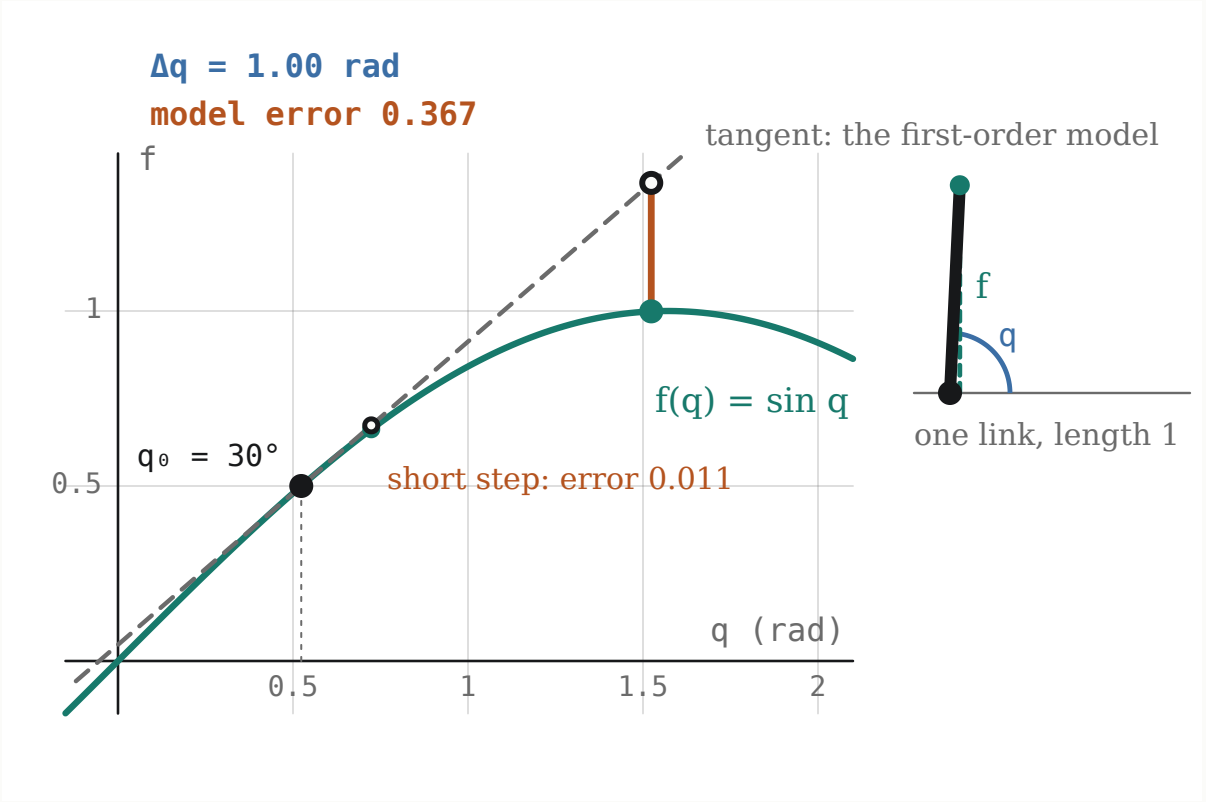
$$f(q + \Delta q) \approx f(q) + f'(q)\,\Delta q$$

$f'(q)$ local gain: output change per radian, at this $q$

$\Delta q$ step in radians; for small steps the model error shrinks like $\Delta q^2$

**WORKED EXAMPLE**

$q = 30°$: $f = 0.5$, $f' = \cos 30° \approx 0.866$. Step $\Delta q = 0.2$ rad: model $\approx 0.673$, actual $\approx 0.662$.



DEFINITION

# Jacobians and Robot Motion

## The Jacobian Matrix

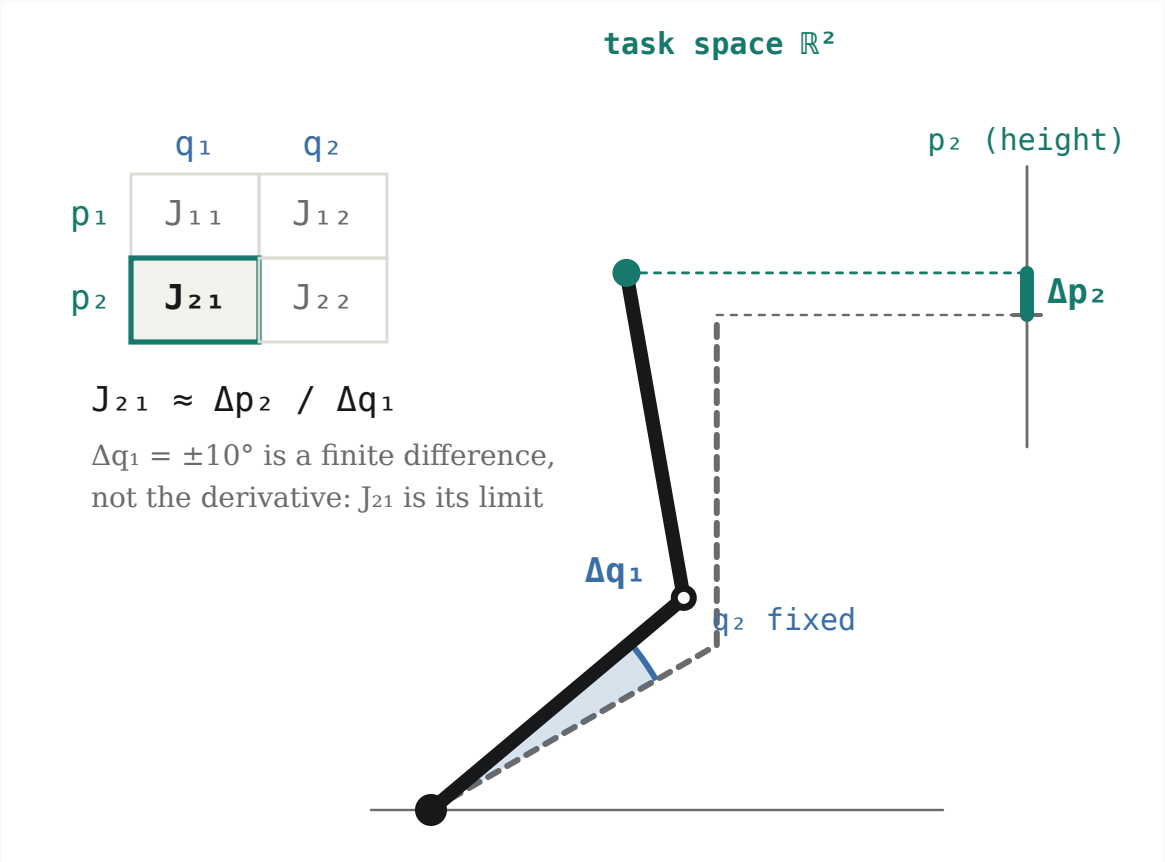
$$J_{ij}(q) = \frac{\partial f_i}{\partial q_j}(q), \qquad J(q) \in \mathbb{R}^{m \times n}$$

row $i$ output $p_i = f_i(q)$, $i = 1, \ldots, m$

column $j$ joint $q_j$, $j = 1, \ldots, n$; the other joints held fixed

$$\Delta p \approx J(q)\, \Delta q$$

$\Delta q \in \mathbb{R}^n$ a small joint change, in radians; $\Delta p \in \mathbb{R}^m$



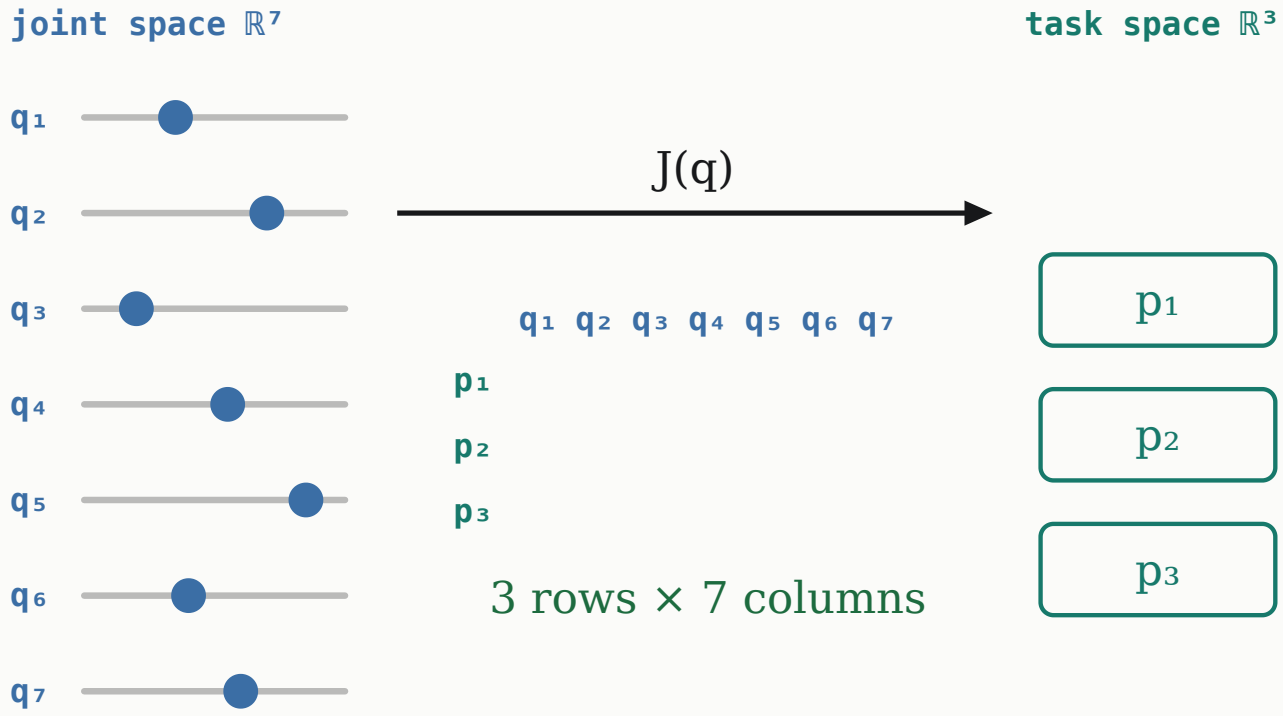
VOTE

# Jacobians and Robot Motion

## What Shape Is This Jacobian?

An arm has 7 joints. The task is only its 3D tip position. What shape is $J$?

- (A) $7 \times 3$
- (B) $3 \times 7$
- (C) $7 \times 7$



PICTURE

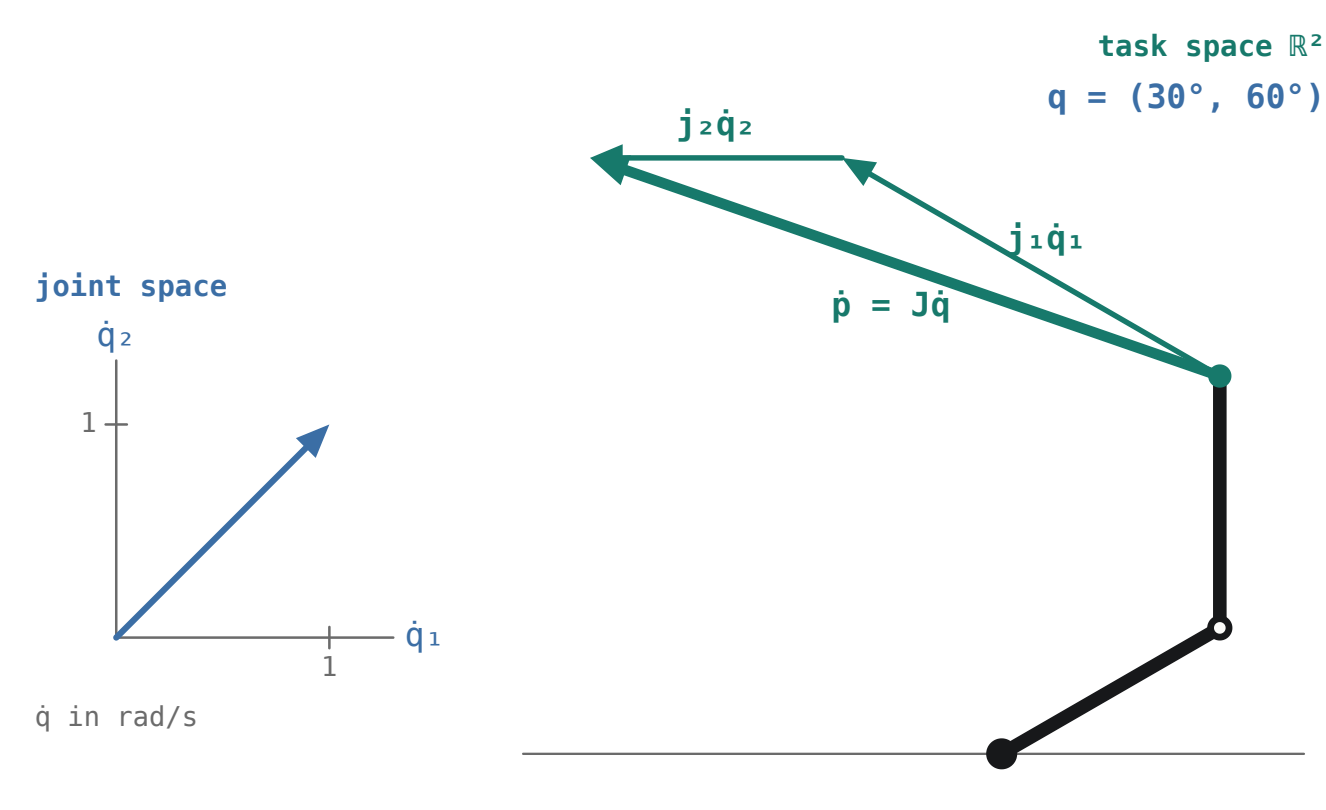
# Jacobians and Robot Motion

## Columns Are Joint Effects

$$J = \begin{bmatrix} j_1 & \cdots & j_n \end{bmatrix}, \quad J\Delta q = \sum_{j=1}^{n} j_j \, \Delta q_j$$

- $j_j \in \mathbb{R}^m$ — column $j$: tip length per radian of joint $j$ alone
- $\Delta q_j$ — how far joint $j$ turns; the contributions add
- $j_j(q)$ — changes with the pose $q$

Drawn: $\dot{p} = J\dot{q}$ at $\dot{q} = (1, 1)$ rad/s, $q = (30°, 60°)$: velocities, not paths.



VOTE

# Jacobians and Robot Motion

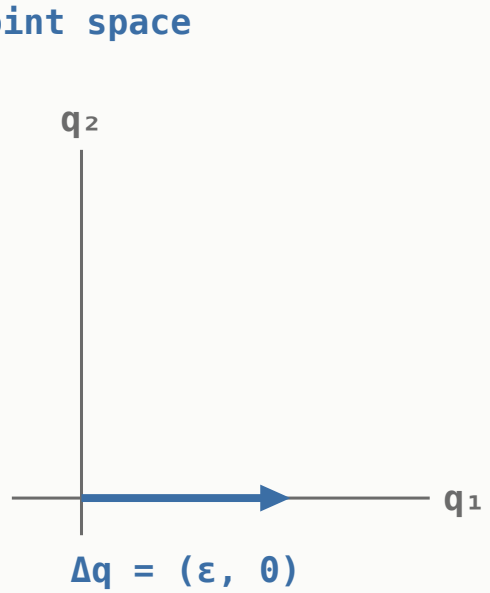
## Which Way Does the Tip Move?

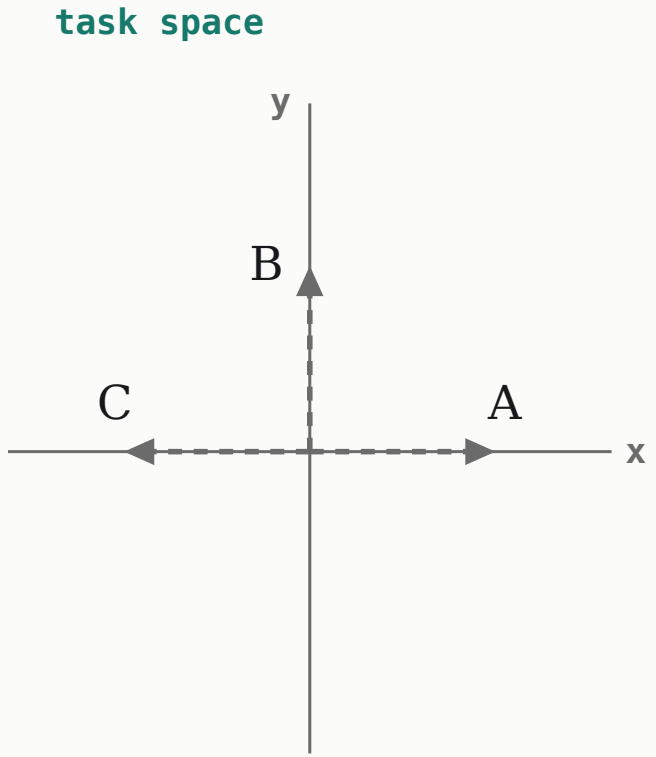
$$J = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}$$

$J$ given; task axes $x$ right, $y$ up

Only joint 1 turns, by a small $\varepsilon > 0$: $\Delta q = (\varepsilon, 0)$. The tip mainly moves…

- A right
- B up
- C left



DERIVATION

# Jacobians and Robot Motion

## Worked Model: The Two-Link Arm

$$p = \begin{bmatrix} l_1 \cos q_1 + l_2 \cos q_{12} \\ l_1 \sin q_1 + l_2 \sin q_{12} \end{bmatrix}$$

$q_{12}$ $q_1 + q_2$

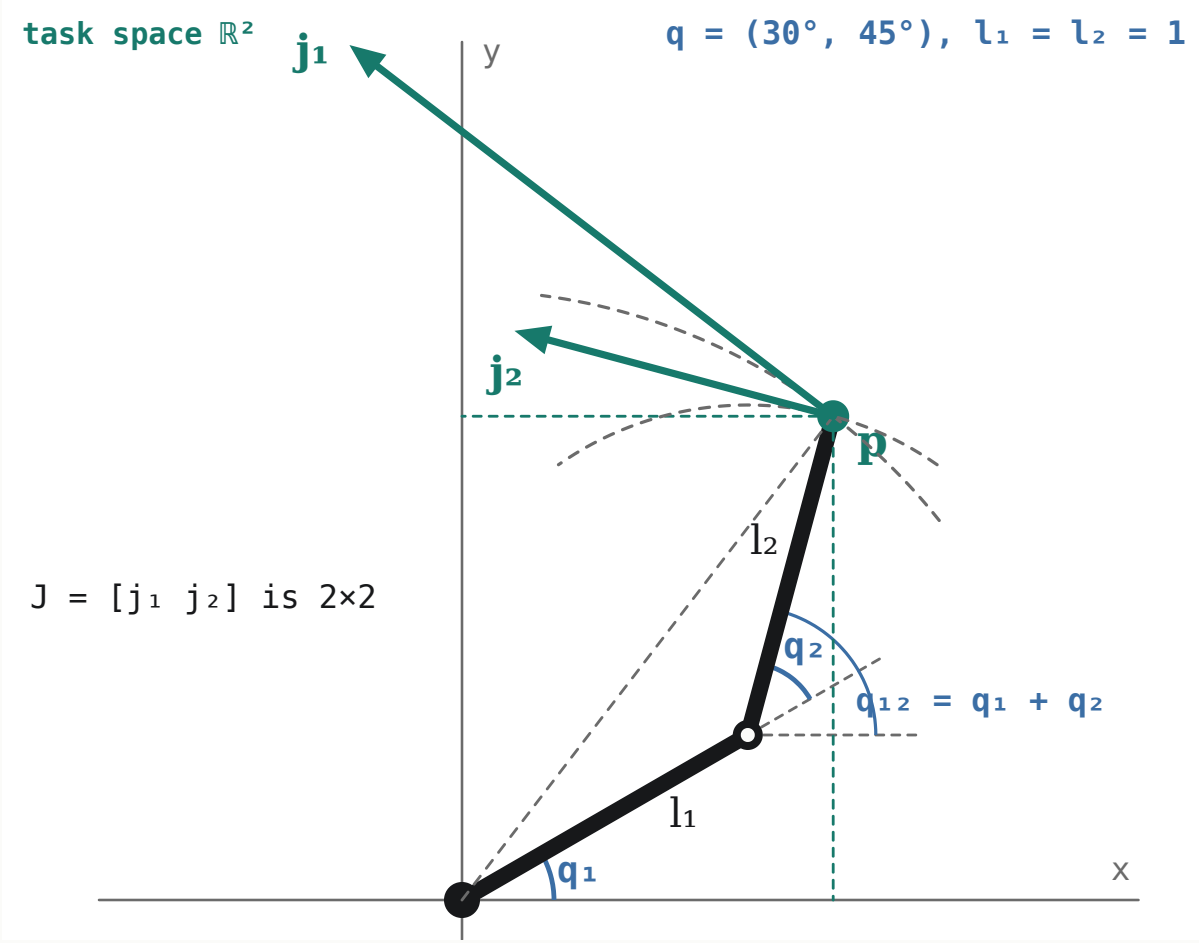
$l_1, l_2 > 0$ lengths; radians

$$j_1 = \begin{bmatrix} -l_1 \sin q_1 - l_2 \sin q_{12} \\ l_1 \cos q_1 + l_2 \cos q_{12} \end{bmatrix}$$

$j_1$ $\partial p / \partial q_1$

$$j_2 = \begin{bmatrix} -l_2 \sin q_{12} \\ l_2 \cos q_{12} \end{bmatrix}$$

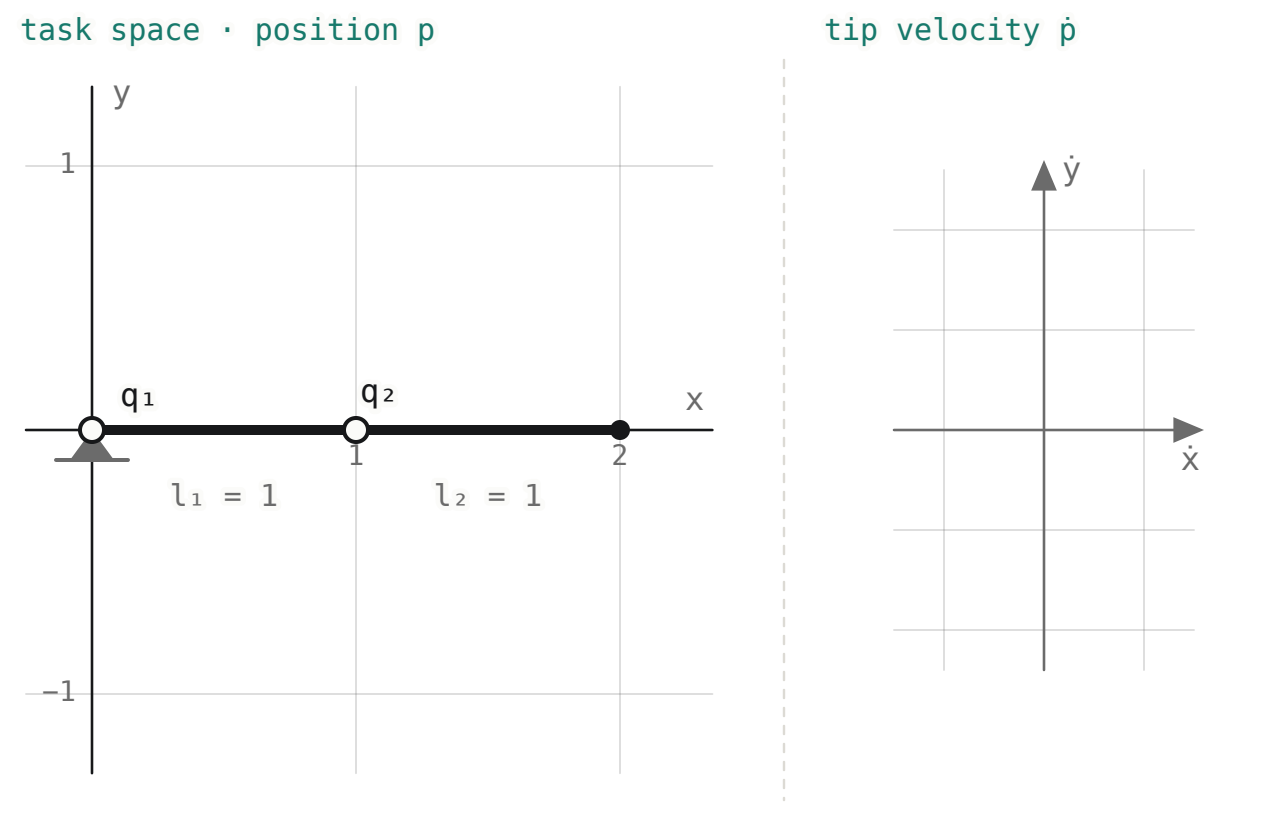
$j_2$ $\partial p / \partial q_2$



VOTE

# Jacobians and Robot Motion

## Which Tip Velocities Are Available?




Straight arm, $q = (0, 0)$: $J = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$. Which nonzero tip velocities $J\dot{q}$ exist?

A. only horizontal, along the arm

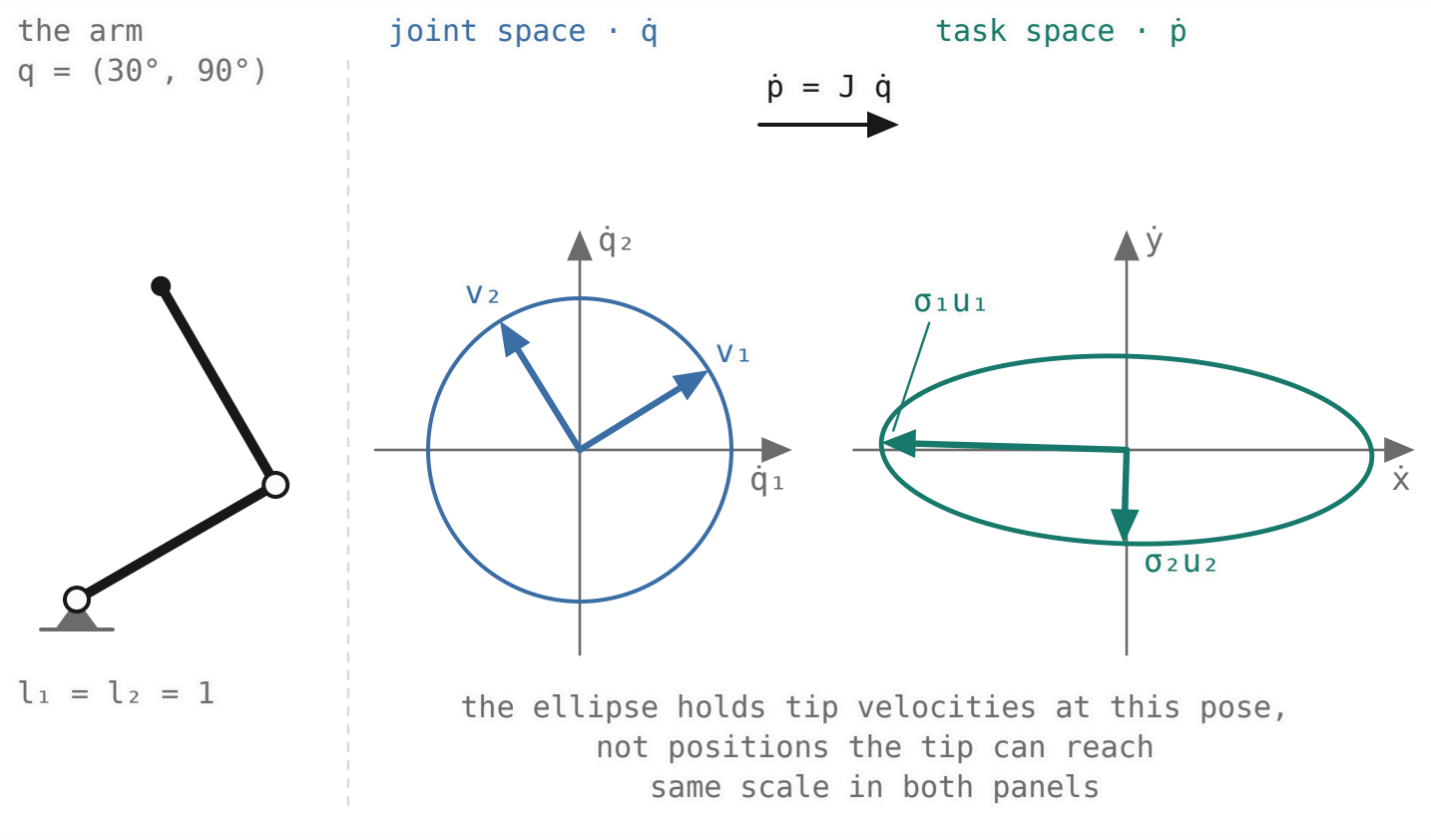
B. only vertical, across the arm

C. any direction in the plane

PICTURE

# Jacobians and Robot Motion

## The SVD of the Jacobian




$$J = U\Sigma V^{\mathsf{T}}, \qquad Jv_i = \sigma_i u_i$$

- $v_i$ unit joint velocity: both joints move together
- $u_i$ unit tip direction in task space
- $\sigma_i$ local gain at this $q$: 1.62 and 0.62

$$\dot{p} = J(q)\,\dot{q}$$

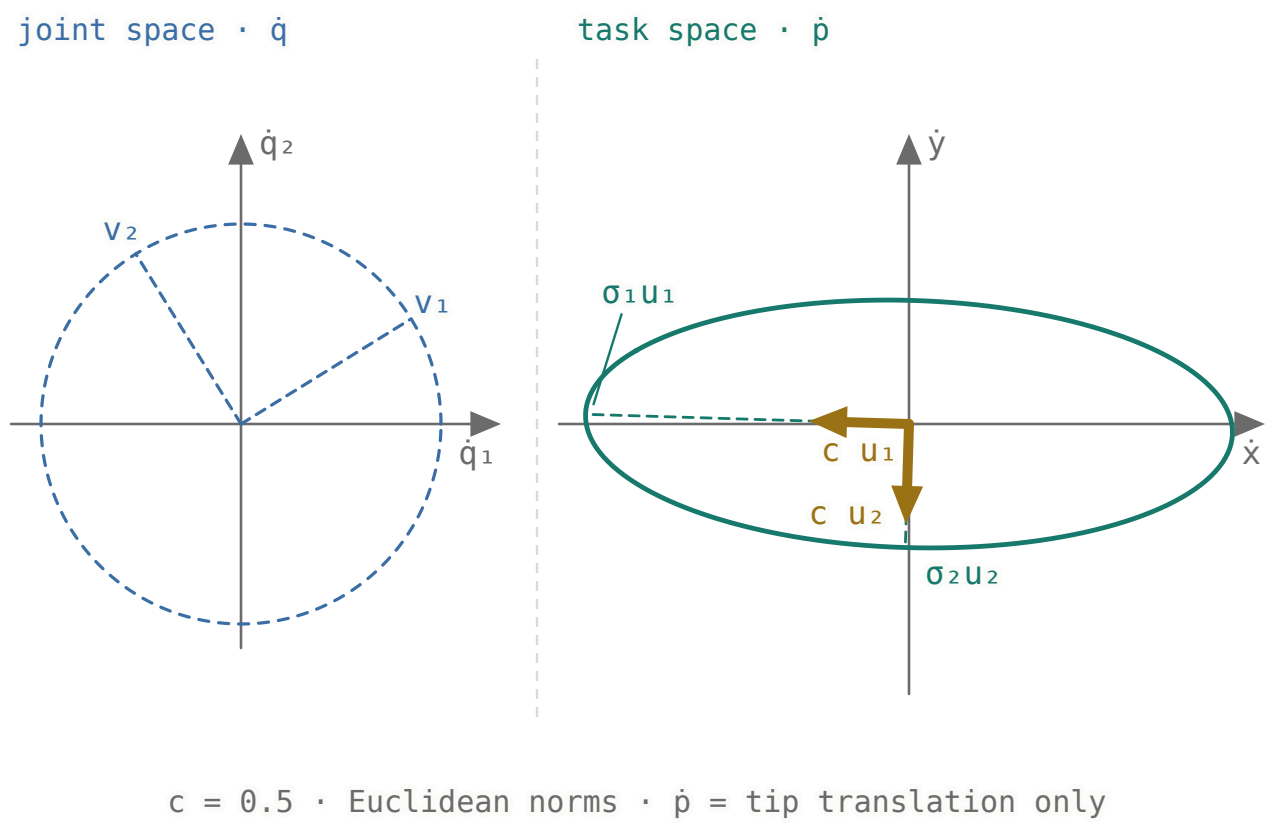
- $\dot{q}$, $\dot{p}$ joint and tip velocities; exact by the chain rule
- $\Delta p \approx J \Delta q$ finite steps: first order only

VOTE

$$J(30°,\ 90°) \approx \begin{bmatrix} -1.37 & -0.87 \\ 0.37 & -0.50 \end{bmatrix}$$

# Jacobians and Robot Motion

## Which Direction Needs Less Joint Speed?




The tip must move at the same speed $c$, along $u_1$ or $u_2$. Which needs the smaller joint speed $\|\dot{q}\|$?

A along $u_1$, the long axis

B along $u_2$, the short axis

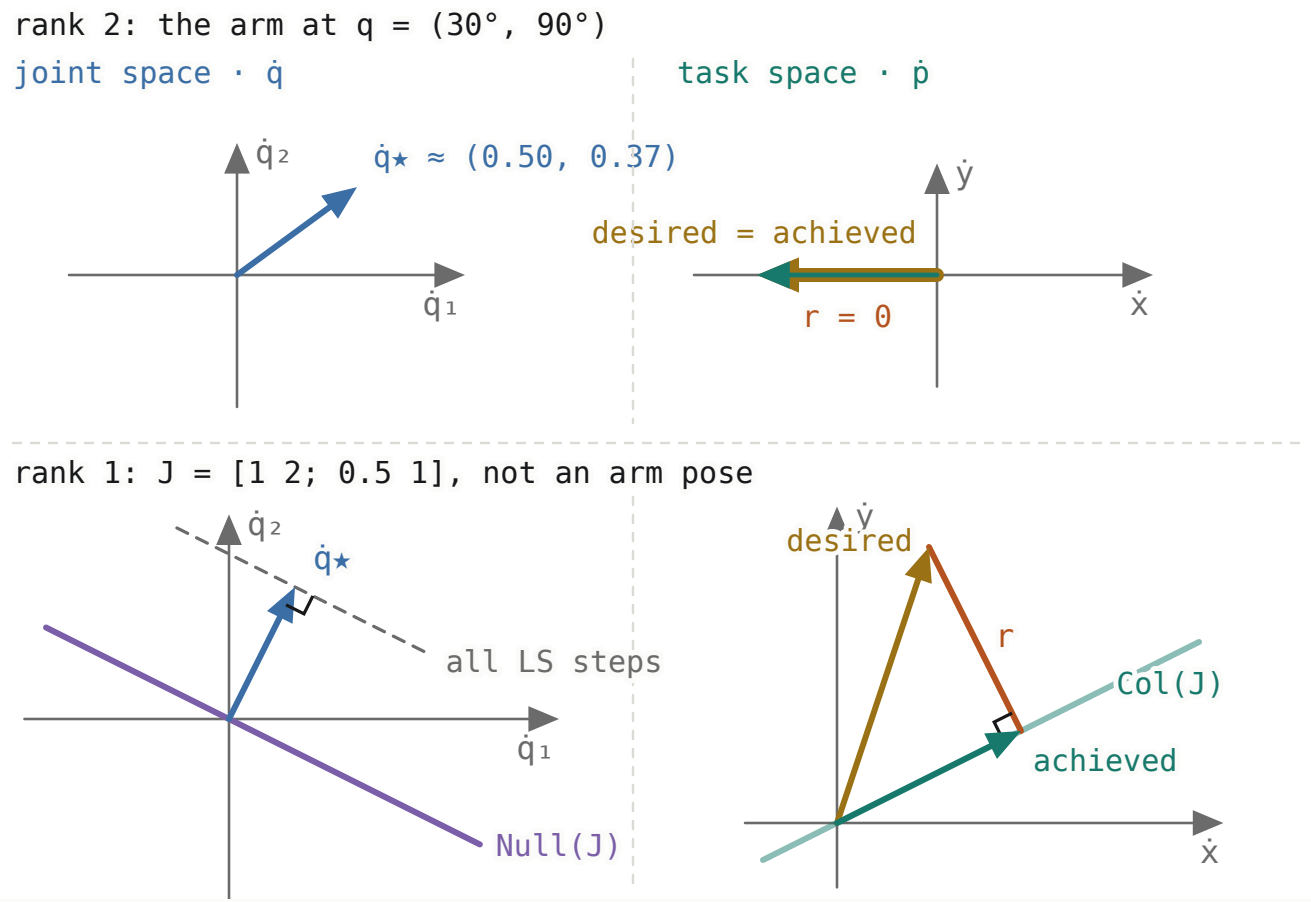
C the same for both

WORKED EXAMPLE

$$J(30°,90°) \approx \begin{bmatrix} -1.37 & -0.87 \\ 0.37 & -0.50 \end{bmatrix}$$

# Jacobians and Robot Motion

## Inverse Differential Kinematics




$$\dot{q}^\star = J^+ \dot{p}_\mathrm{d}$$

- $\dot{p}_\mathrm{d}$ desired tip velocity
- $J^+$ exact if possible, else least squares; min $\|\dot{q}\|$
- $r$ $\dot{p}_\mathrm{d} - J\dot{q}^\star$, desired − achieved

**EXAMPLE**

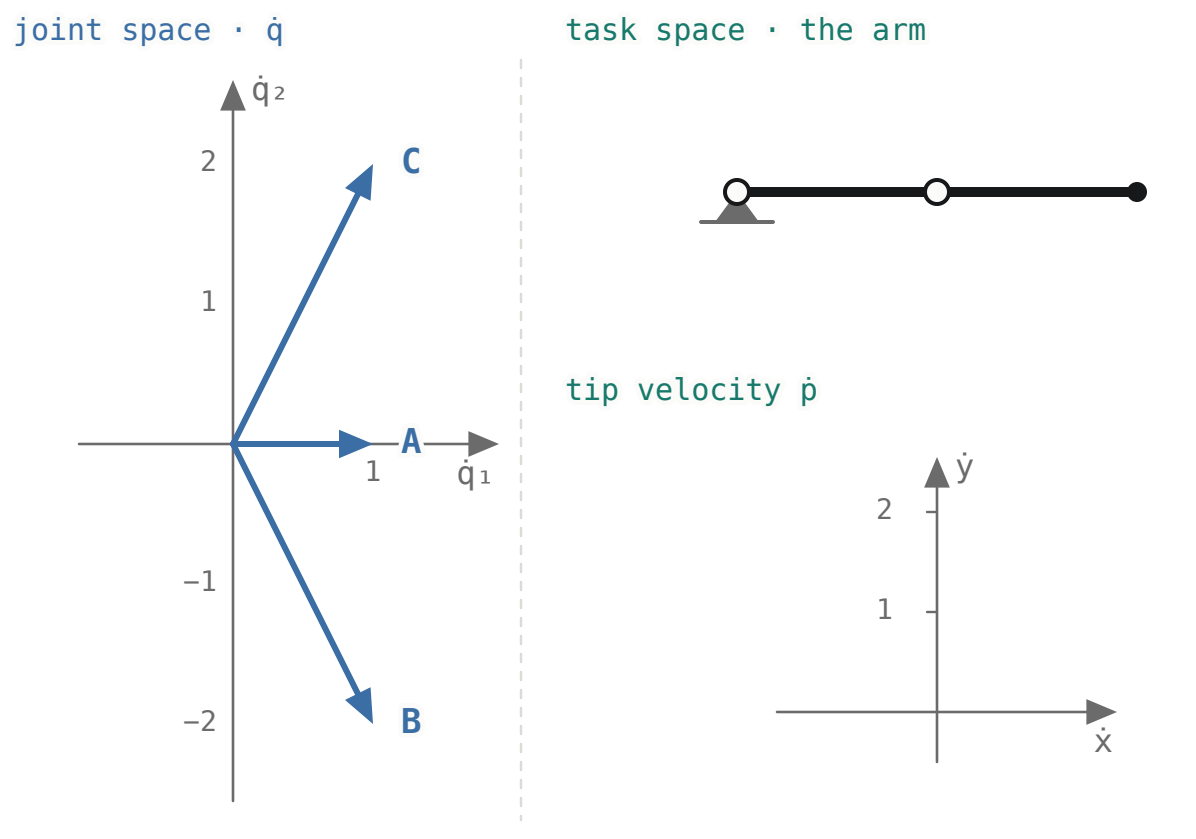
Rank 2: $(-1, 0)$ gets $\approx (0.50, 0.37)$; $r = 0$.

Rank 1: $(0.5, 1.5)$ gets $(0.2, 0.4)$, reaching $(1, 0.5)$; $r = (-0.5, 1)$.

VOTE

# Jacobians and Robot Motion

## Which Joint Velocity Leaves the Tip Still?




The straight arm again. Which joint velocity gives $J\dot{q} = 0$?

A $\dot{q} = (1, 0)$

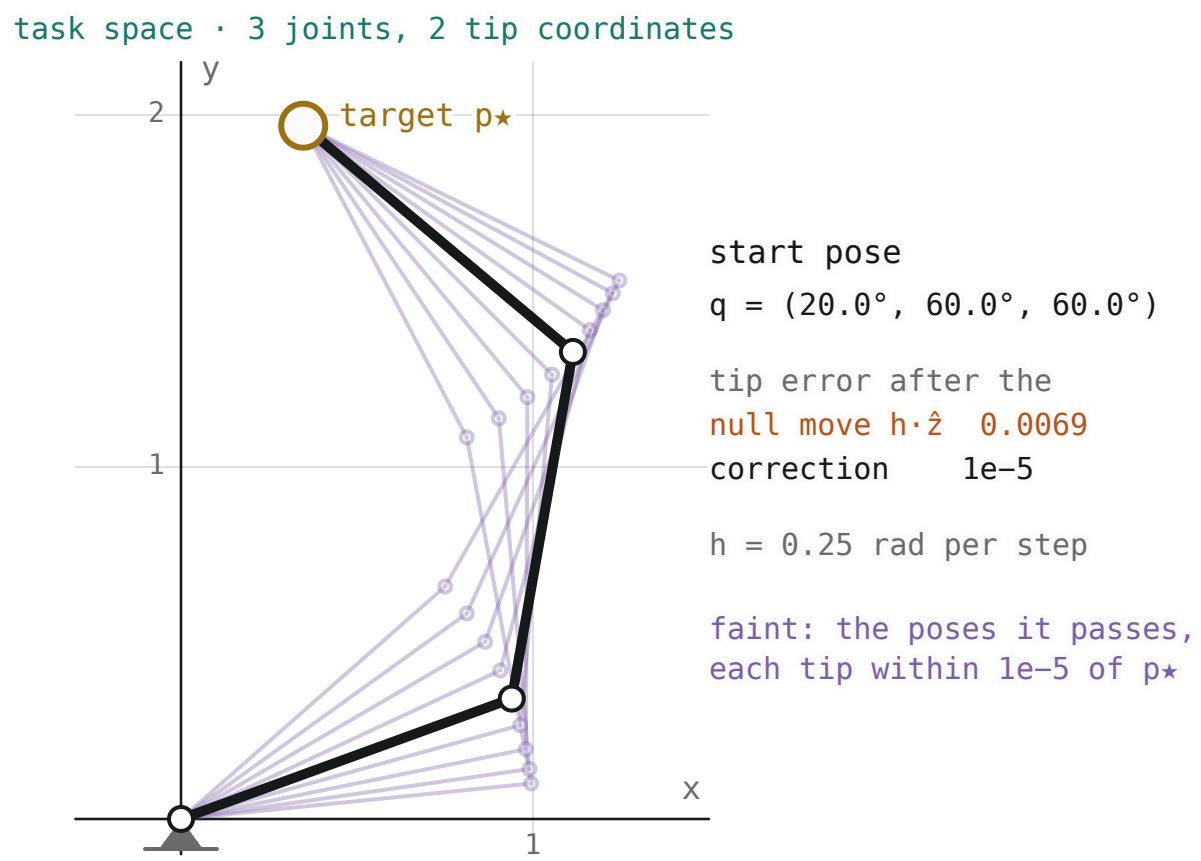
B $\dot{q} = (1, -2)$

C $\dot{q} = (1, 2)$

PICTURE

# Jacobians and Robot Motion

## Redundancy and Null Motion




$$\dot{q} = J^{+}\dot{p}_{\mathrm{d}} + z, \qquad Jz = 0$$

- $J$ $2 \times 3$: two tip coordinates, three joints, rank 2
- $z$ null motion, nullity $n - \operatorname{rank} J = 1$: $z$ contributes exactly zero tip velocity

**EACH FINITE STEP: FIRST ORDER ONLY**

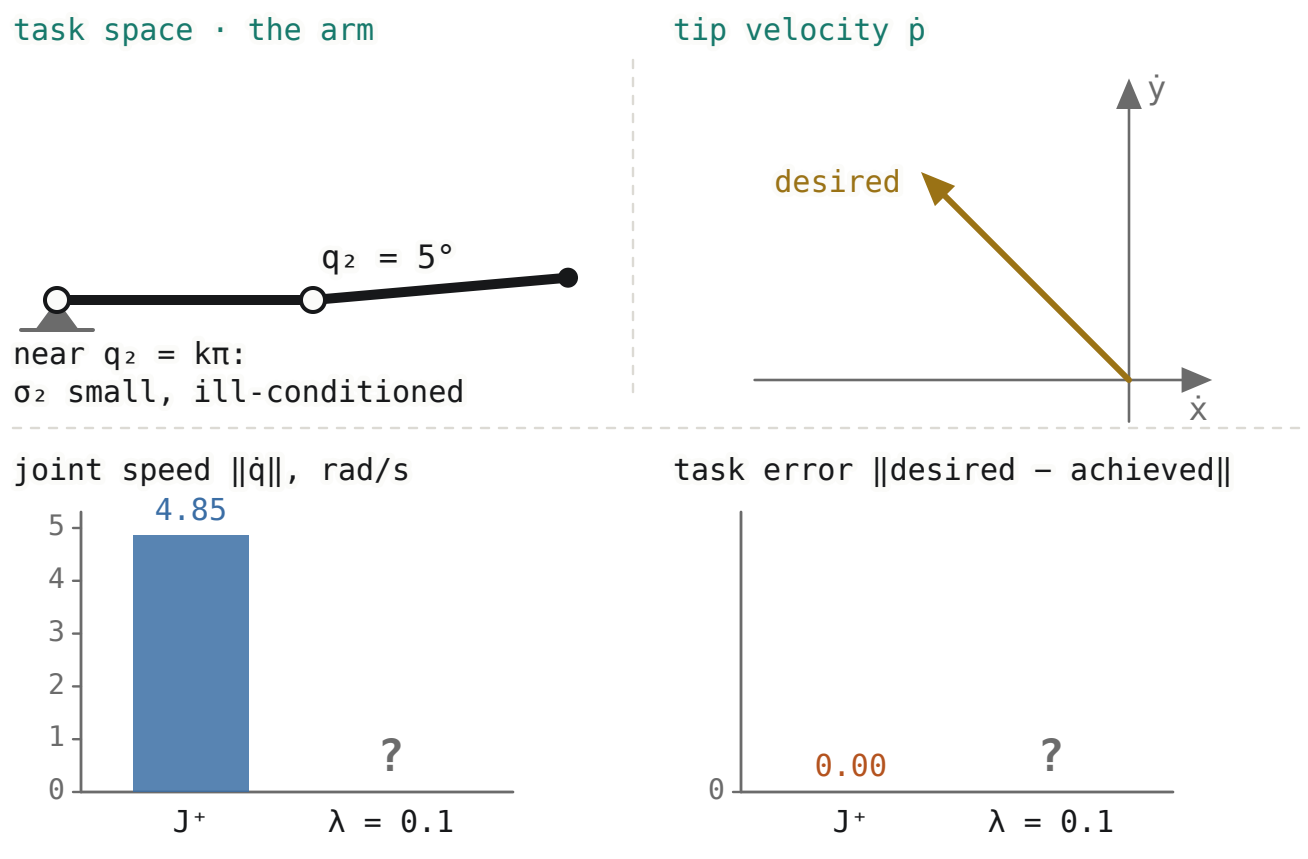
$q \leftarrow q + h\,\hat{z}(q)$, with $\hat{z}$ a unit vector in $\operatorname{Null} J(q)$

then correct: $q \leftarrow q + J^{+}\big(p^{\star} - p(q)\big)$

VOTE

# Jacobians and Robot Motion

## What Does Damping Do Near a Singularity?



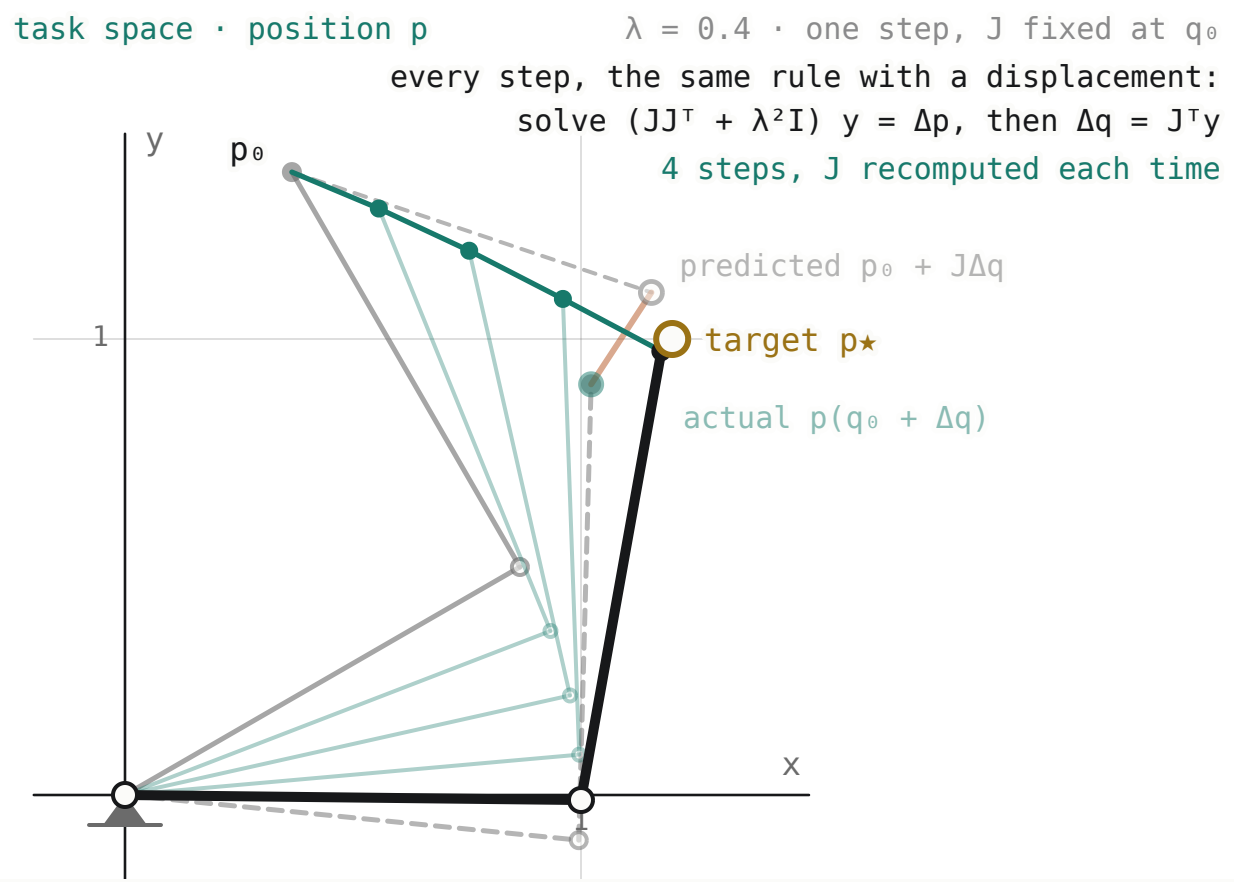

With $\lambda = 0.1$ the damped step minimises $\|J\dot{q} - \dot{p}_{\mathrm{d}}\|^2 + \lambda^2\|\dot{q}\|^2$. What does damping do?

- **A** hits every desired $\dot{p}_{\mathrm{d}}$ exactly
- **B** accepts task error to reduce joint speed
- **C** changes the arm's link lengths

IN PRACTICE

# Jacobians and Robot Motion

## Damping and Relinearization




$$\dot{q}_\lambda = J^\mathsf{T}(JJ^\mathsf{T} + \lambda^2 I)^{-1}\dot{p}_\mathrm{d}, \quad \lambda > 0$$

$J$ $m \times n$ at this $q$

$JJ^\mathsf{T} + \lambda^2 I$ $m \times m$, positive definite

| STEPS | LARGEST MODEL ERROR | FINAL TARGET ERROR |
|---|---|---|
| 1 | 0.24 | 0.20 |
| 4 | 0.024 | 0.036 |
| 8* | 0.0064 | 0.018 |

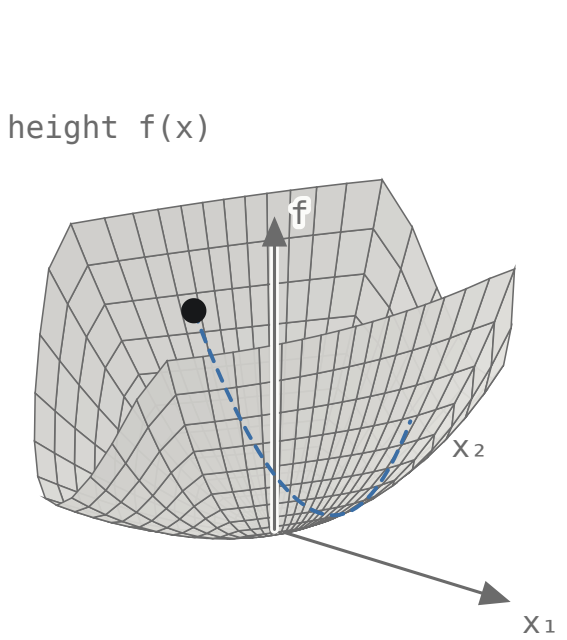
*computed, not animated. Smaller steps cut model error to leading order, not target error as fast; joint and speed limits are separate.

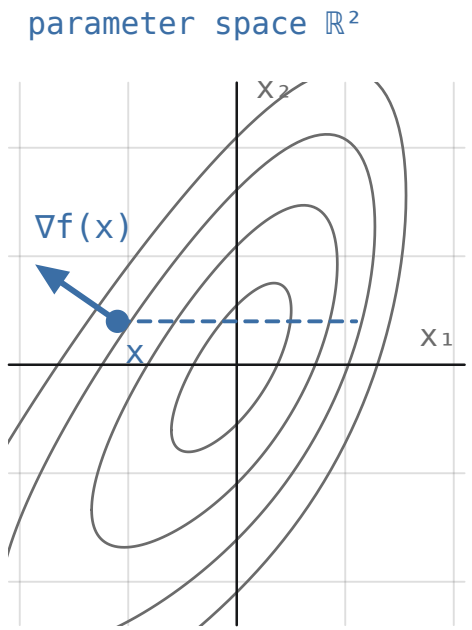
DEFINITION

# Hessians and Local Curvature

## From Vector Outputs to a Scalar Objective

$$f : \mathbb{R}^n \to \mathbb{R}, \qquad \nabla f(x) \in \mathbb{R}^n, \qquad H(x) = \nabla^2 f(x) \in \mathbb{R}^{n \times n}$$




- $f(x)$ one number, the objective at parameters $x$
- $\nabla f(x)$ $n \times 1$, the local slope, pointing uphill
- $H(x)$ $n \times n$, how $\nabla f$ changes as $x$ moves; same letter as the quadratic forms, on purpose
- $J(q)$ vector output $p(q) \in \mathbb{R}^m$: $m \times n$

VOTE

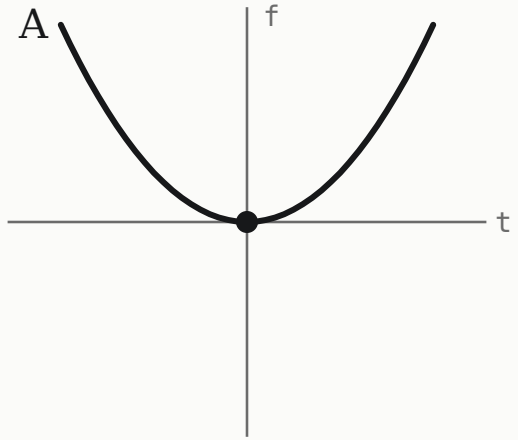
# Hessians and Local Curvature

## Which Curve Has a Local Minimum at 0?

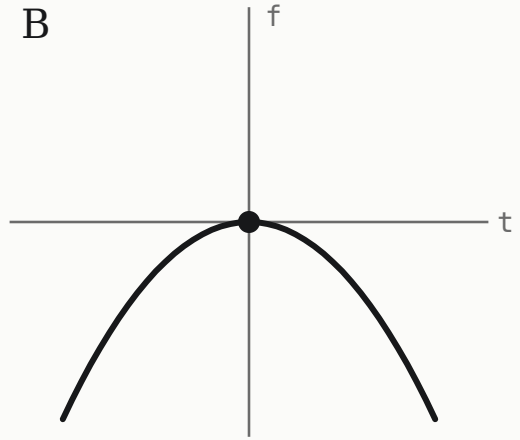
All three have $f'(0) = 0$. Which one has a local minimum at $t = 0$?

**parameter t ∈ ℝ, height f(t)**

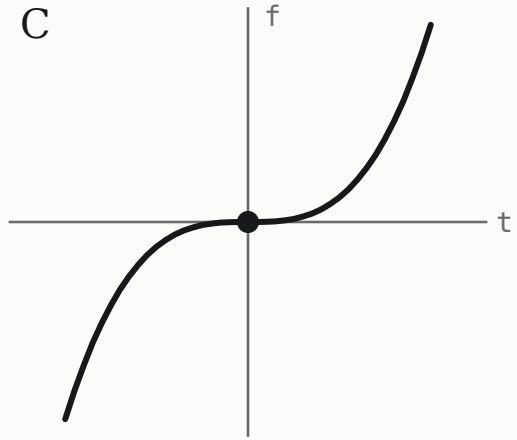
A




B




C




(A) $f(t) = t^2$

(B) $f(t) = -t^2$

(C) $f(t) = t^3$

DEFINITION

# Hessians and Local Curvature

## The Hessian Matrix

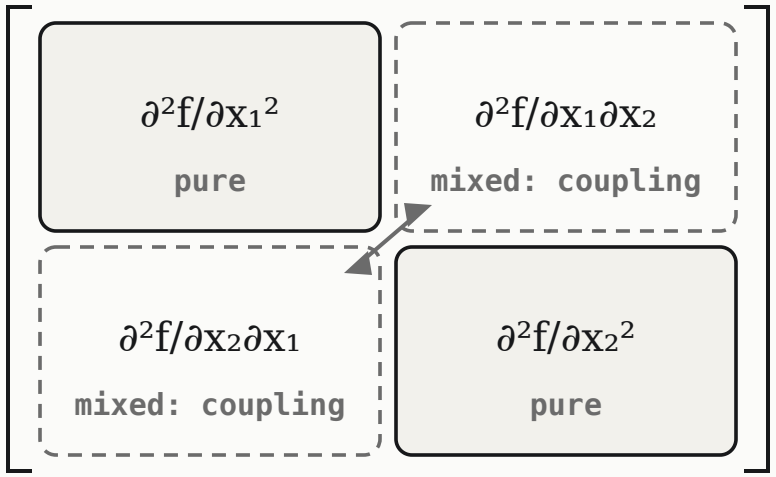
$$H_{ij}(x) = \frac{\partial^2 f}{\partial x_i \, \partial x_j}(x), \qquad i, j = 1, \ldots, n$$

$H_{ij}$ how the slope $\partial f / \partial x_j$ changes when $x_i$ moves

$f \in C^2$ second partials continuous near $x$

then $H_{ij} = H_{ji}$: $H = H^{\mathsf{T}}$, symmetric $n \times n$

H(x) =



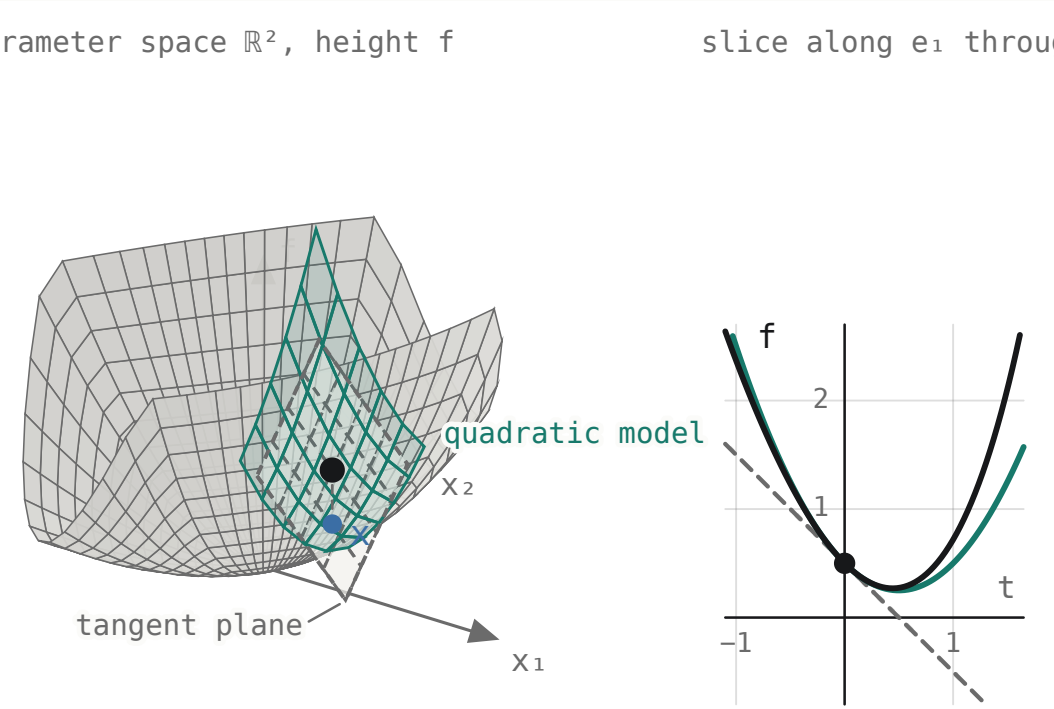

equal if f ∈ C²

MODEL

# Hessians and Local Curvature

## The Second-Order Approximation

$$f(x+\Delta x) \approx f(x) + \nabla f(x)^{\mathsf{T}} \Delta x + \frac{1}{2} \Delta x^{\mathsf{T}} H(x) \Delta x$$




- $x$ fixed expansion point; $f \in C^2$
- $\Delta x$ small step
- $\nabla f$ $n \times 1$, tilt: tangent plane
- $H$ $n \times n$, bend: quadratic model

**WORKED EXAMPLE**

$f = e^{x_1} - 1 - x_1 + \frac{1}{2}(x_2 - x_1)^2$, $x = (0, 1)$:

$f(x) = \frac{1}{2}$, $\nabla f = (-1, 1)$, $H = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$

VOTE

# Hessians and Local Curvature

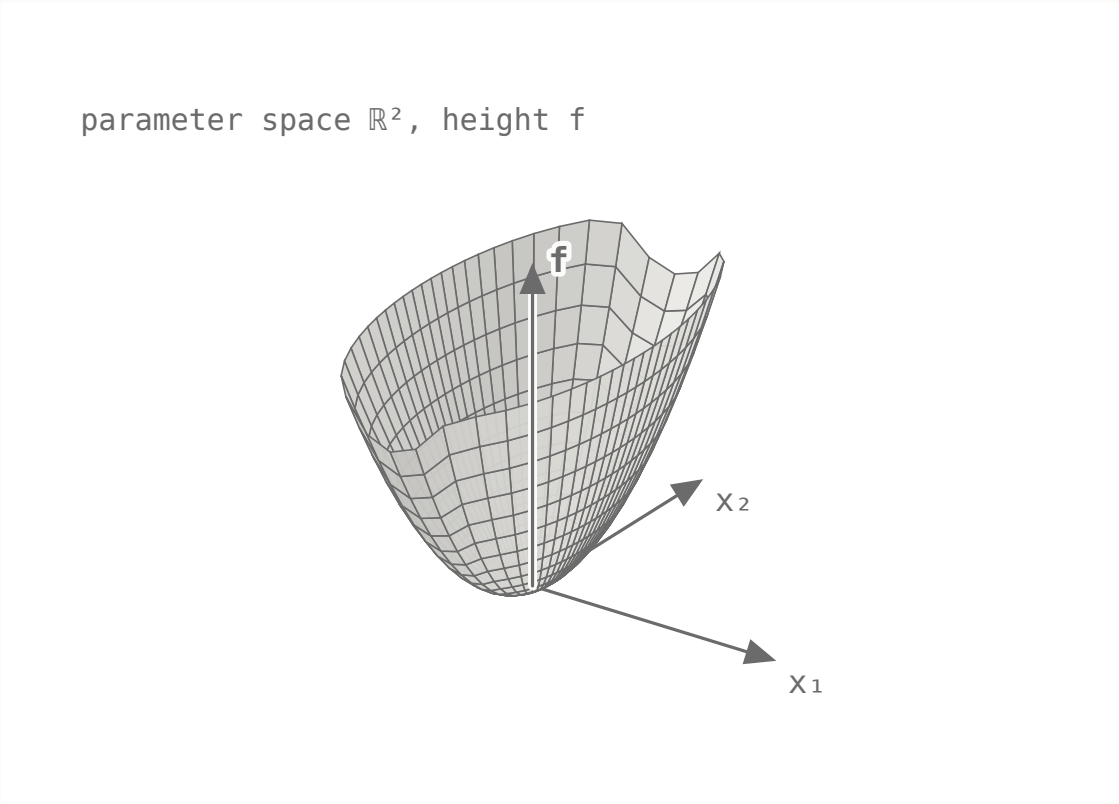
## What Is the Hessian of This Bowl?

$f(x_1, x_2) = 2x_1^2 + \frac{1}{2}x_2^2$. Which matrix is its Hessian?

A $\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$

B $\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$

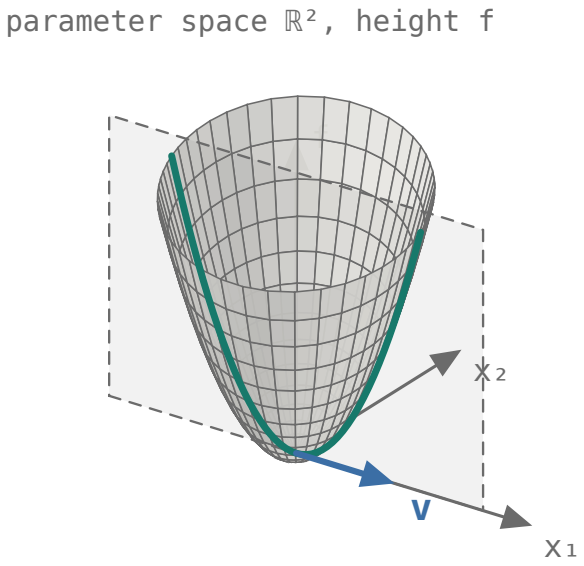
C $\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$

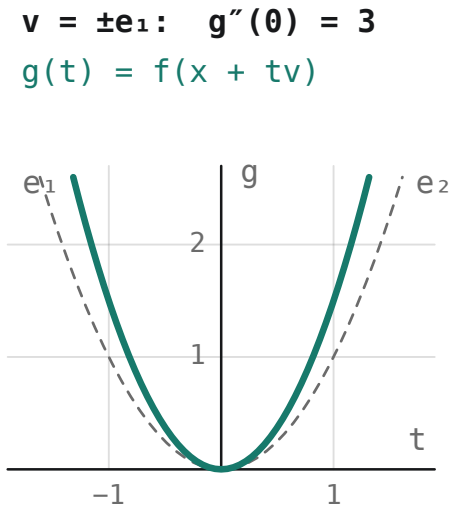


DEFINITION

# Hessians and Local Curvature

## Curvature Along a Direction




$$g(t) = f(x + tv), \qquad g''(0) = v^{\mathsf{T}} H(x)\, v$$

- $v$ direction, unit: $\|v\| = 1$, so directions compare fairly
- $g$ height of $f$ along the line through $x$
- $g''(0)$ second derivative along that line, not the surface's geometric curvature
- why $g'(t) = \nabla f(x + tv)^{\mathsf{T}} v$; differentiate once more
- here $f = \frac{1}{2} x^{\mathsf{T}} H x$, $H = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$, $x = 0$

$$H = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

VOTE

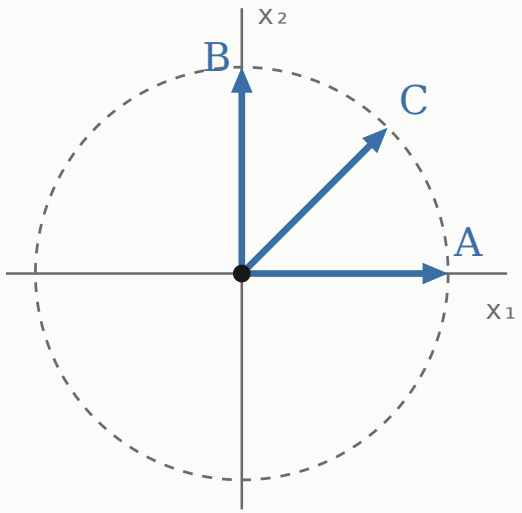
# Hessians and Local Curvature

## Which Direction Curves Up Most?

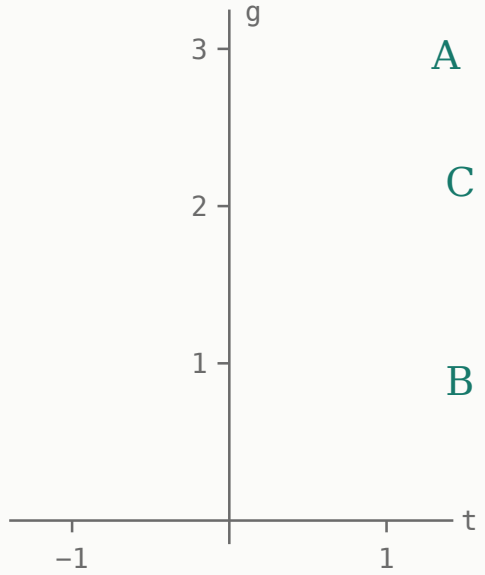
Along which unit $v$ is $g''(0) = v^\top H v$ largest?

- **A** $v = e_1$
- **B** $v = e_2$
- **C** $v = \frac{1}{\sqrt{2}}(1, 1)$, at $45°$

parameter space $\mathbb{R}^2$




g(t) = f(x + tv)

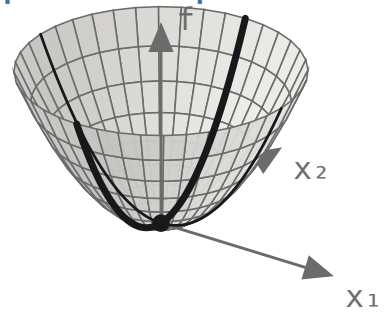


# Hessians and Local Curvature

## Classifying a Stationary Point

f = x₁² + x₂²

parameter space ℝ²

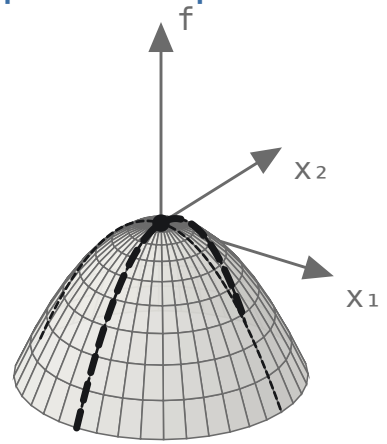



**λ₁, λ₂ > 0 (PD): strict local min**

vᵀHv = +2.00

f = −x₁² − x₂²

parameter space ℝ²

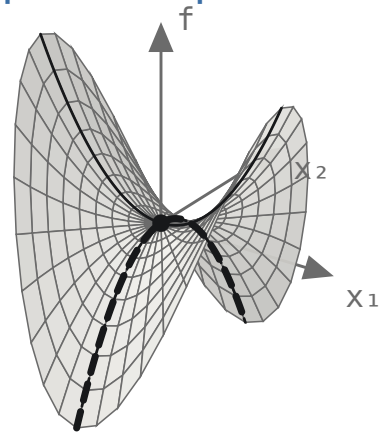



**λ₁, λ₂ < 0 (ND): strict local max**

vᵀHv = −2.00

f = x₁² − x₂²

parameter space ℝ²




**λ₁ > 0 > λ₂ (indefinite): saddle**

vᵀHv = −2.00

$$f(x^\star + \Delta x) \approx f(x^\star) + \tfrac{1}{2}\Delta x^\mathsf{T} H(x^\star)\Delta x$$

Local tests only. $f \in C^2$ on a convex open set: convex $\iff$ $H(x) \succeq 0$ at every $x$.

- $x^\star, \Delta x$ stationary point ($\nabla f = 0$), small step
- $f$ $C^2$ near $x^\star$: $H$ symmetric $n \times n$
- $\lambda_i$ eigenvalues of $H(x^\star)$: their signs decide

VOTE

# Hessians and Local Curvature

## Can a Semidefinite Hessian Decide?

$$f_{\pm}(x, y) = x^2 \pm y^4$$

$(x, y)$ any point; both are $C^2$

at $(0, 0)$ $\nabla f_{\pm} = 0$, $H_{\pm} = \mathrm{diag}(2, 0)$

Can this Hessian alone classify both points?

A. Yes: both are minima

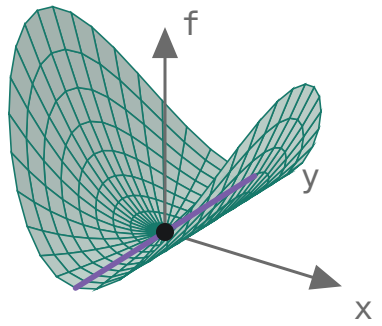
B. Yes: both are saddles

C. No: higher-order terms matter

parameter space $\mathbb{R}^2$

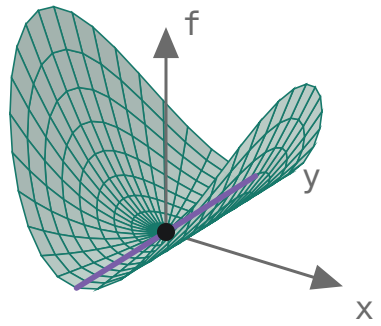
$f_+ = x^2 + y^4$

model at 0: $x^2$




$f_- = x^2 - y^4$

model at 0: $x^2$



DERIVATION

# Hessians and Local Curvature

## The Least-Squares Hessian

$$f(x) = \tfrac{1}{2}\|Ax - b\|^2, \qquad \nabla f(x) = A^\mathsf{T}(Ax - b), \qquad H = A^\mathsf{T}A$$

$H$ Hessian

$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = A$, $b = (1, 0)$: best output, shortest input, curvature?

Project onto $Ax = (1, 2)s$.
$s = 1/5$, $\hat{b} = (1, 2)/5$, $r = (4, -2)/5$

$x_1 + 2x_2 = 1/5$; shortest $x = (1, 2)/25$

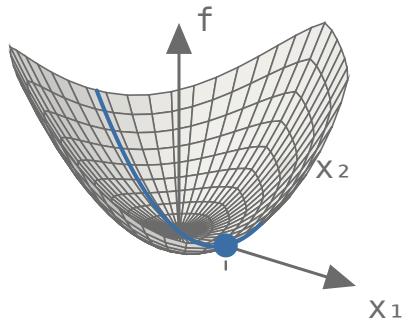
$\lambda(H) = 25, 0$
Flat direction $(2, -1)$; $\min f = 2/5$.

**surfaces: b = 0**

**A =** $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ **full column rank**

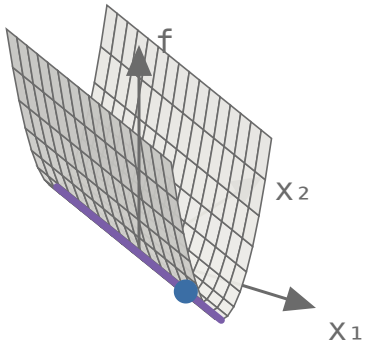
**A =** $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ **rank 1**

f = 0.50 height × 0.5




**λ = 3, 1**
one lowest point

f = 0.00 height × 0.5




**λ = 25, 0**
a line of lowest points

$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ $b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

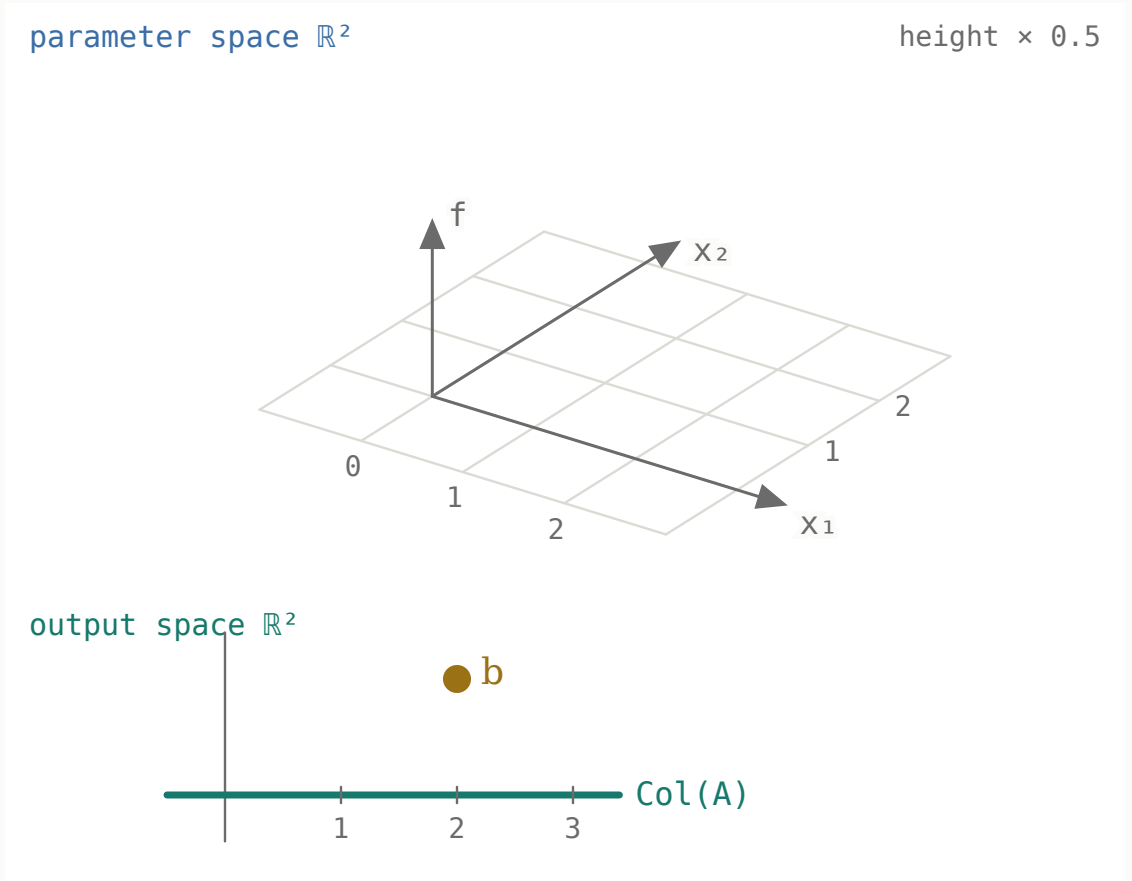
VOTE

# Hessians and Local Curvature

## What Shape Is This Least-Squares Objective?

$f = \frac{1}{2}\|Ax - b\|^2$. Choose a method; find $A\hat{x}, r, A^+b$, a flat direction and the graph shape.

- (A) A bowl with one lowest point
- (B) A trough, bottom: $x_1 + x_2 = 2$
- (C) A saddle



CONNECTION

$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ $b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

# Hessians and Local Curvature

## Closing the Loop: Hessian, Gram Matrix, SVD

$$H = A^\top A = V \Sigma^\top \Sigma V^\top, \quad H_\lambda = A^\top A + \lambda^2 I \succ 0$$

$V, \Sigma^\top\Sigma$ SVD of $A$; diagonal $\sigma_i^2$, then zeros

$H_\lambda$ Hessian of $\frac{1}{2}\|Ax - b\|^2 + \frac{\lambda^2}{2}\|x\|^2$, $\lambda > 0$

Damped LS as earlier, no halves: Hessian $2H_\lambda$, same minimiser.

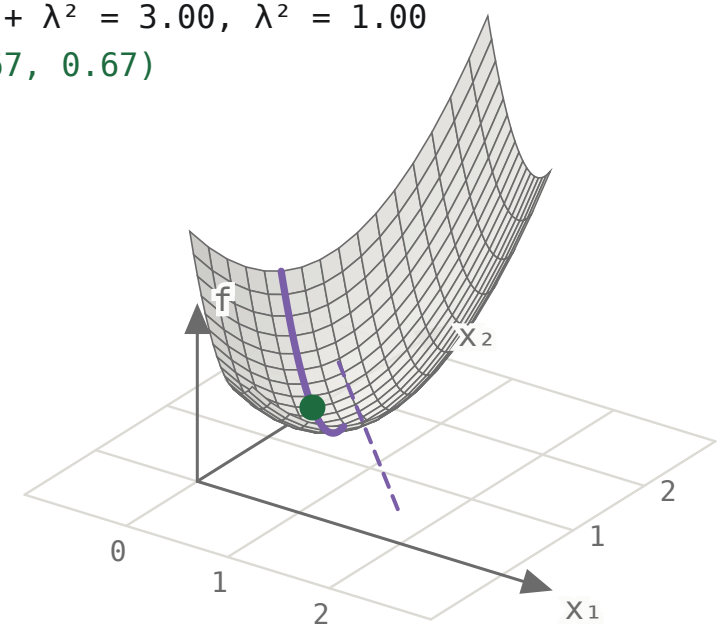
Next lecture: systematic derivatives.

parameter space $\mathbb{R}^2$ height × 0.5

**λ = 1.00**

eigenvalues 2 + λ² = 3.00, λ² = 1.00

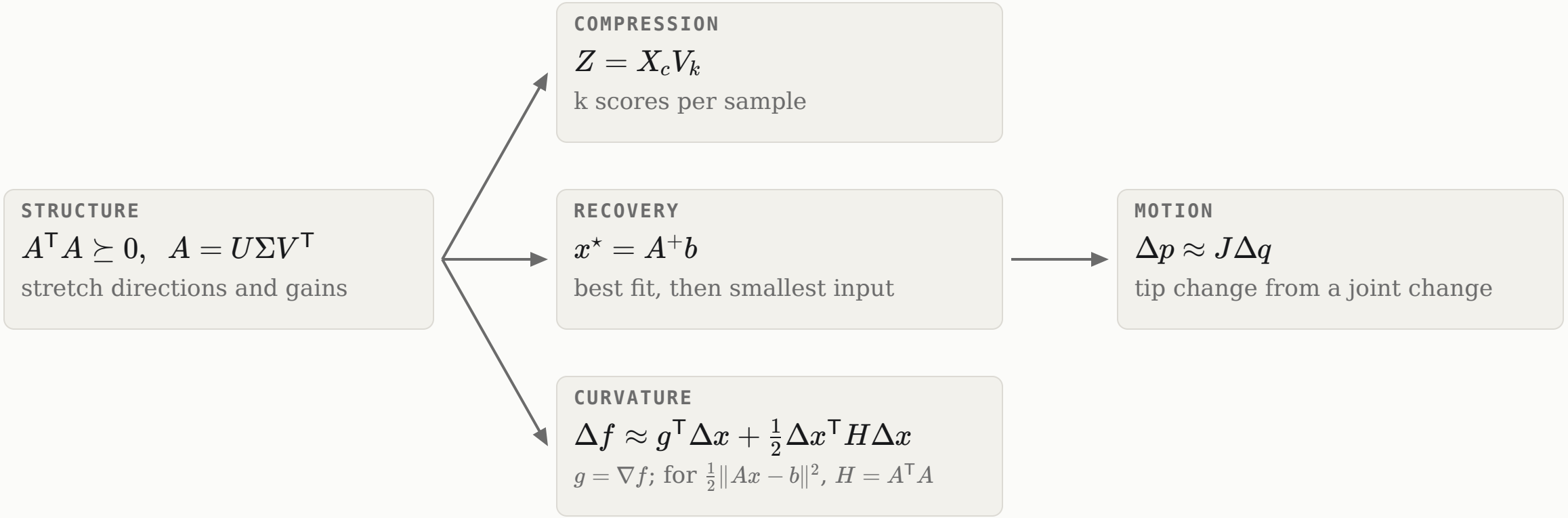
minimiser (0.67, 0.67)



CONCEPT MAP

# Connecting the Concepts

## One Matrix, Five Jobs



VOTE

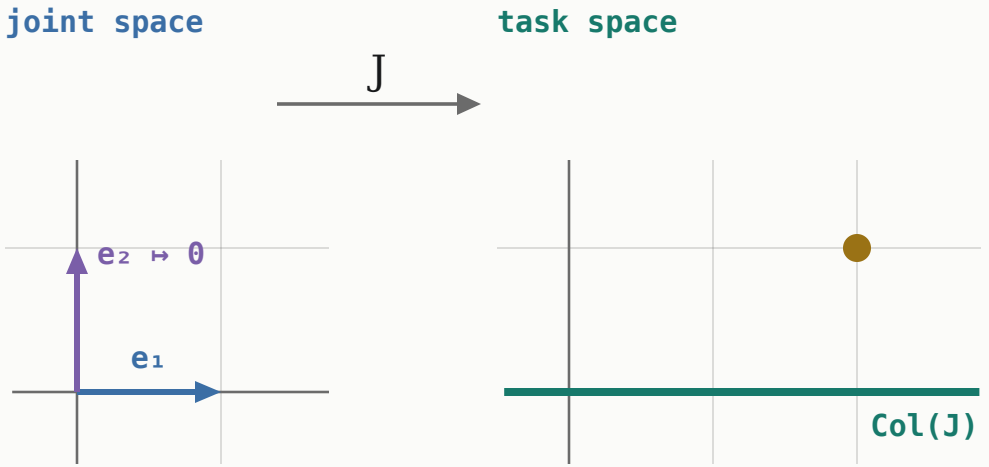
# Connecting the Concepts

## Exit Question: Rank, Step and Curvature

Linear model: $J = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, desired $\Delta p = (2, 1)$.

Which set is right?

- **A** rank 2, minimum-norm step $(1, 1)$, $J^\top J$ PD
- **B** rank 1, minimum-norm LS step $(1, 0)$, $J^\top J$ PSD but not PD
- **C** rank 1, minimum-norm step $(2, 0)$, target exactly reachable



SELECTED REFERENCES

# References


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2. Trefethen & Bau. *Numerical Linear Algebra*, Lec. 4–5, 11–12. SIAM, 1997.
3. Goodfellow et al. *Deep Learning*, 2.7–2.9, 2.12, 4.2–4.3. MIT Press, 2016.
4. Lynch & Park. *Modern Robotics*, 5.1, 5.3–5.4, 6.2–6.3. Cambridge, 2017.
5. Buss, S. R. *Introduction to Inverse Kinematics*. UCSD, 2004.



Course standards: *Notation and Conventions*, symbols, shapes and the two sign conventions; *Textbooks and Reading Guide*, what each source is for and where its notation differs.